

RL Notes

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1 Reproducing Kernel Hilbert Space

Definition. A Hilbert space $\mathcal{H}$ is a vector space equipped with an inner product $\langle \cdot, \cdot \rangle$, which is an operation with the following properties

1. Symmetry

$$\langle u, v \rangle = \langle v, u \rangle \quad \forall u, v \in \mathcal{H}$$

2. Bi-linearity

$$\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle \quad \forall u, v, w \in \mathcal{H}, \forall \alpha, \beta \in \mathbb{R}$$

3. Positive Semi-definiteness

$$\langle u, u \rangle \geq 0 \quad \forall u \in \mathcal{H}$$

$$\langle u, u \rangle = 0 \iff u = 0$$

For simplicity, we will only consider real scalars and define a vector space as a space of functions from $\mathcal{X} \rightarrow \mathbb{R}$. Note that this definition also extends to normal vector spaces like $\mathbb{R}^n$, where we can think of $\mathbb{R}^n$ as a space of all functions that map from $\{1, \dots, n\} \rightarrow \mathbb{R}$.

Definition. A functional L defined on $\mathcal{H}$ is a function that maps vectors $f \in \mathcal{H}$ to a scalar. There are many types of functionals such as an integration functional and differentiation at a point functional. An evaluation functional L_x is one that satisfies

$$L_x(f) = f(x) \quad \forall f \in \mathcal{H}$$

Definition. A linear functional L is a functional that satisfies

1. Additivity

$$L(u + v) = L(u) + L(v) \quad \forall u, v \in \mathcal{H}$$

2. Homogeneity

$$L(\alpha u) = \alpha L(u) \quad \forall u \in \mathcal{H} \text{ and } \alpha \in \mathbb{R}$$

Using this definition, we know that all evaluation functionals are linear. As a side note, the dual space of a vector space V is the set of all linear functionals on V .

Definition. The Riesz Representation Theorem states that every **continuous** linear functional L defined on $\mathcal{H}$ can be defined as an inner product with an unique fixed $g \in \mathcal{H}$, e.g.

$$L(f) = \langle f, g \rangle$$

This means that all continuous evaluation functionals L_x have unique $k_x = k(x, \cdot) \in \mathcal{H}$ such that

$$f(x) = L_x(f) = \langle k(x, \cdot), f(\cdot) \rangle = \langle f(\cdot), k(x, \cdot) \rangle \quad \forall f \in \mathcal{H}$$

where the second equality is due to symmetry of the inner product.

Definition. Moreover, $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a kernel function. A kernel function is a function with the following properties

1. Symmetry

$$k(x, y) = k(y, x)$$

2. A Positive-Semi-definite Gram Matrix

$$\forall n \in \mathbb{N}, \text{ any subset } S \subseteq \{x_1, x_2, \dots, x_n\} \in \mathcal{X}$$

must have a "Gram Matrix" K defined by $K_{ij} = k(x_i, x_j)$ that positive semi-definite. (A matrix $K \in \mathbb{R}^{n \times n}$ is positive semi-definite if $a^T K a \geq 0 \quad \forall a \in \mathbb{R}^n$)

We can show k that is used to characterize the evaluation functional for $\mathcal{H}$ is a kernel function by proving these two properties. Starting with symmetry, since the inner product is symmetric $k_x = k(x, \cdot)$ characterizes an evaluation functional

$$k(x, y) = \langle k(x, \cdot), k(y, \cdot) \rangle = \langle k(y, \cdot), k(x, \cdot) \rangle = k(y, x) \quad \forall x, y \in \mathcal{X}$$

Moreover, for any subset $S \subseteq \{x_1, x_2, \dots, x_n\}$ and its corresponding gram matrix K , we have that

$$\begin{aligned} a^T K a &= \sum_i^n \sum_j^n a_i a_j \langle k(x_i, \cdot), k(x_j, \cdot) \rangle \\ &= \left\langle \sum_i^n a_i k(x_i, \cdot), \sum_j^n a_j k(x_j, \cdot) \right\rangle \\ &= \left\langle \sum_i^n a_i k(x_i, \cdot), \sum_i^n a_i k(x_i, \cdot) \right\rangle \quad (\text{switch variable name}) \\ &\geq 0 \quad (\text{by positive semi-definiteness of the inner norm}) \end{aligned}$$

Note that due to symmetry of the *kernel* function $k(x, \cdot) = k(\cdot, x)$, meaning

$$f(x) = L_x(f) = \langle k(x, \cdot), f(\cdot) \rangle = \langle k(\cdot, x), f(\cdot) \rangle$$

Definition 1. A Reproducing Kernel Hilbert Space (RKHS) is a Hilbert space $\mathcal{H}$ where all evaluation functionals are continuous. Linear functions that are bounded are also continuous and vice versa. Since evaluation functionals are linear, we could replace the "continuous" with "bounded" in the definition above as well.

Definition 2. A Reproducing Kernel Hilbert Space (RKHS) is a Hilbert space $\mathcal{H}$ with a reproducing kernel. A reproducing kernel is a kernel that satisfies the following conditions

1. k has the reproducing property

$$f(x) = \langle k(x, \cdot), f(\cdot) \rangle \quad \forall x \in \mathcal{X}, \forall f \in \mathcal{H}$$

2. $\mathcal{H}$ is the **closure** span of the set of kernel functions centered at all possible points (e.g. $\mathcal{H}_0 \text{span}\{k(x, \cdot) : x \in \mathcal{X}\}$ is dense in $\mathcal{H}$).

$$\mathcal{H} = \overline{\text{span}\{k(x, \cdot) : x \in \mathcal{X}\}}$$

Note that the first condition of definition 2 technically implies the second condition of definition 2. The closure span does not only include finite linear combinations of centered kernel functions but also limit of a sequence of these linear combinations since $\mathcal{H}_0$ is **dense** in $\mathcal{H}$ if it satisfies any of the following equivalent conditions:

1. $\forall f \in \mathcal{H}, \forall \epsilon > 0, \exists f_0 \in \mathcal{H}_0$ such that $\|f - f_0\|_{\mathcal{H}} < \epsilon$
2. $\forall f \in \mathcal{H}, \exists \{f_n\}_{n=1}^\infty \subset \mathcal{H}_0$ such that $\lim_{n \rightarrow \infty} \|f - f_n\|_{\mathcal{H}} = 0$

Assume for contradiction there exists a $f \in \mathcal{H}$ that is not in $\overline{\text{span}\{k(x, \cdot) : x \in \mathcal{X}\}}$. f can be decomposed into $f = u + v$, where $u \in \overline{\text{span}\{k(x, \cdot) : x \in \mathcal{X}\}}$ and $v \in \overline{\text{span}\{k(x, \cdot) : x \in \mathcal{X}\}}^\perp$. This means

$$\begin{aligned} \langle k(x, \cdot), v(\cdot) \rangle &= 0 \quad \forall x \in \mathcal{X} \\ \implies v(x) &= 0 \quad \forall x \in \mathcal{X} \\ \implies v &= 0 \\ f &= u + v = u \in \overline{\text{span}\{k(x, \cdot) : x \in \mathcal{X}\}} \end{aligned}$$

which is a contradiction.

It can be proved that "Definition 1" implies "Definition 2" (with the help of the Riesz Representation Theorem) and that "Definition 2" also implies "Definition 1".

Property of RKHS. For each RKHS, there is a unique reproducing kernel k . Assume there is another reproducing kernel k' . We know that

$$\begin{aligned} f(x) &= \langle k(x, \cdot), f(\cdot) \rangle = \langle k'(x, \cdot), f(\cdot) \rangle \quad \forall x \in \mathcal{X}, \forall f \in \mathcal{H} \\ \implies \langle k(x, \cdot) - k'(x, \cdot), f(\cdot) \rangle &= 0 \quad \forall x \in \mathcal{X}, \forall f \in \mathcal{H} \end{aligned}$$

Since $k(x, \cdot) - k'(x, \cdot)$ is orthogonal to all functions $f \in \mathcal{H}$ including itself (since $k(x, \cdot) - k'(x, \cdot) \in \mathcal{H}$).

$$\|k(x, \cdot) - k'(x, \cdot)\| = 0$$

$$\implies k'(x, \cdot) - k(x, \cdot) = 0 \quad \text{by the definition of the inner product}$$

$$\implies k'(x, \cdot) = k(x, \cdot)$$

$$\implies k'(x, y) = k(x, y) \quad \forall x, y \in \mathcal{X} \quad (\text{equal functions are the same evaluated at all points } y)$$

Property of RKHS. Each kernel function K can only be associated with one Hilbert space $\mathcal{H}$ of functions $f : \mathcal{X} \rightarrow \mathbb{R}$. For any two Hilbert spaces H_1 and H_2 that share the same reproducing kernel k , we know that $H_0 = \text{span}\{k(x, \cdot) : x \in \mathcal{X}\}$ is dense on both Hilbert spaces. For any $f, g \in \mathcal{H}_0$

$$\begin{aligned} f &= \sum_i^n a_i k(x_i, \cdot) \\ g &= \sum_j^m b_j k(x_j, \cdot) \\ \implies \langle f, g \rangle_{H_1} &= \sum_i^n \sum_j^m a_i b_j \langle k(x_i, \cdot), k(x_j, \cdot) \rangle_{H_1} \\ &= \sum_i^n \sum_j^m a_i b_j k(x_i, x_j) \\ &= \sum_i^n \sum_j^m a_i b_j \langle k(x_i, \cdot), k(x_j, \cdot) \rangle_{H_2} \\ &= \langle f, g \rangle_{H_2} \end{aligned}$$

The values of a continuous function on vectors from a vector space V are fully determined by its values a dense subspace $U \subseteq V$, meaning the inner product function is the same for both H_1 and H_2 (inner product is function over vectors and the vectors are functions in this case).

Example of RKHS 1. One such example is $\mathcal{H} = \mathbb{R}^n$ where we can think of $\mathbb{R}^n$ as the vector space of all functions mapping $X = \{1, \dots, n\}$ to $\mathbb{R}$, e.g.

$$\mathcal{H} = \mathbb{R}^n = \{f : \{1, \dots, n\} \rightarrow \mathbb{R}\}$$

$k(x, \cdot)$, the reproducing kernel centered at x that characterizes the evaluation functional L_x is defined as $k(x, \cdot) = \mathbf{1}[\cdot = x] = \delta(x - \cdot)$ (Kronecker Delta as defined in digital signal processing or DSP), e.g.

$$e_x = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

where the 1 is position at index x (one indexed since we defined $X = \{1, \dots, n\}$).

$$f(x) = \langle k(x, \cdot), f(\cdot) \rangle$$

and

$$k(x, y) = \langle k(x, \cdot), k(y, \cdot) \rangle = e_x^T e_y = \delta_{xy}$$

Therefore, the kernel function is just the Kronecker Delta (function of 2 inputs when mentioned in a context outside of DSP).

Example of RKHS 2. Another example is

$$\mathcal{H} = \{f : \mathbb{R} \rightarrow \mathbb{R} : f(t) = \sum_{n=0}^{\infty} a_n \frac{t^n}{\sqrt{n!}} e^{-\frac{t^2}{2}}; \sum_{n=0}^{\infty} a_n^2 < \infty\} (\text{the coefficients } a_k \text{ must be square summable})$$

with the inner product defined as

$$\begin{aligned}\langle f, g \rangle &= \left\langle \sum_{n=0}^{\infty} a_n \frac{t^n}{\sqrt{n!}} e^{-\frac{t^2}{2}}, \sum_{n=0}^{\infty} b_n \frac{t^n}{\sqrt{n!}} e^{-\frac{t^2}{2}} \right\rangle \\ &= \sum_{n=0}^{\infty} a_n b_n\end{aligned}$$

Note Cauchy-Schwarz inequality holds for all Hilbert spaces, we get that

$$\begin{aligned}\langle f, g \rangle &= \sum_{n=0}^{\infty} a_n b_n \\ &\leq \sqrt{\langle f, f \rangle} \sqrt{\langle g, g \rangle} \\ &\leq \sqrt{\sum_{n=0}^{\infty} a_n^2} \sqrt{\sum_{n=0}^{\infty} b_n^2} \\ &\leq \infty\end{aligned}$$

To get the function $k(x, \cdot) \in \mathcal{H}$ associated for the evaluation functional L_x , $k(x, \cdot)$ must satisfy

$$\begin{aligned}\langle k(x, \cdot), f \rangle &= f(x) \\ &= \sum_{n=0}^{\infty} a_n \frac{x^n}{\sqrt{n!}} e^{-\frac{x^2}{2}} \\ &= \left\langle \sum_{n=0}^{\infty} \frac{x^n}{\sqrt{n!}} e^{-\frac{x^2}{2}} \frac{t^n}{\sqrt{n!}} e^{-\frac{t^2}{2}}, \sum_{n=0}^{\infty} a_n \frac{t^n}{\sqrt{n!}} e^{-\frac{t^2}{2}} \right\rangle\end{aligned}$$

meaning

$$\begin{aligned}k(x, y) &= \langle k(x, \cdot), k(y, \cdot) \rangle \\ &= \left\langle \sum_{n=0}^{\infty} \frac{x^n}{\sqrt{n!}} e^{-\frac{x^2}{2}}, \sum_{n=0}^{\infty} \frac{y^n}{\sqrt{n!}} e^{-\frac{y^2}{2}} \right\rangle \\ &= \left\langle \sum_{n=0}^{\infty} \frac{x^n}{\sqrt{n!}} e^{-\frac{x^2}{2}} t^n, \sum_{n=0}^{\infty} \frac{y^n}{\sqrt{n!}} e^{-\frac{y^2}{2}} t^n \right\rangle \\ &= \left\langle \sum_{n=0}^{\infty} \frac{x^n}{\sqrt{n!}} e^{-\frac{x^2}{2}} \frac{y^n}{\sqrt{n!}} e^{-\frac{y^2}{2}} \right\rangle \\ &= \left\langle e^{-\frac{x^2+y^2}{2}} \sum_{n=0}^{\infty} \frac{(xy)^n}{n!} \right\rangle \\ &= \left\langle e^{-\frac{x^2+y^2}{2}} e^{xy} \right\rangle \\ &= \left\langle e^{-\frac{x^2+2xy+y^2}{2}} \right\rangle \\ &= e^{-\frac{(x+y)^2}{2}}\end{aligned}$$

which is known as the Gaussian radial basis function (RBF). Note that

$$\begin{aligned}k(x, x) &= \sum_{n=0}^{\infty} \left(\frac{x^n}{\sqrt{n!}} e^{-\frac{x^2}{2}} \right)^2 \\ &= e^{-x^2} \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} \\ &= e^{-x^2} e^{x^2} \\ &= 1 < \infty\end{aligned}$$

so $k(x, \cdot) \in \mathcal{H}$

Example of RKHS 3. Another example is

$$\mathcal{H} = \{f : \mathbb{R} \rightarrow \mathbb{R} : f(t) = a_0 t^2 + a_1 \sqrt{2}t + a_2\}$$

which is the set of polynomials of degree 0, 1, 2. We can define the inner product as

$$\begin{aligned} \langle f, g \rangle &= \langle a_0 t^2 + a_1 \sqrt{2}t + a_2, b_0 t^2 + b_1 \sqrt{2}t + b_2 \rangle \\ &= \sum_{n=0}^2 a_n b_n \end{aligned}$$

To get the function $k(x, \cdot) \in \mathcal{H}$ associated for the evaluation functional L_x , $k(x, \cdot)$ must satisfy

$$\begin{aligned} \langle k(x, \cdot), f \rangle &= f(x) \\ &= a_0 x^2 + a_1 \sqrt{2}x + a_2 \\ &= \langle x^2 t^2 + \sqrt{2} \sqrt{2}xt + 1 * 1, a_0 t^2 + a_1 \sqrt{2}t + a_2 \rangle \end{aligned}$$

meaning $k(x, t) = x^2 t^2 + 2xt + 1 = (1 + xt)^2$.

Note that the kernel functions in examples 2 and 3 are precisely the Gaussian and polynomial kernels used in ML, where x and t would be vectors. With the ML kernels, rather than taking dot products between features directly, we are essentially taking dot products between those two features' linear functions in other Hilbert spaces.

2 Integral Probability Metric

Integral probability metrics (IPM) are distance functions between probability distributions over a set $\mathcal{X}$. Each IPM is associated with a class of real-valued functions $\mathcal{F}$

$$d_{\mathcal{F}}(P, Q) = \sup_{f \in \mathcal{F}} |\mathbb{E}_{X \sim P}[f(X)] - \mathbb{E}_{Y \sim Q}[f(Y)]|$$

where both X and Y takes values from $\mathcal{X}$

IPM Example 1. Total Variation Distance We get D_{TV} with

$$\mathcal{F} = \{f : \|f\|_{\infty} \leq 1\}$$

$$\begin{aligned} D_{TV}(P, Q) &= \sup_{f \in \mathcal{F}} |\mathbb{E}_{X \sim P}[f(X)] - \mathbb{E}_{X \sim Q}[f(X)]| \quad (\text{note we just switched variable names from Y to X}) \\ &= |\mathbb{E}_{X \sim P} [\mathbb{1} [P(X^{-1}(X)) > Q(X^{-1}(X))]] \\ &\quad - \mathbb{E}_{X \sim Q} [\mathbb{1} [P(X^{-1}(X)) > Q(X^{-1}(X))]]| \\ &= P(A) - Q(A) \\ &= Q(\bar{A}) - P(\bar{A}) \\ &= \frac{1}{2} \sum_{\omega \in \Omega} |P(\omega) - Q(\omega)| \end{aligned}$$

where f essentially maps each random variable X to an indicator of whether the probability of the underlying sample ω has a higher probability under P compared to Q (remember $X : \Omega \rightarrow \mathbb{R}$ maps samples $\omega \in \Omega$ to $X(\omega) \in \mathbb{R}$) and $A = \{\omega : P(\omega) > Q(\omega), \omega \in \Omega\}$ is the event of all samples that have a higher probability under P compared to Q . Assume for contradiction that there is a function that gives an even larger difference in expected values. Then f assigns a value to $X(\omega) \in \mathcal{X}$ less than 1 such that $P(\omega) > Q(\omega)$ or a value larger than 0 such that $P(\omega) \leq Q(\omega)$, both of which would decrease the absolute difference (it would decrease both the max and min of the expression in the absolute value since they are the negative of each other) and create a contradiction.

IPM Example 2. Maximum Mean Discrepancy We get MMD with

$$\mathcal{F} = \{f : \|f\|_{\mathcal{H}} \leq 1\}$$

where $\mathcal{H}$ is some RKHS. If k is the kernel function for the RKHS, $k_x = k(x, \cdot)$ represents the kernel function centered at x

$$\begin{aligned}
& \mathbb{E}_{X \sim P}[f(X)] - \mathbb{E}_{Y \sim Q}[f(Y)] \\
&= \mathbb{E}_{X \sim P}[\langle k_x, f \rangle] - \mathbb{E}_{Y \sim Q}[\langle k_y, f \rangle] \\
&= \langle \mathbb{E}_{X \sim P}[k_x], \mathbb{E}_{X \sim P}[f] \rangle - \langle \mathbb{E}_{Y \sim Q}[k_y], \mathbb{E}_{Y \sim Q}[f] \rangle \\
&= \langle \mathbb{E}_{X \sim P}[k_x], f \rangle - \langle \mathbb{E}_{Y \sim Q}[k_y], f \rangle \quad (\text{since } f \text{ is not dependent on } x) \\
&= \langle \mathbb{E}_{X \sim P}[k_x] - \mathbb{E}_{Y \sim Q}[k_y], f \rangle
\end{aligned}$$

Thus

$$\begin{aligned}
MMD(P, Q) &= \sup_{f \in \mathcal{F}} |\mathbb{E}_{X \sim P}[f(X)] - \mathbb{E}_{Y \sim Q}[f(X)]| \\
&= \sup_{f \in \mathcal{F}} |\langle \mathbb{E}_{X \sim P}[k_x] - \mathbb{E}_{Y \sim Q}[k_y], f \rangle| \\
&= \|\mathbb{E}_{X \sim P}[k_x] - \mathbb{E}_{Y \sim Q}[k_y]\|_{\mathcal{H}}
\end{aligned}$$

where the last line comes from Cauchy-Schwarz, e.g.

$$\langle u, v \rangle \leq \|u\|_{\mathcal{H}} \|v\|_{\mathcal{H}}$$

and the maximum can be attained in our case with

$$\frac{1}{\|\mathbb{E}_{X \sim P}[k_x] - \mathbb{E}_{Y \sim Q}[k_y]\|_{\mathcal{H}}} \mathbb{E}_{X \sim P}[k_x] - \mathbb{E}_{Y \sim Q}[k_y]$$

Continuing

$$\begin{aligned}
MMD(P, Q) &= \|\mathbb{E}_{X \sim P}[k_x] - \mathbb{E}_{Y \sim Q}[k_y]\|_{\mathcal{H}} \\
&= \sqrt{\langle \mathbb{E}_{X \sim P}[k_x] - \mathbb{E}_{Y \sim Q}[k_y], \mathbb{E}_{X \sim P}[k_x] - \mathbb{E}_{Y \sim Q}[k_y] \rangle} \\
&= \sqrt{\langle \mathbb{E}_{X \sim P}[k_x], \mathbb{E}_{X \sim P}[k_x] \rangle - 2\langle \mathbb{E}_{X \sim P}[k_x], \mathbb{E}_{Y \sim Q}[k_y] \rangle + \langle \mathbb{E}_{Y \sim Q}[k_y], \mathbb{E}_{Y \sim Q}[k_y] \rangle} \\
&= \sqrt{\mathbb{E}_{X, X' \sim P}[\langle k_x, k_{x'} \rangle] - 2\mathbb{E}_{X \sim P, Y \sim Q}[\langle k_x, k_y \rangle] + \mathbb{E}_{Y, Y' \sim Q}[\langle k_y, k_{y'} \rangle]} \\
&= \sqrt{\mathbb{E}_{X, X' \sim P}[k(X, X')] - 2\mathbb{E}_{X \sim P, Y \sim Q}[k(X, Y)] + \mathbb{E}_{Y, Y' \sim Q}[k(Y, Y')]}
\end{aligned}$$

where the random variables X, X', Y, Y' are at least pairwise independent. Note that this expression can be estimated using samples.

3 Evaluation

In offline RL, there is a policy π_β (which could technically be a mix or many different policies, like a mixture of gaussians, but a mixture of policies) used to collect a dataset of transitions $\mathcal{D} = \{(s_i, a_i, s'_i, r_i)\}$, which we use to train a policy π . However, we are not actually able to run policies of our choice and can only rely on the data gathered by π_β . As a result, typically off policy algorithms tend to perform poorly. For instance, in Q learning, the target values are set as the $r(s, a) + \max_{a'} Q(s', a')$, which can exploit the noise in the Q value estimator to overestimate the Q values. Even though target networks can help mitigate the issue, the issue will still be present. In a typically online RL setting, we are able to deploy updated policies in the real world to see if the actions that are hypothesized to generate large Q values actually do generate large rewards. If not, the algorithm will learn from the data and decrease the overestimated Q values, which would not be possible with offline RL. As a result, we need algorithms to help us evaluate the policy, similar to using an evaluation dataset in vanilla supervised ML, instead of just relying on the training reward (e.g. the Q values we see during training in Q learning). We typically evaluate the policy by looking at average $V(s)$ all states in the dataset $\mathcal{D}$

3.1 Doubly Robust Estimator

In a contextual bandit, we are given a state and take a single action (e.g. one step RL problem) to get a reward. To evaluate the policy, there are two basic approaches, the first of which is direct estimation, where we learn a Q value estimator $\hat{Q}(s, a)$ (estimates the single step $r(s, a)$ in the contextual bandit case), and

$$\hat{V}_{DM}^\pi = \frac{1}{|\mathcal{D}|} \sum_{s \in \mathcal{D}} \hat{Q}(s, \pi(s))$$

where S is the list of all states in the gathered dataset $\mathcal{D}$.

The other approach is to use importance sampling or the inverse propensity score, where we fit a model for $\hat{\pi}_\beta(a|s)$ (since we do not assume we have access to information about the policy that gathered the data), and

$$\hat{V}_{IPS}^\pi = \frac{1}{|\mathcal{D}|} \sum_{(s,a,r) \in \mathcal{D}} \frac{1}{|\mathcal{D}|} \sum_{(s,a,r) \in \mathcal{D}} \frac{r\pi(a|s)}{\hat{\pi}_\beta(a|s)}$$

Combining the two approaches, we get the doubly robust estimator for a deterministic learned policy π as

$$\hat{V}_{DR}^\pi = \frac{1}{|\mathcal{D}|} \sum_{(s,a,r) \in \mathcal{D}} \frac{(r - \hat{Q}(s, a))\pi(a|s)}{\hat{\pi}_\beta(a|s)} + \hat{Q}(s, \pi(s))$$

Note that this doubly robust estimator tends to have smaller variance than the importance sampled estimator since the ratio of probabilities (which can be very large) is now multiplying a smaller number if $\hat{Q}$ is a good estimator of Q . The doubly robust estimator has larger variance the direct estimate $\hat{V}_{DM}^\pi$. However, $\hat{V}_{DR}^\pi$ tends to have better bias than both $\hat{V}_{IPS}^\pi$ and $\hat{V}_{DM}^\pi$ since the bias also tends to 0 if either $\hat{\pi}_\beta$ is a good approximation of π_β or $\hat{Q}$ is a good approximation of Q (the bias scales linearly with the product of the two errors, see Doubly Robust Policy Evaluation and Learning by Dudik et al. for proofs).

Extending this to the multi-step RL case, we get that

$$\bar{V}_{DR}^{T+1-t} = \hat{V}(s_t) + \frac{\pi(a_t|s_t)}{\hat{\pi}_\beta(a_t|s_t)}(r_t + \gamma \bar{V}_{DR}^{T-t} - \hat{Q}(s_t, a_t))$$

where $\bar{V}_{DR}^{T+1-t}$ is the estimator given $T + 1 - t$ steps remaining. We can solve this for each trajectory using DP through Bellman equations by looping backwards starting with $\bar{V}_{DR}^0$ and solving towards $\bar{V}_{DR}^T$. We can then take the average $\bar{V}_{DR}^T$ for all trajectories in the dataset $\mathcal{D}$ to evaluate π .

3.2 Marginalized Importance Sampling

Besides applying importance weights on the probability of a trajectory, we can also apply importance weights to the state action marginal probabilities and evaluate the policy using

$$\mathbb{E}_{s,a \sim \mu_\gamma^\pi(s,a)} [r(s, a)] = \mathbb{E}_{s,a \sim \mu_\gamma^{\pi_\beta}(s,a)} \left[\frac{\mu_\gamma^\pi(s, a)}{\mu_\gamma^{\pi_\beta}(s, a)} r(s, a) \right] = \mathbb{E}_{s,a \sim \mu_\gamma^{\pi_\beta}(s,a)} [\tau(s, a) r(s, a)]$$

Here,

$$\mu_\gamma^\pi(s, a) = \begin{cases} \lim_{T \rightarrow \infty} \frac{1}{T+1} \sum_{t=0}^T d_t^\pi(s, a) = \lim_{t \rightarrow \infty} d_t^\pi(s, a) & \text{if } \gamma = 1 \\ (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t d_t^\pi(s, a) & \text{if } \gamma \in [0, 1) \end{cases}$$

where the state action marginal probabilities under policy π is expressed as $d^\pi(s, a)$ and $\tau(s, a) = \frac{\mu_\gamma^\pi(s, a)}{\mu_\gamma^{\pi_\beta}(s, a)}$.

Essentially, $\mu_\gamma^{\pi_\beta}$ is the stationary distribution in the non-discounted setting and is the sum of the discounted probabilities in the discounted setting. Note that this sum of discounted probabilities can be described by the same process used in successor features, where at every time step, we have a $(1 - \gamma)$ probability of terminating after entering the state and taking an action (including the first state and action). Then $\mu_\gamma^{\pi_\beta}(s, a)$ represents the probability we terminate after entering state s and taking the action a .

To approximate τ , we can solve the following consistency condition

$$\begin{aligned} \mu_\gamma^\pi(s', a') &= (1 - \gamma) \mu_0(s') \pi(a'|s') + \gamma \int \pi(a'|s') P(s'|s, a) d^\pi(s, a) ds da \\ \iff \mu_\gamma^{\pi_\beta}(s', a') \tau(s', a') &= (1 - \gamma) \mu_0(s') \pi(a'|s') + \gamma \int \pi(a'|s') P(s'|s, a) \mu_\gamma^{\pi_\beta}(s', a') (s, a) \tau(s, a) ds da \end{aligned}$$

The first term being multiplied by $(1 - \gamma)$ represents the probability we terminate after (s', a') if we terminate at the first time step.

For the second term, let's express the probability we reach (s, a) $l^{\pi_\beta}(s, a)$, the probability we take one more step from (s, a) to (s', a') as $\gamma \pi(a'|s') P(s'|s, a)$, and the probability we terminate $(1 - \gamma)$. The second term (integral) being multiplied by γ represents the probability we terminate after (s', a') if we do not terminate at the first time step, which is the sum over all other state action pairs (s, a) of

$$l^{\pi_\beta}(s, a) \gamma \pi(a'|s') P(s'|s, a) (1 - \gamma) = l^{\pi_\beta}(s, a) (1 - \gamma) \gamma \pi(a'|s') P(s'|s, a) = \mu^{\pi_\beta}(s, a) \gamma \pi(a'|s') P(s'|s, a)$$

where the last term is the same as the probability we terminate at (s, a) multiplied by the probability we take one more step to (s', a') . We can also remove the extra factor of γ since we are given that we did not terminate at the first time step using the memoryless property of the geometric distribution.

We can then use a neural net to approximate $\tau(s', a')$ and set the loss function a divergence function between the left and right hand side of the consistency equation using samples to estimate the loss.

3.3 Linear Value Estimation

For any finite state MDP with state space $|S|$ and rewards dependent on the state, let V^π be a vector of length $|S|$, where $V^\pi(s_t)$ is the element in V^π corresponding to state s_t and contains the value function evaluated at s_t under policy π , and let r be a vector of rewards of length $|S|$ where $r(s_t)$ is the element r corresponding to state s_t that contains the reward for state s_t . We know that the MDP must satisfy

$$\begin{aligned} V^\pi(s_t) &= \mathbb{E}_\pi \left[\sum_{t'=t}^{\infty} \gamma^{t'-t} r(s_{t'}) | s_t \right] \\ &= \mathbb{E}_\pi [r(s_t) | s_t] + \gamma \mathbb{E}_\pi [V^\pi(s_{t+1}) | s_t] \\ &= r(s_t) + \gamma \mathbb{E}_\pi [V^\pi(s_{t+1}) | s_t] \quad \forall s_t \in S \end{aligned}$$

Let's denote transition probability transition matrix P^π ($|S| \times |S|$) where $P_{i,j}^\pi$ represents the probability of going from state i to j after one timestep under the policy π (each row corresponds to the outward probabilities of one state). We can then express V^π in a vectorized manner:

$$V^\pi = r + \gamma P^\pi V^\pi$$

We can also solve for V^π to evaluate the MDP for the current policy

$$\begin{aligned} V^\pi &= r + \gamma P^\pi V^\pi \\ \implies V^\pi &= (I - \gamma P^\pi)^{-1} r \end{aligned}$$

If we are given an infinitely large state space, we can no longer store the value function evaluated at every state and can resort to a feature representation of state s_t using a vector of length $|K|$ $\phi(s_t)$. We can then use a linear model for $V^\pi(s_t)$

$$V^\pi(s_t) = \phi(s_t)^T w_v$$

where w_v is some learned parameter vector.

Assuming that the state space is still finite with each state s having a feature representation $\phi(s)$, we can stack all of the $\phi(s)^T$ for all states s into a feature matrix Φ of size $|S| \times |K|$, where each row represents the feature representation for a particular state. Using the same linear model, we get that the MDP must satisfy

$$\begin{aligned} V^\pi &= \Phi w_v = r + \gamma P^\pi V^\pi \\ \implies \Phi w_v &= r + \gamma P^\pi \Phi w_v \end{aligned}$$

Assuming we also use a linear model for the reward r and the expected next state Φ' feature representation.

$$\begin{aligned} r &= \Phi w_r \\ \Phi' &= P^\pi \Phi = \Phi P_\Phi \end{aligned}$$

we get that w_r and P_Φ can be approximated using least squares as follows

$$\begin{aligned} w_r &\approx (\Phi^T \Phi)^{-1} \Phi^T r \\ \Phi P_\Phi &\approx (\Phi^T \Phi)^{-1} \Phi^T P^\pi \Phi = (\Phi^T \Phi)^{-1} \Phi^T \Phi' \end{aligned}$$

One thing to note is that taking the expected value of the next state feature representation and multiplying with w_v or w_r is the same as taking the expected value of the product of the next state feature representation and w_v or w_r since the models are linear.

Using these linear models, we get

$$\begin{aligned} V^\pi &= r + \gamma P^\pi V^\pi \\ \implies \Phi w_v &= \Phi w_r + \gamma \Phi P_\Phi w_v \end{aligned}$$

which is also satisfied by any solution to

$$\begin{aligned} w_v &= w_r + \gamma P_\Phi w_v \\ \implies w_v &= (I - \gamma P_\Phi)^{-1} w_r \end{aligned}$$

Plugging in the least squares approximations for w_r and P_Φ from above, we get that

$$\begin{aligned} w_v &= (I - \gamma P_\Phi)^{-1} w_r \\ &= (I - \gamma (\Phi^T \Phi)^{-1} \Phi^T \Phi')^{-1} (\Phi^T \Phi)^{-1} \Phi^T r \\ &= (I - \gamma (\Phi^T \Phi)^{-1} \Phi^T \Phi')^{-1} (\Phi^T \Phi)^{-1} (\Phi^T \Phi) (\Phi^T \Phi)^{-1} \Phi^T r \\ &= (I (\Phi^T \Phi) - \gamma (\Phi^T \Phi) (\Phi^T \Phi)^{-1} \Phi^T \Phi')^{-1} \Phi^T r \\ &= (\Phi^T \Phi - \gamma \Phi^T \Phi')^{-1} \Phi^T r \end{aligned}$$

Given a dataset $|D|$ comprised of transitions $(s_t, a_t, r(s_t), s_{t+1})$, we can form a Φ matrix with dimensions $|D| \times |K|$ where each row is the feature representation of a state in the dataset and solve for w_v using the least squares equation above. Essentially, we let the states be an approximation to the entire state space (also known as an empirical MDP). Note that each row in Φ' is the expected value of the feature representation at the next timestep given we came from the state corresponding to the same row in Φ , but we just approximate this using a sampled next time step. However, this sampled next time step must come from π and not the behavior policy π_β that collected the data, meaning this LSTD algorithm only works with on policy data. To extend this to work for off policy data, we just need to represent the Q^π function rather than the value function V^π with a linear model, e.g.

$$Q^\pi \Phi w_q$$

where Φ is now a $|S| \times |A| \times |K|$ matrix (one row per state action pair). For the empirical MDP with dataset $\mathcal{D} = (s_t, a_t, r(s_t), s_{t+1})$, we can form Φ using (s_t, a_t) . For Φ' , we can form it using $(s_{t+1}, \pi(s_{t+1}))$, where we sample an action for s_{t+1} from the policy using $\pi(s_{t+1})$.

Besides using the linear models for r and Φ' , we can derive the LSTD algorithm from analyzing the TD algorithm. In TD(0) learning, we loop over all transitions and repeat the following for some learning rate $\alpha \in (0, 1]$

$$\begin{aligned} w_v &= w_v - \alpha \nabla_{w_v} \frac{1}{2} (r + \gamma \text{sg}(\phi(s_{t+1})^T w_v) - \phi(s_t)^T w_v)^2 \\ &= w_v + \alpha \phi(s_t) (r + \gamma \phi(s_{t+1})^T w_v - \phi(s_t)^T w_v) \end{aligned}$$

where sg is the stop gradient operator. Note if the number of states is small, we can let w_v be a vector of length $|S|$ such that $w_v = V^\pi$ and $\phi(s_t)$ will be a one-hot vector of length $|S|$ that is 1 at the index corresponding to s_t . In this case, TD(0) learning becomes

$$V^\pi(s_t) = V^\pi(s_t) + \alpha (r + \gamma V^\pi(s_{t+1}) - V^\pi(s_t))$$

Note that TD(0) learning converges when the gradient is equal to 0, e.g.

$$\begin{aligned} &\sum_{(s_t, a_t, r(s_t), s_{t+1}) \in \mathcal{D}} \phi(s_t) (r(s_t) + \gamma \phi(s_{t+1})^T w_v - \phi(s_t)^T w_v) = 0 \\ \iff &\frac{1}{|D|} \sum_{(s_t, a_t, r(s_t), s_{t+1}) \in \mathcal{D}} r(s_t) \phi(s_t) + \gamma \phi(s_t) \phi(s_{t+1})^T w_v - \phi(s_t) \phi(s_t)^T w_v = 0 \end{aligned}$$

Given sufficient samples, the above condition approximates

$$\mathbb{E}_\pi [r(s_t) \phi(s_t) + \gamma \phi(s_t) \phi(s_{t+1})^T w_v - \phi(s_t) \phi(s_t)^T w_v] = 0$$

meaning instead of waiting for TD(0) learning to converge with gradient updates, we can just solve the following:

$$\begin{aligned} &\mathbb{E}_\pi [r(s_t) \phi(s_t) + \gamma \phi(s_t) \phi(s_{t+1})^T w_v - \phi(s_t) \phi(s_t)^T w_v] = 0 \\ \implies &\mathbb{E}_\pi [\phi(s_t) \phi(s_t)^T w_v - \gamma \phi(s_t) \phi(s_{t+1})^T w_v] = \mathbb{E}_\pi [r(s_t) \phi(s_t)] \\ \implies &\frac{1}{|D|} \Phi^T \Phi w_v - \gamma \Phi^T \Phi' w_v = \frac{1}{|D|} \Phi^T r \\ \implies &w_v = (\Phi^T \Phi - \gamma \Phi^T \Phi')^{-1} \Phi^T r \end{aligned}$$

which is the same LSTD algorithm from before.

Note that the MDP value function for the given policy π satisfies

$$\begin{aligned} &V^\pi(s_t) = r(s_t) + \gamma \mathbb{E}_\pi [V^\pi(s_{t+1}) | s_t] \quad \forall s_t \in S \\ \mathbb{E}_\pi [r(s_t) + \gamma \phi(s_{t+1})^T w_v - \phi(s_t)^T w_v | s_t] &= 0 \quad \forall s_t \in S \\ \implies \mathbb{E}_\pi [r(s_t) \phi(s_t) + \gamma \phi(s_t) \phi(s_{t+1})^T w_v - \phi(s_t) \phi(s_t)^T w_v] &= 0 \end{aligned}$$

which is also exactly the condition for TD learning convergence.

Note that even though

$$\begin{aligned} \mathbb{E}_\pi [r(s_t) + \phi(s_{t+1})^T w_v - \phi(s_t)^T w_v | s_t] &= 0 \quad \forall s \implies \mathbb{E}_\pi [r(s_t) + \phi(s_{t+1})^T w_v - \phi(s_t)^T w_v] = 0 \\ \mathbb{E}_\pi [r(s_t) + \phi(s_{t+1})^T w_v - \phi(s_t)^T w_v] &= 0 \not\implies \mathbb{E}_\pi [r(s_t) + \phi(s_{t+1})^T w_v - \phi(s_t)^T w_v | s_t] = 0 \quad \forall s \end{aligned}$$

One example is an MDP with two states where $\phi(s') = -\phi(s)$, $r(s') = -r(s)$, and at each time step, we are guaranteed to go to the other state; thus, there can be degenerate solutions where the value function is not consistent with the MDP when the feature vectors are linearly dependent.

Note that this is a more classical method that does not address distribution shift since we are assuming simple linear function approximators. Recent work uses more expressive function approximators like deep neural nets, which typically work better in practice but are also have higher variance and produce more irrational values for these out of distribution states.

4 Sampling and Approximation Error

4.1 Concentration Inequalities

Hoeffdings Inequality

Hoeffdings inequality says for

$$\hat{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

where the X_i 's are mutually independent and satisfy $b_- \leq X_i \leq b_+$ almost surely, then

$$\begin{aligned} P(\hat{X}_n \geq \mu + \epsilon) &\leq e^{-2n\epsilon^2/(b_+ - b_-)^2} \\ P(\hat{X}_n \leq \mu - \epsilon) &\leq e^{-2n\epsilon^2/(b_+ - b_-)^2} \\ \implies P(\hat{X}_n - \mu \geq \epsilon \cup \hat{X}_n - \mu \leq -\epsilon) &= P(|\hat{X}_n - \mu| \geq \epsilon) \leq 2e^{-2n\epsilon^2/(b_+ - b_-)^2} \end{aligned}$$

where the last line comes from the Union bound. If we want $P(|\hat{X}_n - \mu| \geq \epsilon)$ to be bounded by some probability δ , we want

$$\begin{aligned} \delta &\geq 2e^{-2n\epsilon^2/(b_+ - b_-)^2} \geq P(|\hat{X}_n - \mu| \geq \epsilon) \\ \iff -2n\epsilon^2/(b_+ - b_-)^2 &\leq \log \frac{\delta}{2} \\ \iff \epsilon &\geq (b_+ - b_-) \sqrt{\frac{\log \frac{\delta}{2}}{2n}} \\ \iff n &\geq \frac{(b_+ - b_-)^2 \log \frac{\delta}{2}}{2\epsilon^2} \end{aligned}$$

Thus, for any $\epsilon \geq (b_+ - b_-) \sqrt{\frac{\log \frac{\delta}{2}}{2n}}$, we have that $P(|\hat{X}_n - \mu| \geq \epsilon) \leq \delta$. Using the smallest possible ϵ , we get that

$$P(|\hat{X}_n - \mu| \geq (b_+ - b_-) \sqrt{\frac{\log \frac{\delta}{2}}{2n}}) \leq \delta$$

Vectorized Hoeffdings Inequality

There is a similar vectorized form of Hoeffdings inequality (not actually called this but just referring to it as such). Given a categorical random variable Z that takes on values d different values $\{1, \dots, d\}$, we can create a vector q of length d such that $q_i = P(Z = i)$. If we take N samples of Z and estimate q using $\hat{q}$ where $\hat{q}_i = \frac{N(i)}{N}$, then 2 times the TVD between q and $\hat{q}$ can be bounded as

$$P\left(\|q - \hat{q}\|_1 \geq \sqrt{d} \left(\frac{1}{\sqrt{N}} + \epsilon\right)\right) \leq e^{-N\epsilon^2}$$

For $P\left(\|q - \hat{q}\|_1 \geq \sqrt{d} \left(\frac{1}{\sqrt{N}} + \epsilon\right)\right) \leq \delta$, we want

$$\begin{aligned} e^{-N\epsilon^2} &\leq \delta \\ \iff -N\epsilon^2 &\leq \log \delta \\ \iff \epsilon &\geq \sqrt{\frac{\log \frac{1}{\delta}}{N}} \\ \iff N &\geq \frac{1}{\epsilon^2} \log \frac{1}{\delta} \end{aligned}$$

Thus, for any $\epsilon \geq \sqrt{\frac{1}{N} \log \frac{1}{\delta}}$, we have that $P\left(\|q - \hat{q}\|_1 \geq \sqrt{d} \left(\frac{1}{\sqrt{N}} + \epsilon\right)\right) \leq e^{-N\epsilon^2}$. Using the smallest ϵ such that this is guaranteed (to get the tighter bound), we get that

$$\begin{aligned}
& P\left(\|q - \hat{q}\|_1 \geq \sqrt{d} \left(\frac{1}{\sqrt{N}} + \sqrt{\frac{\log \frac{1}{\delta}}{N}}\right)\right) \leq \delta \\
& \iff P\left(\|q - \hat{q}\|_1 \geq \sqrt{\frac{d}{N}} + \sqrt{\frac{d \log \frac{1}{\delta}}{N}}\right) \leq \delta \leq \delta \\
& \iff P\left(\|q - \hat{q}\|_1 \geq \frac{1}{\log \frac{1}{\delta}} \sqrt{\frac{d \log \frac{1}{\delta}}{N}} + \sqrt{\frac{d \log \frac{1}{\delta}}{N}}\right) \leq \delta \\
& \iff P\left(\|q - \hat{q}\|_1 \geq \left(1 + \frac{1}{\log \frac{1}{\delta}}\right) \sqrt{\frac{d \log \frac{1}{\delta}}{N}}\right) \leq \delta \\
& \iff P\left(\|q - \hat{q}\|_1 \geq c \sqrt{\frac{d \log \frac{1}{\delta}}{N}}\right) \leq \delta
\end{aligned}$$

roughly for some constant c when δ is fixed.

4.2 Preliminaries

When estimating error bounds and performing theoretical analysis, we will assume oracle exploration, where we are given a dataset $\mathcal{D}$ with N samples for all possible state action pairs (s, a) , meaning we sample $s' \sim P(s'|s, a)$ N to get a sample $(s', a, s, r) \in \mathcal{D}$ N times for every possible (s, a) pair. We will also assume a finite state and action space with tabular estimation (e.g. no neural net function approximator) for the Q function, which can be expressed as

$$\begin{aligned}
Q^\pi(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(s'|s, a)}[V^\pi(s')] \\
&= r(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s') \\
&\iff Q^\pi = r + \gamma P V^\pi
\end{aligned}$$

where Q^π is a vector of length $|S||A|$ and P is $|S||A| \times |S|$ such that

$$P_{i,j} = P(s_j|(s, a)_i)$$

meaning the i th row of P contains the probability of transitioning to state s_j for all possible states given the previous state action pair was the state action pair corresponding to the i th index.

Note that we can also express V^π as

$$\begin{aligned}
V^\pi(s) &= \mathbb{E}_{a \sim \pi(a|s)}[Q^\pi(s, a)] \\
&= \sum_a \pi(a|s) Q^\pi(s, a) \\
&\iff V^\pi = \Pi Q^\pi
\end{aligned}$$

where Π is $|S| \times |S||A|$ such that

$$\Pi_{i,j} = \mathbb{1}[s_j = s_i] \pi(s_j|a_j)$$

where $s_j = (s, a)_j[0]$ and $a_j = (s, a)_j[1]$. This means the i th row contains $\pi(a|s_i)$ for all actions a in the indices that correspond to state-action pair entries in Q^π where the state in the pair is i th state. If the state in the pair is not the i th state, the entry is 0. Notice that that each row sums to 1.

Using this substitution for V^π , we get

$$\begin{aligned}
Q^\pi &= r + \gamma P V^\pi \\
&= r + \gamma P \Pi Q^\pi \\
&= r + \gamma P^\pi Q^\pi
\end{aligned}$$

where $P^\pi = P\Pi$ is $|S||A| \times |S||A|$ and satisfies

$$P_{i,j}^\pi = P((s, a)_j | (s, a)_i)$$

each row of P^π contains the outward transition probabilities under policy π (since a_j depends on π) $P^\pi((s, a)_j | (s, a)_i)$ over all possible $(s, a)_j$ pairs for the fixed state-action pair $(s, a)_i$.

Since we do not know P^π in reality, we approximate it using

$$\hat{P}^\pi = \hat{P}\Pi$$

where each row of $\hat{P}^\pi$ is an approximation of P^π using our dataset

$$\hat{P}_{i,j}^\pi = \hat{P}^\pi((s, a)_j | (s, a)_i) = \frac{\pi(a_j | s_j) N(s_j, s_i, a_i)}{N(s_i, a_i)}$$

Similarly, we also may not know r , so we can also approximate it using a vector $\hat{r}$ of length $|S||A|$ such that

$$\hat{r}_i = r((s, a)_i) = \frac{\sum_{(s, a, s', r) \in \mathcal{D}} \delta((s, a) = (s, a)_i) r(s, a)}{N(s, a)_i}$$

For policy evaluation, we want to find a Q^π that satisfies

$$\begin{aligned} Q^\pi &= r + \gamma P^\pi Q^\pi \\ \iff Q^\pi &= (I - \gamma P^\pi)^{-1} r \end{aligned}$$

Since we do not know the transition probabilities P (and hence do not know P^π) or the real rewards r we estimate Q^π using $\hat{Q}^\pi$ using

$$\hat{Q}^\pi = (I - \gamma \hat{P}^\pi)^{-1} \hat{r}$$

Note that instead of directly solve for Q^π , we can also repeatedly apply the Bellman operator $\mathcal{B}$

$$Q_{k+1}^\pi = \mathcal{B}Q_k^\pi = r + \gamma P^\pi Q_k^\pi$$

until convergence to get Q^π . Just as before, since we do not know P^π or r exactly, we approximate them using $\hat{P}^\pi$ and $\hat{r}$ respectively and perform the empirical Bellman operator $\hat{\mathcal{B}}$

$$\hat{Q}_{k+1}^\pi = \hat{\mathcal{B}}\hat{Q}_k^\pi = \hat{r} + \gamma \hat{P}^\pi \hat{Q}_k^\pi$$

until convergence to get $\hat{Q}$.

Note that in an infinite horizon case with some discount factor γ

$$\sum_{t=0}^{\infty} \gamma^t c = \frac{c}{1 - \gamma}$$

However, if we are in the finite horizon setting and have $\gamma = 1$,

$$\sum_{t=0}^{T-1} c = Tc$$

Therefore, $\frac{1}{1-\gamma}$ is analogous to the value of the horizon T in the finite horizon case (e.g. smaller γ means smaller horizon since the impact of c farther in the future decreases faster and vice versa).

4.3 Error in Evaluation

Policy Evaluation Bound.

Due to finite samples when constructing $\hat{P}^\pi$, we get sampling error when doing policy evaluation for some arbitrary policy π by solving for $\hat{Q}$ using $\hat{P}^\pi$. For now, we will assume we know the real rewards for simplicity, so $\hat{r} = r$. We can represent this sampling error as

$$\|Q^\pi - \hat{Q}^\pi\|_\infty$$

The simulation tells us that

$$\begin{aligned}
Q^\pi - \hat{Q}^\pi &= (I - \gamma \hat{P}^\pi)^{-1} (I - \gamma \hat{P}^\pi) Q^\pi - (I - \gamma \hat{P}^\pi)^{-1} r \\
&= (I - \gamma \hat{P}^\pi)^{-1} (I - \gamma \hat{P}^\pi) Q^\pi - (I - \gamma \hat{P}^\pi)^{-1} (I - \gamma P^\pi) Q^\pi \\
&= (I - \gamma \hat{P}^\pi)^{-1} ((I - \gamma \hat{P}^\pi) - (I - \gamma P^\pi)) Q^\pi \\
&= \gamma (I - \gamma \hat{P}^\pi)^{-1} (P^\pi - \hat{P}^\pi) Q^\pi \\
&= \gamma (I - \gamma \hat{P}^\pi)^{-1} (P \Pi - \hat{P} \Pi) Q^\pi \\
&= \gamma (I - \gamma \hat{P}^\pi)^{-1} (P - \hat{P}) \Pi Q^\pi \\
&= \gamma (I - \gamma \hat{P}^\pi)^{-1} (P - \hat{P}) V^\pi
\end{aligned}$$

Moreover, another lemma tells us that

$$\begin{aligned}
\|(I - \gamma \hat{P}^\pi)^{-1} v\|_\infty &= \frac{1}{(1 - \gamma)} \left((1 - \gamma) \|(I - \gamma \hat{P}^\pi)^{-1} v\|_\infty \right) \\
&= \frac{1}{(1 - \gamma)} \left(\|(I - \gamma \hat{P}^\pi)^{-1} v\|_\infty - \gamma \|(I - \gamma \hat{P}^\pi)^{-1} v\|_\infty \right) \\
&\leq \frac{1}{(1 - \gamma)} \left(\|(I - \gamma \hat{P}^\pi)^{-1} v\|_\infty - \gamma \|P^\pi (I - \gamma \hat{P}^\pi)^{-1} v\|_\infty \right) \quad (1) \\
&\leq \frac{1}{(1 - \gamma)} \|(I - \gamma \hat{P}^\pi)^{-1} v - \gamma P^\pi (I - \gamma \hat{P}^\pi)^{-1} v\|_\infty \quad (2) \\
&\leq \frac{1}{(1 - \gamma)} \|(I - \gamma \hat{P}^\pi)(I - \gamma \hat{P}^\pi)^{-1} v\|_\infty \\
&= \frac{1}{(1 - \gamma)} \|v\|_\infty
\end{aligned}$$

(1) is because each row of P^π sums to 1 and taking the maximum of a set of weighted average of some elements is always at most the maximum of those elements. Equality only holds when all the weight is on the largest element for at least one of the weighted averages. (2) is from the reverse triangle inequality. Combining the two, we get that

$$\begin{aligned}
\|Q^\pi - \hat{Q}^\pi\|_\infty &= \|\gamma (I - \gamma \hat{P}^\pi)^{-1} (P - \hat{P}) V^\pi\|_\infty \\
&= \gamma \|(I - \gamma \hat{P}^\pi)^{-1} (P - \hat{P}) V^\pi\|_\infty \\
&\leq \frac{\gamma}{1 - \gamma} \|(P - \hat{P}) V^\pi\|_\infty \\
&= \frac{\gamma}{1 - \gamma} \max_{(s,a)} (P_{(s,a)} - \hat{P}_{(s,a)})^T V^\pi \quad ((s,a) \text{ subscript means row corresponding to } (s,a) \text{ as a vector}) \\
&\leq \frac{\gamma}{1 - \gamma} \max_{(s,a)} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \|V^\pi\|_\infty \quad (\text{by Hölder's inequality})
\end{aligned}$$

We know that

$$\begin{aligned}
V^\pi(s) &= \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s, a) | s_0 = s \right] \\
&\leq \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_{max} | s_0 = s \right] \\
&= \sum_{t=0}^{\infty} \gamma^t R_{max} \mathbb{E}[1 | s_0 = s] \\
&= \frac{1}{1 - \gamma} R_{max} \\
\implies \|V^\pi\|_\infty &\leq \frac{1}{1 - \gamma} R_{max}
\end{aligned}$$

where $R_{max} = \max_{(s,a)} r(s, a)$ Therefore,

$$\begin{aligned}\|Q^\pi - \hat{Q}^\pi\|_\infty &\leq \frac{\gamma}{1-\gamma} \max_{(s,a)} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \|V^\pi\|_\infty \\ &\leq \frac{\gamma}{(1-\gamma)^2} R_{max} \max_{(s,a)} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1\end{aligned}$$

Moreover, for any particular state action pair (s,a) ,

$$\|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 = \sum_{s'} |P(s'|s, a) - \hat{P}(s'|s, a)|$$

is greater than or equal to

$$\left(1 + \frac{1}{\log \frac{|S||A|}{\delta}}\right) \sqrt{\frac{|S| \log \frac{|S||A|}{\delta}}{N}}$$

with probability less than or equal $\frac{\delta}{|S||A|}$ using the oracle exploration assumption and information from the "Concentration Inequalities" section. Thus, we have that

$$\begin{aligned}&P\left(\max_{(s,a)} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \geq \left(1 + \frac{1}{\log \frac{|S||A|}{\delta}}\right) \sqrt{\frac{|S| \log \frac{|S||A|}{\delta}}{N}}\right) \\ &= P\left(\cup_{(s,a)} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \geq \left(1 + \frac{1}{\log \frac{|S||A|}{\delta}}\right) \sqrt{\frac{|S| \log \frac{|S||A|}{\delta}}{N}}\right) \\ &\leq |S||A| \frac{\delta}{|S||A|} \quad (\text{by the Union Bound}) \\ &= \delta\end{aligned}$$

Setting

$$\left(1 + \frac{1}{\log \frac{|S||A|}{\delta}}\right) \sqrt{\frac{|S| \log \frac{|S||A|}{\delta}}{N}} = c_2 \sqrt{\frac{|S| \log \frac{1}{\delta}}{N}}$$

and replacing $\max_{(s,a)} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1$ with $c_2 \sqrt{\frac{|S| \log \frac{1}{\delta}}{N}}$, we have that

$$\begin{aligned}&P\left(\|Q^\pi - \hat{Q}^\pi\|_\infty \geq \frac{\gamma}{(1-\gamma)^2} R_{max} c_2 \sqrt{\frac{|S| \log \frac{1}{\delta}}{N}}\right) \leq \delta \\ \implies &P\left(\|Q^\pi - \hat{Q}^\pi\|_\infty \leq \frac{\gamma}{(1-\gamma)^2} R_{max} c_2 \sqrt{\frac{|S| \log \frac{1}{\delta}}{N}}\right) \geq 1 - \delta\end{aligned}$$

Substituting

$$\epsilon = \frac{\gamma}{(1-\gamma)^2} R_{max} c_2 \sqrt{\frac{|S| \log \frac{1}{\delta}}{N}}$$

we have that for any policy π

$$P\left(\|Q^\pi - \hat{Q}^\pi\|_\infty \leq \epsilon\right) \geq 1 - \delta \quad (\text{Policy Evaluation Bound})$$

Optimal Q Value Bound.

Using this we can also construct an error bound for the Q function under the optimal policy using the real and approximate transition probabilities

$$\begin{aligned}\|Q^* - \hat{Q}^*\|_\infty &= \|\sup_{\pi} Q^\pi - \sup_{\pi} \hat{Q}^\pi\| \\ &\leq \sup_{\pi} \|Q^\pi - \hat{Q}^\pi\|\end{aligned}$$

The last line is true since

$$\|\sup_{\pi} Q^{\pi} - \sup_{\pi} \hat{Q}^{\pi}\| \leq \|Q^{\arg \sup_{\pi} Q^{\pi}} - \hat{Q}^{\arg \sup_{\pi} Q^{\pi}}\|$$

and

$$\|\sup_{\pi} Q^{\pi} - \sup_{\pi} \hat{Q}^{\pi}\| \leq \|Q^{\arg \sup_{\pi} \hat{Q}^{\pi}} - \hat{Q}^{\arg \sup_{\pi} \hat{Q}^{\pi}}\|$$

This means that

$$\|Q^* - \hat{Q}^*\|_{\infty} \geq \epsilon \implies \sup_{\pi} \|Q^{\pi} - \hat{Q}^{\pi}\| \geq \epsilon$$

Since

$$(A \implies B) \iff (A \subseteq B) \implies P(A) \leq P(B)$$

and the **Policy Evaluation Bound** holds for all policies (including the arg sup policy), we have that

$$\begin{aligned} P\left(\|Q^* - \hat{Q}^*\|_{\infty} \geq \epsilon\right) &\leq P\left(\sup_{\pi} \|Q^{\pi} - \hat{Q}^{\pi}\| \geq \epsilon\right) = P\left(\|Q^{\arg \sup_{\pi} \|Q^* - \hat{Q}^*\|_{\infty}} - \hat{Q}^{\arg \sup_{\pi} \|Q^* - \hat{Q}^*\|_{\infty}}\| \geq \epsilon\right) \leq \delta \\ &\implies P\left(\|Q^* - \hat{Q}^*\|_{\infty} \leq \epsilon\right) \geq 1 - \delta \quad (\text{Optimal Q Value Bound}) \end{aligned}$$

Optimal Policy Bound.

Since

$$\begin{aligned} \|Q^* - Q^{\hat{\pi}^*}\|_{\infty} &= \|Q^* - \hat{Q}^* + \hat{Q}^* - Q^{\hat{\pi}^*}\|_{\infty} \\ &\leq \|Q^* - \hat{Q}^*\|_{\infty} + \|\hat{Q}^* - Q^{\hat{\pi}^*}\|_{\infty} \quad (\text{triangle inequality}) \\ &= \|Q^* - \hat{Q}^*\|_{\infty} + \|\hat{Q}^{\hat{\pi}^*} - Q^{\hat{\pi}^*}\|_{\infty} \end{aligned}$$

Therefore

$$\begin{aligned} P\left(\|Q^* - Q^{\hat{\pi}^*}\|_{\infty} \geq 2\epsilon\right) &\leq P\left(\|Q^* - \hat{Q}^*\|_{\infty} + \|\hat{Q}^{\hat{\pi}^*} - Q^{\hat{\pi}^*}\|_{\infty} \geq 2\epsilon\right) \\ &\leq P\left(\|Q^* - \hat{Q}^*\|_{\infty} \geq \epsilon \cup \|\hat{Q}^{\hat{\pi}^*} - Q^{\hat{\pi}^*}\|_{\infty} \geq \epsilon\right) \quad (3) \\ &\leq P\left(\|Q^* - \hat{Q}^*\|_{\infty} \geq \epsilon\right) + P\left(\|\hat{Q}^{\hat{\pi}^*} - Q^{\hat{\pi}^*}\|_{\infty} \geq \epsilon\right) \quad (\text{by the Union Bound}) \\ &= \leq 2\delta \\ &\implies P\left(\|Q^* - \hat{Q}^*\|_{\infty} \leq 2\epsilon\right) \geq 1 - 2\delta \quad (\text{Optimal Policy Bound}) \end{aligned}$$

where (3) is true because

$$\|Q^* - Q^{\hat{\pi}^*}\|_{\infty} \geq 2\epsilon \implies \|Q^* - \hat{Q}^*\|_{\infty} \geq \epsilon \cup \|\hat{Q}^{\hat{\pi}^*} - Q^{\hat{\pi}^*}\|_{\infty} \geq \epsilon$$

Bellman Backup Bound.

When performing policy evaluation using the Bellman operator instead of directly solving for Q^{π} , we also get sampling error from using the empirical Bellman operator

$$\begin{aligned} \|B^{\pi} Q^{\pi} - \hat{B}^{\pi} Q^{\pi}\|_{\infty} &= \|r + \gamma P^{\pi} Q^{\pi} - (\hat{r} + \gamma \hat{P}^{\pi} Q^{\pi})\|_{\infty} \\ &= \|(r - \hat{r}) + \gamma(P^{\pi} - \hat{P}^{\pi})Q^{\pi}\|_{\infty} \\ &= \|(r - \hat{r}) + \gamma(P\Pi - \hat{P}\Pi)Q^{\pi}\|_{\infty} \\ &= \|(r - \hat{r}) + \gamma(P - \hat{P})\Pi Q^{\pi}\|_{\infty} \\ &= \|(r - \hat{r}) + \gamma(P - \hat{P})V^{\pi}\|_{\infty} \\ &\leq \|r - \hat{r}\|_{\infty} + \gamma\|(P - \hat{P})V^{\pi}\|_{\infty} \quad (\text{triangle inequality}) \\ &\leq \|r - \hat{r}\|_{\infty} + \gamma\|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \|V^{\pi}\|_{\infty} \quad (\text{from the Policy Evaluation Bound. section}) \\ &\leq \|r - \hat{r}\|_{\infty} + \frac{\gamma}{1 - \gamma} R_{max} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \quad (\text{from the Policy Evaluation Bound. section}) \end{aligned}$$

Notice that the value function has the same Bellman backup bound

$$\begin{aligned}
\|B^\pi V^\pi - \hat{B}^\pi V^\pi\|_\infty &= \|\Pi r + \gamma \Pi P^\pi Q^\pi - (\Pi \hat{r} + \gamma \Pi \hat{P}^\pi Q^\pi)\|_\infty \\
&= \|\Pi (r + \gamma P^\pi Q^\pi - (\hat{r} + \gamma \hat{P}^\pi Q^\pi))\|_\infty \\
&\leq \|r + \gamma P^\pi Q^\pi - (\hat{r} + \gamma \hat{P}^\pi Q^\pi)\|_\infty \\
&= \|B^\pi Q^\pi - \hat{B}^\pi Q^\pi\|_\infty
\end{aligned}$$

where the second to last step is because taking the maximum of a set of weighted average of some elements is always at most the maximum of those elements. This makes sense since the bound in expectation is typically looser than the original but the bounds are equally tight when the π puts all the probability on the action that achieves the bound.

From the **Policy Evaluation Bound**. section, we also know that

$$\begin{aligned}
&P \left(\max_{(s,a)} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \geq \left(1 + \frac{1}{\log \frac{2|S||A|}{\delta}}\right) \sqrt{\frac{|S| \log \frac{2|S||A|}{\delta}}{N}} \right) \leq \frac{\delta}{2} \\
\Rightarrow &P \left(\frac{\gamma}{1-\gamma} R_{max} \max_{(s,a)} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \geq \frac{\gamma}{1-\gamma} R_{max} \left(1 + \frac{1}{\log \frac{2|S||A|}{\delta}}\right) \sqrt{\frac{|S| \log \frac{2|S||A|}{\delta}}{N}} \right) \leq \frac{\delta}{2}
\end{aligned}$$

Moreover, by Hoeffding's Inequality and the oracle exploration assumption, we get that

$$\begin{aligned}
&P \left(|r(s, a) - \hat{r}(s, a)| \geq (R_{max} - (-R_{max})) \sqrt{\frac{\log \frac{2}{\delta}}{2N}} \right) \leq \frac{\delta}{2} \\
\Rightarrow &P \left(|r(s, a) - \hat{r}(s, a)| \geq 2R_{max} \sqrt{\frac{\log \frac{4}{\delta}}{2N}} \right) \leq \frac{\delta}{2}
\end{aligned}$$

If we want the probability to be bounded by $\frac{\delta}{2|S||A|}$, we get that

$$P \left(|r(s, a) - \hat{r}(s, a)| \geq 2R_{max} \sqrt{\frac{\log \frac{4|S||A|}{\delta}}{2N}} \right) \leq \frac{\delta}{2|S||A|}$$

Using this, we can bound

$$\begin{aligned}
&P \left(\|r - \hat{r}\|_\infty \geq 2R_{max} \sqrt{\frac{\log \frac{4|S||A|}{\delta}}{2N}} \right) \\
&= P \left(\max_{(s,a)} |r(s, a) - \hat{r}(s, a)| \geq 2R_{max} \sqrt{\frac{\log \frac{4|S||A|}{\delta}}{2N}} \right) \\
&= P \left(\cup_{(s,a)} |r(s, a) - \hat{r}(s, a)| \geq 2R_{max} \sqrt{\frac{\log \frac{4|S||A|}{\delta}}{2N}} \right) \\
&\leq |S||A| \frac{\delta}{2|S||A|} \quad (\text{by the Union Bound}) \\
&= \frac{\delta}{2}
\end{aligned}$$

Combining the two bounds above and the fact that

$$\begin{aligned}
&\|B^\pi Q^\pi - \hat{B}^\pi Q^\pi\|_\infty \geq 2R_{max} \sqrt{\frac{\log \frac{2|S||A|}{\delta}}{2N}} + \frac{\gamma}{1-\gamma} R_{max} \left(1 + \frac{1}{\log \frac{|S||A|}{\delta}}\right) \sqrt{\frac{|S| \log \frac{|S||A|}{\delta}}{N}} \\
\Rightarrow &\|r - \hat{r}\|_\infty + \frac{\gamma}{1-\gamma} R_{max} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \geq 2R_{max} \sqrt{\frac{\log \frac{2|S||A|}{\delta}}{2N}} + \frac{\gamma}{1-\gamma} R_{max} \left(1 + \frac{1}{\log \frac{|S||A|}{\delta}}\right) \sqrt{\frac{|S| \log \frac{|S||A|}{\delta}}{N}}
\end{aligned}$$

$$\begin{aligned}
& P \left(\|B^\pi Q^\pi - \hat{B}^\pi Q^\pi\|_\infty \geq 2R_{max} \sqrt{\frac{\log \frac{4|S||A|}{\delta}}{2N}} + \frac{\gamma}{1-\gamma} R_{max} \left(1 + \frac{1}{\log \frac{2|S||A|}{\delta}} \right) \sqrt{\frac{|S| \log \frac{2|S||A|}{\delta}}{N}} \right) \\
& \leq P \left(\|r - \hat{r}\|_\infty + \frac{\gamma}{1-\gamma} R_{max} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \geq 2R_{max} \sqrt{\frac{\log \frac{2|S||A|}{\delta}}{2N}} + \frac{\gamma}{1-\gamma} R_{max} \left(1 + \frac{1}{\log \frac{2|S||A|}{\delta}} \right) \sqrt{\frac{|S| \log \frac{2|S||A|}{\delta}}{N}} \right) \\
& \leq P \left(\|r - \hat{r}\|_\infty \geq 2R_{max} \sqrt{\frac{\log \frac{2|S||A|}{\delta}}{2N}} \cup \frac{\gamma}{1-\gamma} R_{max} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \geq \frac{\gamma}{1-\gamma} R_{max} \left(1 + \frac{1}{\log \frac{2|S||A|}{\delta}} \right) \sqrt{\frac{|S| \log \frac{2|S||A|}{\delta}}{N}} \right) \\
& \leq P \left(\|r - \hat{r}\|_\infty \geq 2R_{max} \sqrt{\frac{\log \frac{2|S||A|}{\delta}}{2N}} \right) \\
& \quad + P \left(\frac{\gamma}{1-\gamma} R_{max} \|P_{(s,a)} - \hat{P}_{(s,a)}\|_1 \geq \frac{\gamma}{1-\gamma} R_{max} \left(1 + \frac{1}{\log \frac{2|S||A|}{\delta}} \right) \sqrt{\frac{|S| \log \frac{2|S||A|}{\delta}}{N}} \right) \quad (\text{by the Union Bound}) \\
& \leq \frac{\delta}{2} + \frac{\delta}{2} \\
& = \delta \\
& \Rightarrow P \left(\|B^\pi Q^\pi - \hat{B}^\pi Q^\pi\|_\infty \leq 2R_{max} \sqrt{\frac{\log \frac{4|S||A|}{\delta}}{2N}} + \frac{\gamma}{1-\gamma} R_{max} \left(1 + \frac{1}{\log \frac{2|S||A|}{\delta}} \right) \sqrt{\frac{|S| \log \frac{2|S||A|}{\delta}}{N}} \right) \geq 1 - \delta
\end{aligned}$$

Under the oracle exploration assumption, we have that $N = \frac{|\mathcal{D}|}{|S||A|}$, meaning we can rewrite the bound as

$$\begin{aligned}
P \left(\|B^\pi Q^\pi - \hat{B}^\pi Q^\pi\|_\infty \leq 2R_{max} \sqrt{\frac{|S||A| \log \frac{4|S||A|}{\delta}}{2|\mathcal{D}|}} + \frac{\gamma}{1-\gamma} R_{max} \left(1 + \frac{1}{\log \frac{2|S||A|}{\delta}} \right) \sqrt{\frac{|S|^2 |A| \log \frac{2|S||A|}{\delta}}{|\mathcal{D}|}} \right) & \geq 1 - \delta \\
\Rightarrow P \left(\|B^\pi Q^\pi - \hat{B}^\pi Q^\pi\|_\infty \leq \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right) & \geq 1 - \delta
\end{aligned}$$

where C depends on $\sqrt{\log \frac{1}{\delta}}$ (and also $\frac{1}{\sqrt{\log \frac{1}{\delta}}}$ but this is very small for small delta), .

4.4 Model Free Error

5 Distributional Shift

5.1 Explicit Policy Constraints

One way to make sure we are taking too many out of distribution action is to add a KL constraint to policy iteration for every state s

$$\begin{aligned}
\pi(a|s) &= \arg \max_{\pi} \mathbb{E}_{a \sim \pi(a|s)} [Q^{\pi_{old}}(s, a)] \\
&\text{subject to } D_{KL}(\pi || \pi_{\beta}) \leq \epsilon \\
&\sum_a \pi(a|s) = 1 \\
&\pi(a|s) \geq 0 \quad \forall a
\end{aligned}$$

However, this constraint can be counterproductive in many cases. For instance, if π^β was constructed to be a uniform policy, we are essentially forcing learned policy to be more random.

Moreover, we typically do not have access to π_β and typically need to use behavioral cloning to learn a neural net that approximates π_β or use a technique that only relies on samples of π_β .

Regardless, the problem (where the variables being optimized are the values of $\pi(a|s)$ for all actions a) above can

be rewritten as

$$\begin{aligned}
& \arg \max_{\pi} \sum_a \pi(a|s) Q^{\pi_{\text{old}}}(s, a) \\
& \text{subject to } \epsilon - \sum_a \pi(a|s) \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} \geq 0 \\
& \sum_a \pi(a|s) = 1 \\
& \pi(a|s) \geq 0 \quad \forall a
\end{aligned}$$

We know that $\pi = \pi_{\beta}$ strictly satisfies the KL constraint and satisfies the other two linear constraints and the objective is convex (it is linear in $\pi(a|s)$), Slater's condition tells us strong duality holds. Taking the dual, we can transform the problem into

$$\arg \min_{\lambda > 0, \vec{\lambda}', \nu} \arg \max_{\pi} \sum_a \pi(a|s) Q^{\pi_{\text{old}}}(s, a) + \lambda \left(\epsilon - \sum_a \pi(a|s) \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} \right) + \nu \left(\sum_a \pi(a|s) - 1 \right) + \sum_a \lambda'_a \pi(a|s)$$

Note that if we just convert the KL constraint into a penalty term, the solution we will solve for also applies for problems when we have a KL penalty weighted by λ and not an explicit KL constraint (the epsilon term does not affect the form of the final solution). We can also directly solve the KKT conditions (since strong duality holds). Solving the KKT conditions, we can start by taking the gradient of the Lagrangian

$$\mathcal{L}(\pi, \lambda, \vec{\lambda}', \nu) = \sum_a \pi(a|s) Q^{\pi_{\text{old}}}(s, a) + \lambda \left(\epsilon - \sum_a \pi(a|s) \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} \right) + \nu \left(\sum_a \pi(a|s) - 1 \right) + \sum_a \lambda'_a \pi(a|s)$$

with respect to $\pi(a|s)$ and setting it equal to $\vec{0}$. For each action, setting the partial derivative equal to 0 gives us

$$\begin{aligned}
Q^{\pi_{\text{old}}}(s, a) + \lambda \left(-\pi(a|s) \frac{\pi_{\beta}(a|s)}{\pi(a|s)} \frac{1}{\pi_{\beta}(a|s)} + \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} \right) + \nu(1) + \lambda'_a &= 0 \\
\implies Q^{\pi_{\text{old}}}(s, a) - \lambda \left(1 + \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} \right) + \nu + \lambda'_a &= 0 \\
\implies Q^{\pi_{\text{old}}}(s, a) - \lambda - \lambda \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} + \nu + \lambda'_a &= 0 \\
\implies \lambda \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} &= Q^{\pi_{\text{old}}}(s, a) - \lambda + \nu + \lambda'_a \\
\implies \log \frac{\pi(a|s)}{\pi_{\beta}(a|s)} &= \frac{Q^{\pi_{\text{old}}}(s, a) - \lambda + \nu + \lambda'_a}{\lambda} \\
&= \frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda} - 1 + \frac{\nu + \lambda'_a}{\lambda} \\
\implies \pi(a|s) &= \pi_{\beta}(a|s) \exp \left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda} - 1 + \frac{\nu + \lambda'_a}{\lambda} \right) \\
&= \exp \left(-1 + \frac{\nu + \lambda'_a}{\lambda} \right) \pi_{\beta}(a|s) \exp \left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda} \right)
\end{aligned}$$

By complementary slackness, we know that $\lambda'_a = 0$ for all actions where $\pi(a|s) > 0$, so $\pi(a|s) = 0$ or

$$\begin{aligned}
\pi(a|s) &= \exp \left(-1 + \frac{\nu}{\lambda} \right) \pi_{\beta}(a|s) \exp \left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda} \right) \\
&= \frac{1}{Z(s)} \pi_{\beta}(a|s) \exp \left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda} \right)
\end{aligned}$$

where

$$Z(s) = \frac{1}{\exp \left(-1 + \frac{\nu}{\lambda} \right)}$$

KKT conditions also include satisfying all equality constraints

$$\begin{aligned}
& \sum_a \pi(a|s) = 1 \\
\Rightarrow & \sum_a \frac{1}{Z(s)} \pi_\beta(a|s) \exp\left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda}\right) = 1 \\
\Rightarrow & Z(s) = \sum_a \pi_\beta(a|s) \exp\left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda}\right)
\end{aligned}$$

Thus, we get that the optimal next policy given by constrained policy iteration is in the form of

$$\frac{1}{Z(s)} \pi_\beta(a|s) \exp\left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda}\right)$$

where $Z(s)$ is some normalization constant. Notice as λ goes to 0, we go back the greedy policy that tries to maximize the Q values (e.g. as ϵ goes to ∞). As λ goes to ∞ , the learned policy equals the behavior policy (as ϵ goes to 0).

Note that the original problem

$$\begin{aligned}
\pi'(a|s) &= \arg \max_{\pi} \mathbb{E}_{a \sim \pi(a|s)} [Q^{\pi_{\text{old}}}(s, a)] \\
&\text{subject to } D_{KL}(\pi || \pi_\beta) \leq \epsilon \\
&\sum_a \pi(a|s) = 1 \\
&\pi(a|s) \geq 0 \quad \forall a
\end{aligned}$$

could have been rewritten as

$$\begin{aligned}
\pi'(a|s) &= \arg \max_{\pi} \mathbb{E}_{a \sim \pi(a|s)} [Q^{\pi_{\text{old}}}(s, a)] - V^{\pi_{\text{old}}}(s) = \mathbb{E}_{a \sim \pi(a|s)} [Q^{\pi_{\text{old}}}(s, a) - V^{\pi_{\text{old}}}(s)] = \mathbb{E}_{a \sim \pi(a|s)} [A^{\pi_{\text{old}}}(s, a)] \\
&\text{subject to } D_{KL}(\pi || \pi_\beta) \leq \epsilon \\
&\sum_a \pi(a|s) = 1 \\
&\pi(a|s) \geq 0 \quad \forall a
\end{aligned}$$

since $V^{\pi_{\text{old}}}(s)$ is a constant that does not depend on π , so the solution can also be written in the following form

$$\frac{1}{Z(s)} \pi_\beta(a|s) \exp\left(\frac{A^{\pi_{\text{old}}}(s, a)}{\lambda}\right)$$

To train a policy such that it matches the optimal policy given by the next step of constrained policy iteration, we set the optimal learned policy π_θ as policy π_θ that minimizes the D_{KL} with the theoretical optimal policy given by constrained policy iteration

$$\theta = \arg \min_{\theta} D_{KL}(\pi(\cdot|s) || \pi_\theta(\cdot|s)) = \sum_a \pi(a|s) \log\left(\frac{\pi(a|s)}{\pi_\theta(a|s)}\right) = \sum_a \pi(a|s) \log(\pi(a|s)) + \sum_a \pi(a|s) \log\left(\frac{1}{\pi_\theta(a|s)}\right)$$

Since the first term does not depend on θ , we can exclude it from the loss. We can also turn the loss into a reward by setting the objective to maximize the negation of the second term to get that

$$\begin{aligned}
\theta &= \arg \max_{\theta} \sum_a \pi(a|s) \log(\pi_\theta(a|s)) \\
&= \arg \max_{\theta} \sum_a \frac{1}{Z(s)} \pi_\beta(a|s) \exp\left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda}\right) \log(\pi_\theta(a|s)) \\
&= \arg \max_{\theta} \mathbb{E}_{(s) \sim \pi_\beta(a|s)} \left[\frac{1}{Z(s)} \exp\left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda}\right) \log(\pi_\theta(a|s)) \right]
\end{aligned}$$

Summing this objective across all states, we get the following objective to maximize

$$\theta = \arg \max_{\theta} \mathbb{E}_{(s, a) \sim \pi_\beta} \left[\frac{1}{Z(s)} \exp\left(\frac{Q^{\pi_{\text{old}}}(s, a)}{\lambda}\right) \log(\pi_\theta(a|s)) \right]$$

which can be approximated using samples from π_β and optimized using gradient ascent. Note this is essentially weighted behavioral cloning.

We can then perform fitted policy iteration on $Q_\phi(s, a)$ and $\pi_\theta(a|s)$ by alternating the following steps on transitions from the dataset

$$\begin{aligned}\phi &\leftarrow \phi - \alpha \nabla_\phi \left(\mathbb{E}_{(s,a,s') \sim D} \left[\left(Q_\phi(s, a) - (r(s, a) + \gamma \mathbb{E}_{a' \sim \pi_\theta(a'|s')} [Q_\phi(s', a')]) \right)^2 \right] \right) \\ \theta &\leftarrow \theta + \alpha \nabla_\theta \left(\mathbb{E}_{(s,a) \sim \pi_\beta} \left[\log \pi_\theta(a|s) \frac{1}{Z(s)} \exp \left(\frac{1}{\lambda} A^{\pi_{\text{old}}}(s, a) \right) \right] \right)\end{aligned}\quad (2)$$

Even though the KL constraint will hold at convergence, we will still query out of distribution actions a' during intermediate steps.

Instead of solving the problem using only policy iteration, we first do Q -learning using expectile loss as a preliminary step, where the expectile (basically squared quantile loss) loss for expectile τ is defined as

$$L_2^\tau(x) = \begin{cases} (1 - \tau)x^2 & \text{if } x > 0 \\ \tau x^2 & \text{else} \end{cases}$$

where we essentially punish under prediction more when τ is large and vice versa.

Setting $\tau = 0.5$ is just MSE to learn the mean while setting $\tau > 0.5$ lets us learn an upper quantile value. In this Q -learning, we also learn a separate value function network $V_\psi(s)$ instead of just a $Q_\phi(s, a)$ network since depending on s' and a' from the dataset increases the variance (and reduces bias but the paper mentioned lucky states from the variance being more of an issue). where we learn $Q_\phi(s, a)$ and $V_\psi(s)$

$$\begin{aligned}\psi &\leftarrow \psi - \alpha \nabla_\psi \left(\mathbb{E}_{(s,a) \sim D} [L_2^\tau(Q_\phi(s, a) - V_\psi(s))] \right) \\ \phi &\leftarrow \phi - \alpha \nabla_\phi \left(\mathbb{E}_{(s,a,s') \sim D} [(r(s, a) + \gamma V_\psi(s') - Q_\phi(s, a))^2] \right).\end{aligned}$$

This is essentially alternating between training the value function to be the max of the Q function under the support region of the behavior policy and updating the Q function with this maximum

$$\begin{aligned}V(s) &\leftarrow \max_{a \in \Omega(s)} Q(s, a) \\ Q(s, a) &\leftarrow r(s, a) + \max_{a' \in \Omega(s')} Q(s', a')\end{aligned}$$

where

$$\Omega(s) = \{a : \pi_\beta(a|s) \geq \epsilon\}$$

Notice that in continuous spaces, we typically do not see an exact state more than once, so theoretically, using the expectile loss is the same as MSE on $Q(s, a)$ in the tabular setting and if the network was overfitted (the weight does not do anything). However, we are depending on the deep networks ability to generalize between similar states to make the expectile loss work as intended.

Once we finish training this Q function, we will essentially have a Q function with an implicit policy with only in support actions that maximizes the Q value. However, this policy is implicit and to learn an explicit policy, we can run a few gradient steps of step 2 of (2) from the implicit policy constraint method from above to learn an explicit policy we can use.

5.2 Conservative Q-Learning

5.2.1 Policy Evaluation

Another way to mitigate the distributional shift is to make sure our function approximators underestimate the true value, especially for pairs of out-of-distribution actions (so that we are less inclined to take out-of-distribution actions). One method this can be achieved is through Conservative Q-Learning, which can be both used as an algorithm for policy evaluation in conjunction with an RL algorithm to improve a policy or for Q-learning directly. When performing policy evaluation for policy π , we want to find the Q function for π , which we can do by repeating

the following (Bellman operator is a contraction):

$$\begin{aligned}
Q^{k+1} &= \arg \min_Q \mathbb{E}_{s,a,s' \sim \mathcal{D}} \left[\left(Q(s,a) - (r(s,a) + \gamma \mathbb{E}_{a' \sim \pi_k(s')} [Q^k(s',a')]) \right)^2 \right] \\
&= \arg \min_Q \frac{1}{2} \mathbb{E}_{s,a \sim \mathcal{D}} \left[\left(Q(s,a) - \hat{B}^\pi \hat{Q}^k(s,a) \right)^2 \right] \\
&= \frac{1}{2} \sum_{(s,a)} d^{\pi_\beta}(s) \pi_\beta(a|s) \left(Q(s,a) - \hat{B}^\pi \hat{Q}^k(s,a) \right)^2
\end{aligned}$$

where we are assuming the tabular case. $d^{\pi_\beta}(s)$ is the discounted state marginal that only sums up to $\frac{1}{1-\gamma}$ instead of 1

$$d^{\pi_\beta}(s) = \sum_{i=0}^{\infty} \gamma^i p^{\pi_\beta}(s)$$

and $\hat{B}$ is the empirical Bellman operator. Note that we assume $\mathcal{D}$ is sampled from $d^{\pi_\beta}(s) \pi_\beta(a|s)$. However, in practice, we just directly gather use the samples in the dataset without accounting for the state marginal discount. The typical Bellman B^π operator does the following

$$B^\pi Q = r + \gamma P^\pi Q$$

where P^π is $|S||A| \times |S||A|$, each row of P^π contains the true outward transition probabilities $P^\pi(s',a'|s,a)$ over all (s',a') for the fixed state-action pair (s,a) corresponding to that row.

The empirical Bellman operator performs

$$\hat{B}^\pi Q = \hat{r} + \gamma \hat{P}^\pi Q$$

where each row of $\hat{P}^\pi$ is an approximation of P^π using our dataset. $\hat{P}^\pi(s',a'|s,a)$ is estimated as

$$\hat{P}^\pi(s',a'|s,a) = \frac{N(s',a',s,a)}{N(s,a)}$$

in a dataset collected using policy π , and

$$\hat{r}(\tilde{s}, \tilde{a}) = \frac{1}{N(\tilde{s}, \tilde{a})} \sum_{(s,a,s',r) \in \mathcal{D}} \delta((\tilde{s}, \tilde{a})) r$$

Note that in offline RL, we do not have a dataset collected by a policy of our choice, but for a fixed (s',a,s) in the dataset, we can generate a' by querying our latest policy to get modified samples, which is what we did for off-policy RL algorithms, such as actor critic and Q-learning. However, this is not too important since we typically do not use $\hat{P}^\pi(s',a'|s,a)$ (which is not even possible in continuous settings) in practice. We typically assume the simpler tabular case and use the concept of $\hat{P}^\pi$ to reason about the sampling error for theoretical analysis.

Since we are assuming a tabular setting, we can achieve no approximation error, e.g. we are able to set $Q^{k+1} = \hat{B}^\pi \hat{Q}^k(s,a) \quad \forall (s,a)$ (which is not typically the case in practice since Q is usually represented as a function approximator such as a neural network).

The update rule is a convex problem since the quadratic term in

$$\frac{1}{2} \sum_{(s,a)} d^{\pi_\beta}(s) \pi_\beta(a|s) \left(Q(s,a) - \hat{B}^\pi \hat{Q}^k(s,a) \right)^2$$

can be expressed as

$$(Q^T \vec{1})^2 = Q^T (\vec{1} \vec{1}^T) Q$$

and $(\vec{1} \vec{1}^T)$ is PSD.

Therefore, we can solve for the optimal $Q(s,a)$ for each iteration in the tabular case by setting the gradient to $\vec{0}$.

$$\begin{aligned}
\vec{0} &= \nabla_Q \frac{1}{2} \mathbb{E}_{s,a \sim \mathcal{D}} \left[\left(Q(s,a) - \hat{B}^\pi \hat{Q}^k(s,a) \right)^2 \right] \\
&= \nabla_Q \frac{1}{2} \sum_{(s,a)} d^{\pi_\beta}(s) \pi_\beta(a|s) \left(Q(s,a) - \hat{B}^\pi \hat{Q}^k(s,a) \right)^2 \\
&= d^{\pi_\beta}(s) \pi_\beta(a|s) Q - d^{\pi_\beta}(s) \pi_\beta(a|s) \hat{B}^\pi \hat{Q}^k \\
\implies Q &= \hat{B}^\pi \hat{Q}^k
\end{aligned}$$

where $d^{\pi_\beta}(s)$ is the discounted state marginal that only sums to $\frac{1}{1-\gamma}$ instead of 1

$$d^{\pi_\beta}(s) = \sum_{i=0}^{\infty} \gamma^i p^{\pi_\beta}(s)$$

This tells us that element-wise expression for the vector Q of length $|S||A|$ is given by

$$Q(s, a) = \hat{B}^\pi \hat{Q}^k(s, a)$$

5.2.2 Conservative Evaluation Equation 1

Since overestimation is common in offline RL with expressive function approximators for out-of-distribution actions and function approximation error, we want to use a more conservative update by adding a penalty to minimize the Q values at each iteration by minimizing the expected Q value under some arbitrary discounted state action marginal distribution $\mu(s, a) = d^{\pi_\beta}(s)\mu(a|s)$. This gives us the following update rule for learning $\hat{Q}^\pi$

$$\hat{Q}^{k+1} = \arg \min_Q \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] + \frac{1}{2} \mathbb{E}_{s, a \sim \mathcal{D}} \left[\left(Q(s, a) - \hat{B}^\pi \hat{Q}^k(s, a) \right)^2 \right]$$

Notice that $\mu(s, a)$ has the same discounted state marginal as the dataset, since the distribution shift in Q learning arises when we query our function approximator for Q for out-of-distribution actions. For example, in Q learning we set the target as $r(s, a) + Q(s', a')$ where a' comes from querying our latest policy and not the dataset. Similarly, in the Q iteration, we set the target as $r(s, a) + \max_{a'} Q(s', a')$, where a' also does not come directly from the data set. However, the states always come from the dataset.

Similar to the original Q learning update rule, the problem is convex (the quadratic term did not change from above; thus, we can solve for the optimal $Q(s, a)$ by setting

$$\begin{aligned} \vec{0} &= \nabla_Q \left(\alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] + \frac{1}{2} \mathbb{E}_{s, a \sim \mathcal{D}} \left[\left(Q(s, a) - \hat{B}^\pi \hat{Q}^k(s, a) \right)^2 \right] \right) \\ &= \nabla_Q \left(\alpha \sum_{(s, a)} d^{\pi_\beta}(s) \mu(a|s) Q(s, a) + \frac{1}{2} \sum_{(s, a)} d^{\pi_\beta}(s) \pi_\beta(a|s) \left(Q(s, a) - \hat{B}^\pi \hat{Q}^k(s, a) \right)^2 \right) \\ &= \alpha d^{\pi_\beta}(s) \mu(a|s) + d^{\pi_\beta}(s) \pi_\beta(a|s) Q - d^{\pi_\beta}(s) \pi_\beta(a|s) \hat{B}^\pi \hat{Q}^k \\ \implies \hat{Q}^{k+1} &= \hat{B}^\pi \hat{Q}^k - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} \end{aligned}$$

This tells us that the estimated Q value at each iteration is upper bounded by the empirical Bellman targets

$$\hat{Q}^{k+1} \leq \hat{B}^\pi \hat{Q}^k$$

To account for sampling error, we can the **Bellman Backup Bound**. derived above

$$\begin{aligned} &P \left(\|B^\pi Q^\pi - \hat{B}^\pi Q^\pi\|_\infty \leq \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right) \geq 1 - \delta \\ \iff &P \left(\hat{B}^\pi Q^\pi - \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \leq B^\pi Q^\pi \leq \hat{B}^\pi Q^\pi + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right) \geq 1 - \delta \\ \implies &P \left(B^\pi Q^\pi \leq \hat{B}^\pi Q^\pi + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right) \geq P \left(\hat{B}^\pi Q^\pi - \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \leq B^\pi Q^\pi \leq \hat{B}^\pi Q^\pi + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right) \geq 1 - \delta \end{aligned}$$

where where C depends on $\sqrt{\log \frac{1}{\delta}}$. This means that

$$\begin{aligned}
\hat{Q}^{k+1} &= \hat{B}^\pi \hat{Q}^k - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} \\
&\leq B^\pi \hat{Q}^k - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \\
&= R + \gamma P^\pi \hat{Q}^k - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \\
&= R - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} + \gamma P^\pi \hat{Q}^k
\end{aligned}$$

with probability at least $1 - \delta$. Assuming, we are given a modified MDP with reward $R - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}}$ with a Q value represented by $\tilde{Q}$, given the same Q value initialization of such that

$$\hat{Q}^0 = \tilde{Q}^0$$

we have that

$$\hat{Q}^k \leq \tilde{Q}^k \quad \forall k$$

with high probability given a small δ . For the proof by induction, we will assume the bound above always holds for all steps k (the probability of the assumption decreases exponentially in the number of steps). We can prove this using induction.

Base Case: $k=1$

$$\hat{Q}^1 = \hat{B}^\pi \hat{Q}^0 - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} \leq \tilde{Q}^1 = B^\pi \tilde{Q}^0 - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}}$$

For the inductive step, in iteration $k+1$, we have that

$$\begin{aligned}
\hat{Q}^{k+1} &= \hat{B}^\pi \hat{Q}^k - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} \\
&\leq B^\pi \hat{Q}^k - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \quad (\text{w.h.p.}) \\
&\leq B^\pi \tilde{Q}^k - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \quad (2) \\
&= R - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} + \gamma P^\pi \tilde{Q}^k \\
&= \tilde{Q}^{k+1}
\end{aligned}$$

where (2) is true because of the inductive hypothesis and P^π has all positive entries

$$B^\pi \hat{Q}^k = R + \gamma P^\pi \hat{Q}^k \leq R + \gamma P^\pi \tilde{Q}^k = B^\pi \tilde{Q}^k$$

This means that at convergence $\hat{Q} \leq \tilde{Q}$. Another way we can see this is true is by using $\tilde{Q}$ (the converged Q value for the modified MDP) as the initial $\tilde{Q}^0$

$$\begin{aligned}
\hat{Q}^1 &= \hat{B}^\pi \tilde{Q} \leq B^\pi \tilde{Q} - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \\
&= \tilde{Q}
\end{aligned}$$

Since $\hat{Q}^1 \leq \tilde{Q}$, we also know that for all timesteps k , $\hat{Q}^k \leq \tilde{Q}$, which can also be proved using induction. In the inductive step

$$\begin{aligned}
\hat{Q}^{k+1} &= \hat{B}^\pi \hat{Q}^k \\
&\leq B^\pi \hat{Q}^k - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \quad (\text{w.h.p.}) \\
&\leq B^\pi \tilde{Q} - \alpha \frac{\mu(a|s)}{\pi_\beta(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \quad (\text{w.h.p.}) \\
&= \tilde{Q}
\end{aligned}$$

where the second to last step uses both the inductive hypothesis (e.g. $\tilde{Q}$ is element-wise larger than $\hat{Q}^k$) and that $\gamma\hat{P}^\pi$ has all positive entries (so Bellman operator is a monotonically increasing function). Since the converged $\hat{Q}$ should be the same for all initializations, and we found an initialization where $\hat{Q}$ must converge to something smaller than $\tilde{Q}$, we know that $\hat{Q} \leq \tilde{Q}$ with high probability, meaning the delta bound holds for all iterations of using the empirical Bellman operator on the original MDP until convergence (with probability decreasing exponentially with the number of steps it takes $\hat{Q}$ to converge).

Solving for the $\tilde{Q}$ convergence condition, we have that

$$\begin{aligned}\hat{Q} &\leq \tilde{Q} \\ &= (I - \gamma P^\pi)^{-1} \left(R - \alpha \frac{\mu}{\pi_\beta} + \frac{R_{max}}{1 - \gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right) \\ &= Q^\pi + (I - \gamma P^\pi)^{-1} \left(-\alpha \frac{\mu}{\pi_\beta} + \frac{R_{max}}{1 - \gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right)\end{aligned}$$

We know that $(I - \gamma P^\pi)^{-1}$ must have all non-negative entries. Any MDP with state action transition probabilities P^π and reward function R has a Q function Q' that satisfies

$$Q' = (I - \gamma P^\pi)^{-1} R$$

Any reward R that is non-negative element-wise must produce an element-wise non-negative Q' . If we set R to any basis vector of $\mathbb{R}^{|S||A|}$, $e_i \in \{e_1, \dots, e_{|S||A|}\}$, we have

$$Q' = (I - \gamma P^\pi)^{-1} e_i = (I - \gamma P^\pi)_i^{-1}$$

where $(I - \gamma P^\pi)_i^{-1}$ is the i th column of $(I - \gamma P^\pi)^{-1}$, which must be positive elementwise since e_i is nonnegative elementwise. Since this holds for all $i \in \{1, \dots, |S||A|\}$, all columns of $(I - \gamma P^\pi)^{-1}$ must be non-negative element-wise.

As a result

$$\begin{aligned}\alpha \min_{s,a} \frac{\mu}{\pi_\beta} &\geq \frac{R_{max}}{1 - \gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \\ \iff \alpha &\geq \left\| \frac{\pi_\beta}{\mu} \right\|_\infty \frac{R_{max}}{1 - \gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \\ \implies -\alpha \frac{\mu}{\pi_\beta} + \frac{R_{max}}{1 - \gamma} \frac{C}{\sqrt{|\mathcal{D}|}} &\leq 0 \\ \implies \hat{Q} &\leq Q^\pi + (I - \gamma P^\pi)^{-1} \left(-\alpha \frac{\mu}{\pi_\beta} + \frac{R_{max}}{1 - \gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right) \\ &\leq Q^\pi\end{aligned}$$

This means that we can always choose some α that is sufficiently large such that $\hat{Q}$ underestimates Q with high probability. When there is no sampling error, $\alpha \geq 0$ will guarantee underestimation.

5.2.3 Conservative Evaluation Equation 2

To achieve a tighter estimate $\hat{Q}$ that is closer to Q but only underestimates Q in expectation, we can modify the update rule to

$$\hat{Q}^{k+1} = \arg \min_Q \alpha \left(\mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] - \mathbb{E}_{s \sim \mathcal{D}, a \sim \hat{\pi}_\beta(a|s)} [Q(s, a)] \right) + \frac{1}{2} \mathbb{E}_{s, a \sim \mathcal{D}} \left[\left(Q(s, a) - \hat{B}^\pi \hat{Q}^k(s, a) \right)^2 \right]$$

Similar to the section for "Equation 1", the Q learning update rule is convex in Q. Setting the gradient equal to 0, we get

$$\begin{aligned}
\vec{0} &= \nabla_Q \left(\alpha \left(\mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] - \mathbb{E}_{s \sim \mathcal{D}, a \sim \hat{\pi}_\beta(a|s)} [Q(s, a)] \right) + \frac{1}{2} \mathbb{E}_{s, a \sim \mathcal{D}} \left[\left(Q(s, a) - \hat{B}^\pi \hat{Q}^k(s, a) \right)^2 \right] \right) \\
&= \nabla_Q \left(\alpha \sum_{(s, a)} d^{\pi_\beta}(s) \mu(a|s) Q(s, a) - \alpha \sum_{(s, a)} d^{\pi_\beta}(s) \hat{\pi}_\beta(a|s) Q(s, a) + \frac{1}{2} \sum_{(s, a)} d^{\pi_\beta}(s) \pi_\beta(a|s) \left(Q(s, a) - \hat{B}^\pi \hat{Q}^k(s, a) \right)^2 \right) \\
&= \alpha d^{\pi_\beta}(s) \mu(a|s) - \alpha d^{\pi_\beta}(s) \pi_\beta(a|s) + d^{\pi_\beta}(s) \pi_\beta(a|s) Q - d^{\pi_\beta}(s) \pi_\beta(a|s) \hat{B}^\pi \hat{Q}^k \\
\implies \hat{Q}^{k+1} &= \hat{B}^\pi \hat{Q}^k - \alpha \left(\frac{\mu}{\pi_\beta} - \vec{1} \right)
\end{aligned}$$

Unlike with "Equation 1", it will not be the case that the next Q value iterate is underestimated for all (s, a) , e.g.

$$\hat{Q}^{k+1} \leq \hat{B}^\pi \hat{Q}^k$$

since $\frac{\mu(a|s)}{\pi_\beta(a|s)}$ will less than 1 for some (s, a) unless $\mu = \pi_\beta$. However, the expected value of the next Q value iterate, e.g. the next value iterate, is lower bounded for $\mu = \pi$ (when μ is equal to the policy we are trying to evaluate).

$$\begin{aligned}
\hat{V}^{k+1}(s) &= \mathbb{E}_{a \sim \pi(a|s)} [\hat{Q}^{k+1}(s, a)] \\
&= \mathbb{E}_{a \sim \pi(a|s)} \left[\hat{B}^\pi \hat{Q}^k - \alpha \left(\frac{\pi(a|s)}{\pi_\beta(a|s)} - 1 \right) \right] \\
&= \mathbb{E}_{a \sim \pi(a|s)} \left[\hat{B}^\pi \hat{Q}^k(s, a) - \alpha \left(\frac{\pi(a|s)}{\pi_\beta(a|s)} - 1 \right) \right] \\
&= \hat{B}^\pi \hat{V}^k(s) - \alpha \mathbb{E}_{a \sim \pi(a|s)} \left[\frac{\pi(a|s)}{\pi_\beta(a|s)} - 1 \right] \\
&= \hat{B}^\pi \hat{V}^k(s) - \alpha \mathbb{E}_{a \sim \pi(a|s)} \left[\frac{\pi(a|s) - \pi_\beta(a|s)}{\pi_\beta(a|s)} \right] \\
&= \hat{B}^\pi \hat{V}^k(s) - \alpha \sum_a (\pi(a|s) - \pi_\beta(a|s) + \pi_\beta(a|s)) \left(\frac{\pi(a|s) - \pi_\beta(a|s)}{\pi_\beta(a|s)} \right) \\
&= \hat{B}^\pi \hat{V}^k(s) - \alpha \sum_a \frac{(\pi(a|s) - \pi_\beta(a|s))^2}{\pi_\beta(a|s)} - \alpha \sum_a (\pi(a|s) - \pi_\beta(a|s)) \\
&= \hat{B}^\pi \hat{V}^k(s) - \alpha \sum_a \frac{(\pi(a|s) - \pi_\beta(a|s))^2}{\pi_\beta(a|s)} \\
&= \hat{B}^\pi \hat{V}^k(s) - \alpha \sum_a \frac{(\pi(a|s) - \pi_\beta(a|s))^2}{\pi_\beta(a|s)} \\
&\leq \hat{B}^\pi \hat{V}^k(s)
\end{aligned}$$

for $\alpha \geq 0$ where

$$\begin{aligned}
\hat{B}^\pi Q &= \hat{r} + \gamma \hat{P}^\pi Q \\
\hat{B}^\pi V &= \Pi \hat{r} + \gamma \Pi \hat{P}^\pi Q \\
\implies \mathbb{E}_{a \sim \pi(a|s)} [\hat{B}^\pi Q] &= \Pi \hat{r} + \gamma \Pi \hat{P}^\pi Q = \hat{B}^\pi V
\end{aligned}$$

With the basic update rule with no extra terms to induce conservativeness, V converges to

$$V = \Pi(I - \gamma P^\pi)^{-1} r$$

The value function Bellman back up bound is the same as the one for the Q function, so if we incorporate sampling error, we have that at convergence (we can use induction just as before but just add a multiplicative factor of Π in the front of each step)

$$\hat{V} \leq V + \Pi(I - \gamma P^\pi)^{-1} \left(-\alpha \frac{\pi}{\pi_\beta} - \vec{1} + \frac{R_{max}}{1 - \gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right)$$

Thus,

$$\begin{aligned}
\alpha \min_s \mathbb{E}_{a \sim \pi(a|s)} \left[\frac{\pi(a|s)}{\pi_\beta(a|s)} - 1 \right] &\geq \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \\
&\iff \alpha \geq \|\mathbb{E}_{a \sim \pi(a|s)}\|_\infty \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \\
\implies -\alpha \mathbb{E}_{a \sim \pi(a|s)} + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} &\leq 0 \\
\implies \hat{V} &\leq V^\pi + V + (I - \gamma P^\pi)^{-1} \left(-\alpha \mathbb{E}_{a \sim \pi(a|s)} \left[\frac{\pi(a|s)}{\pi_\beta(a|s)} - 1 \right] + \frac{R_{max}}{1-\gamma} \frac{C}{\sqrt{|\mathcal{D}|}} \right) \\
&\leq V^\pi
\end{aligned}$$

This means that we can always choose some α that is sufficiently large such that $\hat{V}$ underestimates V with high probability with $\mu = \pi$. When there is no sampling error, any $\alpha \geq 0$ will guarantee underestimation in expectation for $\mu = \pi$.

5.2.4 Conservative Q Learning

For off policy learning, we can perform full policy evaluation using either of the previous two equations and make a step of policy improvement.

Instead of setting μ equal to the existing policy to achieve underestimation, we can also set it equal to some implicit policy $\pi_{\hat{Q}^k}(a|s)$ that is the maximizer for the existing $\hat{Q}^k$, such as $\pi_{\hat{Q}^k}(a|s) \propto \exp(\hat{Q}^k(s, a))$ the Boltzmann distribution based on the Q -values. Using the derivation from "Conservative Evaluation Equation 2" with $\alpha = 1$ we have that

$$\begin{aligned}
\hat{Q}^{k+1}(s, a) &= \hat{B}^\pi \hat{Q}^k(s, a) - \left(\frac{\mu(s, a)}{\pi_\beta(s, a)} - 1 \right) \\
\implies \mathbb{E}_{\hat{\pi}^{k+1}(a|s)}[\hat{Q}^{k+1}(s, \mathbf{a})] &= \mathbb{E}_{\hat{\pi}^{k+1}(a|s)} \left[\mathcal{B}^\pi \hat{Q}^k(s, \mathbf{a}) \right] - \mathbb{E}_{\hat{\pi}^{k+1}(a|s)} \left[\frac{\pi_{\hat{Q}^k}(a|s)}{\hat{\pi}_\beta(a|s)} - 1 \right] \\
&= \mathbb{E}_{\hat{\pi}^{k+1}(a|s)} \left[\mathcal{B}^\pi \hat{Q}^k(s, \mathbf{a}) \right] - \mathbb{E}_{\pi_{\hat{Q}^k}(a|s)} \left[\frac{\pi_{\hat{Q}^k}(a|s)}{\hat{\pi}_\beta(a|s)} - 1 \right] \\
&\quad + \sum \left(\hat{\pi}_{\hat{Q}^k}(a|s) - \hat{\pi}^{k+1}(a|s) \right) \frac{\hat{\pi}_{\hat{Q}^k}(a|s)}{\hat{\pi}_\beta(a|s)} \\
&\leq \mathbb{E}_{\hat{\pi}^{k+1}(a|s)} \left[\mathcal{B}^\pi \hat{Q}^k(s, \mathbf{a}) \right] - \mathbb{E}_{\pi_{\hat{Q}^k}(a|s)} \left[\frac{\pi_{\hat{Q}^k}(a|s)}{\hat{\pi}_\beta(a|s)} - 1 \right] + 2D_{TV}(\hat{\pi}_{\hat{Q}^k} || \hat{\pi}^{k+1}) \frac{\hat{\pi}_{\hat{Q}^k}(a|s)}{\hat{\pi}_\beta(a|s)}
\end{aligned}$$

Therefore, when

$$\mathbb{E}_{\pi_{\hat{Q}^k}(a|s)} \left[\frac{\pi_{\hat{Q}^k}(a|s)}{\hat{\pi}_\beta(a|s)} - 1 \right] \geq \|2D_{TV}(\hat{\pi}_{\hat{Q}^k} || \hat{\pi}^{k+1}) \frac{\hat{\pi}_{\hat{Q}^k}(a|s)}{\hat{\pi}_\beta(a|s)}\|_\infty$$

the expected value of the next Q value iterate, e.g. the next value iterate, is lower bounded for $\mu = \pi_{\hat{Q}^k}$. Note that this D_{TV} bound technically holds for any μ , but we use $\mu = \pi_{\hat{Q}^k}$ as an example. Essentially, the less the policy being evaluated differs from μ , the higher chance we have that the expected value of the next Q value lower bounds the true Q value in expectation. This bound does not account for sampling error, but we can do so in a similar fashion to the "Conservative Evaluation Equation 2" section.

Note that the policy we are trying to optimize typically just tries to maximize the existing Q function, we can just preset μ to $\pi_{\hat{Q}^k}$ the policy that maximizes $\hat{Q}^k$ instead and just learn the Q function like in Q learning rather than performing policy evaluation then policy improvement back and forth. In practice, we actually preset μ to maximize the Q value at the next iteration and also add a regularizer $\mathcal{R}(\mu)$ to make the algorithm easier to implement, giving us

$$\hat{Q}^{k+1} = \arg \min_Q \max_\mu \alpha \left(\mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)}[Q(s, a)] - \mathbb{E}_{s \sim \mathcal{D}, a \sim \hat{\pi}_\beta(a|s)}[Q(s, a)] \right) + \frac{1}{2} \mathbb{E}_{s, a, s' \sim \mathcal{D}} \left[(Q(s, a) - \mathcal{B}^{\hat{\pi}^k} Q^k(s, a))^2 \right] + \mathcal{R}(\mu)$$

Notice that if $\mathcal{R}(\mu) = c\mathbb{E}_{s \sim \mathcal{D}} [-D_{KL}(\mu, \rho)]$, the inner maximization over μ (which is done over a fixed Q) gives us

$$\begin{aligned} & \arg \max_{\mu} \alpha \left(\mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] - \mathbb{E}_{s \sim \mathcal{D}, a \sim \hat{\pi}_{\beta}(a|s)} [Q(s, a)] \right) + \frac{1}{2} \mathbb{E}_{s, a, s' \sim \mathcal{D}} \left[(Q(s, a) - \mathcal{B}^{\hat{\pi}^k} Q^k(s, a))^2 \right] + \mathcal{R}(\mu) \\ &= \arg \max_{\mu} \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a) - \frac{c}{\alpha} D_{KL}(\mu, \rho)] \end{aligned}$$

This is the same optimization problem from the "Distributional Shift" section since $\mu(a|s)$ distribution can be optimized separately for each state s . The only difference is the objective does not contain the $\lambda\epsilon$ term in the objective but ϵ does not impact the solution if λ is given (λ depends on ϵ), which is fixed to $\frac{c}{\alpha}$ in this case; thus, for any fixed state s

$$\mu(a|s) = \frac{1}{Z(s)} \rho(a|s) \exp(\alpha Q(s, a)) = \frac{\exp(\frac{\alpha}{c} Q(s, a))}{\sum_a \rho(a|s) \exp(\frac{\alpha}{c} Q(s, a))} \rho(a|s)$$

where $\frac{1}{Z(s)}$ is a normalizing constant. Notice that when ρ is uniform

$$\begin{aligned} \mathbb{E}_{s \sim \mathcal{D}} [-D_{KL}(\mu, \rho)] &= -\mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} \left[\log \left(\frac{\mu(a|s)}{\rho(a|s)} \right) \right] \\ &= \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [-\log(\mu(a|s))] + \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [\log(\rho(a|s))] \\ &= \mathbb{E}_{s \sim \mathcal{D}} [\mathcal{H}(\mu)] + \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} \left[\log \left(\frac{1}{|A|} \right) \right] \\ &= \mathbb{E}_{s \sim \mathcal{D}} [\mathcal{H}(\mu)] + \log \left(\frac{1}{|A|} \right) \end{aligned}$$

Note we can transform infinite action spaces into finite ones with something like a tanh transform. Since the second term is a constant, we can ignore it when taking the $\arg \min_Q$ or $\arg \max_{\mu}$, meaning $\mathbb{E}_{s \sim \mathcal{D}} [-D_{KL}(\mu, \rho)]$ has the same effect as $\mathbb{E}_{s \sim \mathcal{D}} [\mathcal{H}(\mu(\cdot|s))]$ as the regularizer. Therefore, when ρ is uniform and $c = \alpha$, we get that

$$\begin{aligned} & \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] + \mathcal{R}(\mu) \\ &= \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] - \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} \left[\log \left(\frac{\exp(Q(s, a))}{\sum_{a'} \exp(Q(s, a'))} \right) \right] \\ &= \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] - \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [\log(\exp(Q(s, a)))] + \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} \left[\log \left(\sum_{a'} \exp(Q(s, a')) \right) \right] \\ &= \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] - \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] + \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} \left[\log \left(\sum_{a'} \exp(Q(s, a')) \right) \right] \\ &= \alpha \mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} \left[\log \left(\sum_{a'} \exp(Q(s, a')) \right) \right] \\ &= \alpha \mathbb{E}_{s \sim \mathcal{D}} \left[\log \left(\sum_{a'} \exp(Q(s, a')) \right) \right] \end{aligned}$$

Plugging this into the original objective, we get that

$$\begin{aligned} & \hat{Q}^{k+1} \\ &= \arg \min_Q \max_{\mu} \alpha \left(\mathbb{E}_{s \sim \mathcal{D}, a \sim \mu(a|s)} [Q(s, a)] - \mathbb{E}_{s \sim \mathcal{D}, a \sim \hat{\pi}_{\beta}(a|s)} [Q(s, a)] \right) + \frac{1}{2} \mathbb{E}_{s, a, s' \sim \mathcal{D}} \left[(Q(s, a) - \mathcal{B}^{\hat{\pi}^k} Q^k(s, a))^2 \right] + \mathcal{R}(\mu) \\ &= \arg \min_Q \max_{\mu} \alpha \left(\mathbb{E}_{s \sim \mathcal{D}} \left[\log \left(\sum_{a'} \exp(Q(s, a')) \right) \right] - \mathbb{E}_{s \sim \mathcal{D}, a \sim \hat{\pi}_{\beta}(a|s)} [Q(s, a)] \right) + \frac{1}{2} \mathbb{E}_{s, a, s' \sim \mathcal{D}} \left[(Q(s, a) - \mathcal{B}^{\hat{\pi}^k} Q^k(s, a))^2 \right] \\ &= \arg \min_Q \max_{\mu} \alpha \mathbb{E}_{s \sim \mathcal{D}} \left[\log \left(\sum_{a'} \exp(Q(s, a')) \right) - \mathbb{E}_{a \sim \hat{\pi}_{\beta}(a|s)} [Q(s, a)] \right] + \frac{1}{2} \mathbb{E}_{s, a, s' \sim \mathcal{D}} \left[(Q(s, a) - \mathcal{B}^{\hat{\pi}^k} Q^k(s, a))^2 \right] \end{aligned}$$

which is typically the algorithm used in practice.

6 Model Based Offline Policy Optimization

We can also train a dynamics model and use model. This model can be used for Dyna style methods, where we use states collected in the real world as starting points for short model based rollouts that we add to a replay buffer. We then use some off-policy RL algorithm to train a policy using both real and model based data.

Just as before, we still have the out of distribution actions problem where the learned policy tricks the model into taking actions that are overestimated to have high Q values. However, we now also have an out of distribution state problem where we take some out of distribution action to fool the model into going into some out of distribution state that leads to another out of distribution action with high Q values, and so and so forth, which compounds the issue.

To deal with this problem, we can modify the reward function to assign a penalty $\lambda u(s, a)$ that represents uncertainty (e.g., how uncertain or wrong the model is), so that the model does not exploit errors in the model with out-of-distribution actions. Note that λ is just a constant.

6.1 Preliminaries

Let $\hat{T}$ be the learned dynamics model and T be the real dynamics model. Let us define the difference in the expected value function under a policy π of the next state under the learned and real dynamics models given the current state and action be

$$G_M^\pi(s, a) = \mathbb{E}_{s' \sim \hat{T}(s, a)} [V_M^\pi(s')] - \mathbb{E}_{s' \sim T(s, a)} [V_M^\pi(s')]$$

where subscript M represents a function evaluated over the MDP with the true dynamics and subscript $\hat{M}$ represents a function evaluated over the MDP with modeled dynamics. Let $W_j(\pi)$ represent the expected reward with transition probabilities given by $\hat{T}$ in the first j time steps and T for all remaining time steps under policy π , and let $\eta_M(\pi)$ and $\eta_{\hat{M}}(\pi)$ and represent the total expected reward under the MDP with the true and modeled dynamics under policy π , then

$$\begin{aligned} \eta_M(\pi) &= W_0(\pi) \\ \eta_{\hat{M}}(\pi) &= W_\infty(\pi) \\ \implies \eta_{\hat{M}}(\pi) - \eta_M(\pi) &= \sum_{j=0}^{\infty} W_{j+1}(\pi) - W_j(\pi) \quad (\text{telescoping sum}) \end{aligned}$$

Moreover,

$$\begin{aligned} &W_j \\ &= \mathbb{E}_{\substack{a_t \sim \pi(s_t) \\ t \leq j: s_t \sim \hat{T}(s_t, a_t) \\ t \geq j+1: s_t \sim T(s_t, a_t)}} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \\ &= \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} \left[\mathbb{E}_{s_{j+1} \sim T(s_t, a_t), a_{j+1}, s_{j+2} \dots \sim (\pi(a_t|s_t), T(s_t, a_t))} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0, a_0, \dots, s_j, a_j \right] \right] = \\ &= \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} \left[\sum_{t=0}^j \gamma^t r(s_t, a_t) + \mathbb{E}_{s_{j+1} \sim T(s_t, a_t), a_{j+1}, s_{j+2} \dots \sim (\pi(a_t|s_t), T(s_t, a_t))} \left[\sum_{t=j+1}^{\infty} \gamma^t r(s_t, a_t) \right] \right] \\ &= \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} \left[\sum_{t=0}^j \gamma^t r(s_t, a_t) + \mathbb{E}_{s_{j+1} \sim T(s_t, a_t)} \left[\mathbb{E}_{a_{j+1}, s_{j+2} \dots \sim (\pi(a_t|s_t), T(s_t, a_t))} \left[\sum_{t=j+1}^{\infty} \gamma^t r(s_t, a_t) | s_j + 1 \right] \right] \right] \\ &= \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} \left[\sum_{t=0}^j \gamma^t r(s_t, a_t) + \gamma^{j+1} \mathbb{E}_{s_{j+1} \sim T(s_t, a_t)} [V_M^\pi(s_{j+1})] \right] \end{aligned}$$

Similarly,

$$\begin{aligned} &W_{j+1} \\ &= \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} \left[\sum_{t=0}^j \gamma^t r(s_t, a_t) + \gamma^{j+1} \mathbb{E}_{s_{j+1} \sim \hat{T}(s_t, a_t)} [V_M^\pi(s_{j+1})] \right] \end{aligned}$$

Taking the difference, we see that

$$\begin{aligned}
W_{j+1} - W_j &= \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} \left[\sum_{t=0}^j \gamma^t r(s_t, a_t) + \gamma^{j+1} \mathbb{E}_{s_{j+1} \sim T(s_t, a_t)} [V_M^\pi(s_{j+1})] \right] \\
&\quad - \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} \left[\sum_{t=0}^j \gamma^t r(s_t, a_t) + \gamma^{j+1} \mathbb{E}_{s_{j+1} \sim \hat{T}(s_t, a_t)} [V_M^\pi(s_{j+1})] \right] \\
&= \gamma^{j+1} \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} \left[\mathbb{E}_{s_{j+1} \sim T(s_t, a_t)} [V_M^\pi(s_{j+1})] - \mathbb{E}_{s_{j+1} \sim \hat{T}(s_t, a_t)} [V_M^\pi(s_{j+1})] \right] \\
&= \gamma^{j+1} \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} [G_M^\pi(s_j, a_j)] \\
\implies \eta_{\hat{M}}(\pi) - \eta_M(\pi) &= \sum_{j=0}^{\infty} W_{j+1}(\pi) - W_j(\pi) \\
&= \sum_{j=0}^{\infty} \gamma^{j+1} \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} [G_M^\pi(s_j, a_j)] \\
&= \gamma \sum_{j=0}^{\infty} \gamma^j \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} [G_M^\pi(s_j, a_j)] \\
&= \gamma \sum_{j=0}^{\infty} \gamma^j \mathbb{E}_{s_0 \sim p_0(s_0), (a_0, \dots, s_j, a_j) \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} [G_M^\pi(s_j, a_j)] \\
&= \gamma \sum_{j=0}^{\infty} \gamma^j \mathbb{E}_{s_t, a_t \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} [G_M^\pi(s_j, a_j)] \\
&= \gamma \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t|s_t), \hat{T}(s_t, a_t))} [G_M^\pi(s_t, a_t)]
\end{aligned}$$

This gives us

$$\begin{aligned}
\eta_M(\pi) &= \eta_{\hat{M}}(\pi) - \gamma \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} [G_M^\pi(s_t, a_t)] \\
&= \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t|s_t), \hat{T}(s_t, a_t))} [r(s_t, a_t)] - \gamma \sum_{t=0}^{\infty} \mathbb{E}_{s_t, a_t \sim (\pi(a_t|s_t), \hat{T}(s_t, a_t))} [\gamma^t G_M^\pi(s_t, a_t)] \\
&= \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t|s_t), \hat{T}(s_t, a_t))} [r(s_t, a_t) - \gamma G_M^\pi(s_t, a_t)] \\
&\geq \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t|s_t), \hat{T}(s_t, a_t))} [r(s_t, a_t) - \gamma |G_M^\pi(s_t, a_t)|]
\end{aligned}$$

6.2 Model Based Offline Policy Optimization (MOPO)

Using the derivation from the preliminaries, we know if we set the penalty $\lambda u(s, a) \geq \gamma |G_M^\pi(s_t, a_t)|$ for some constant λ , any total expected reward under any policy under the modeled dynamics would underestimate the expected total reward of the policy under the true dynamics. Moreover, for a set of functions $\mathcal{F}$ that contains V_M^π , we know that

$$|G_M^\pi(s_t, a_t)| = |\mathbb{E}_{s' \sim \hat{T}(s, a)} [V_M^\pi(s')] - \mathbb{E}_{s' \sim T(s, a)} [V_M^\pi(s')]| \leq \sup_{f \in \mathcal{F}} \left| \mathbb{E}_{s' \sim \hat{T}(s, a)} [f(s')] - \mathbb{E}_{s' \sim T(s, a)} [f(s')] \right|$$

From the "Integral Probability Metric" section, we know if we let

$$\mathcal{F} = \{f : \|f\|_\infty \leq \|V_M^\pi\|_\infty \leq \frac{R_{max}}{1-\gamma}\}$$

then we could set

$$\begin{aligned} \frac{R_{max}}{(1-\gamma)\gamma} \lambda u(s, a) &= \frac{R_{max}}{1-\gamma} D_{TV}(\hat{T}(s, a) || T(s, a)) \\ &\geq \sup_{f \in \mathcal{F}} \left| \mathbb{E}_{s' \sim \hat{T}(s, a)} [f(s')] - \mathbb{E}_{s' \sim T(s, a)} [f(s')] \right| \\ &\geq |G_M^\pi(s_t, a_t)| \end{aligned}$$

Similarly, if

$$\mathcal{F} = \{f : \|f\|_{\mathcal{H}} \leq \nu\}$$

where

$$\nu \geq \|V_M^\pi\|_{\mathcal{H}}$$

for some *RKHS*, then we could set

$$\begin{aligned} \frac{\nu}{\gamma} \lambda u(s, a) &= \nu MMD(\hat{T}(s, a), T(s, a)) \\ &\geq \sup_{f \in \mathcal{F}} \left| \mathbb{E}_{s' \sim \hat{T}(s, a)} [f(s')] - \mathbb{E}_{s' \sim T(s, a)} [f(s')] \right| \\ &\geq |G_M^\pi(s_t, a_t)| \end{aligned}$$

In general, if $\lambda u(s, a) \geq d_{\mathcal{F}}(\hat{T}, T(s, a))$ and $V_M^\pi \in \mathcal{F}$, we can set the penalty to be

$$\lambda u(s, a) \geq c d_{\mathcal{F}}(\hat{T}(s, a), T(s, a)) \geq |G_M^\pi(s_t, a_t)|$$

such that the penalty $\lambda u(s, a)$ upper bounds $|G_M^\pi(s_t, a_t)|$ and will always produce an underestimation of the true total expected return if for any policy if we modify the reward function to be

$$\tilde{r}(s, a) = r(s, a) - \lambda u(s, a)$$

However, this is typically intractable (like with $MMD(\hat{T}(s, a), T(s, a))$ and $D_{TV}(\hat{T}(s, a) || T(s, a))$).

If we let $\tilde{M}$ represent the MDP that is the same as the original MDP M with the true dynamics but the modified reward function $\tilde{r}$

$$\begin{aligned} \eta_M(\pi) &\geq \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t | s_t), \hat{T}(s_t, a_t))} [r(s_t, a_t) - \gamma |G_M^\pi(s_t, a_t)|] \\ &\geq \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t | s_t), \hat{T}(s_t, a_t))} [r(s_t, a_t) - \lambda u(s, a)] \\ &= \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t | s_t), \hat{T}(s_t, a_t))} [r(s_t, a_t)] - \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t | s_t), \hat{T}(s_t, a_t))} [\lambda u(s, a)] \\ &= \eta_{\tilde{M}}(\pi) - \epsilon_u(\pi) \\ &= \eta_{\tilde{M}}(\pi) \end{aligned}$$

where

$$\epsilon_u(\pi) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t | s_t), \hat{T}(s_t, a_t))} [u(s, a)]$$

represent the total expected penalty under policy π and the modeled dynamics, we expect $\epsilon(\pi^\beta)$ to be small since $\hat{T}$ is trained over the data generated by $\epsilon(\pi^\beta)$. As a result, π^β will not produce out of distribution actions, meaning $\hat{T}$ will be relatively accurate (close to T) for trajectories under π^β and the uncertainty penalty will be generally be small. If we let

$$\delta_{min} = \min_{\pi} \epsilon_u(\pi)$$

for any $\delta \geq \delta_{min}$, we can define the optimal policy that gives total expected penalty at most δ as

$$\pi^\delta = \arg \max_{\pi : \epsilon_u(\pi) \leq \delta} \eta_M(\pi)$$

We know that for any policy π from "Preliminaries" that

$$\begin{aligned}
\eta_{\hat{M}}(\pi) - \eta_M(\pi) &= \gamma \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t|s_t), \hat{T}(s_t, a_t))} [G_{\hat{M}}^{\pi}(s_t, a_t)] \\
\Rightarrow |\eta_{\hat{M}}(\pi) - \eta_M(\pi)| &= \gamma \left| \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t|s_t), \hat{T}(s_t, a_t))} [G_{\hat{M}}^{\pi}(s_t, a_t)] \right| \\
&\leq \gamma \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{s_t, a_t \sim (p_0(s_0), \pi(a_t|s_t), \hat{T}(s_t, a_t))} [\lambda u(s_t, a_t)] \\
&= \lambda \epsilon_u(\pi)
\end{aligned}$$

Let us denote the learned optimal policy under $\tilde{M}$ as $\hat{\pi}$. The total expected reward under the true dynamics (the objective at test time) of $\hat{\pi}$ satisfies

$$\begin{aligned}
\eta_M(\hat{\pi}) &\geq \eta_{\tilde{M}}(\hat{\pi}) \quad (\text{modified reward produces underestimation}) \\
&\geq \eta_{\tilde{M}}(\pi) \quad (\text{definition of } \hat{\pi}) \\
&= \eta_{\hat{M}}(\pi) - \lambda \epsilon_u(\pi) \\
&\geq \eta_M(\pi) - 2\epsilon_u(\pi) \quad (\text{from above}) \\
\Rightarrow \eta_M(\hat{\pi}) &\geq \sup_{\pi} \{\eta_M(\pi) - 2\lambda \epsilon_u(\pi)\} \\
&\geq \eta_M(\hat{\pi}) \geq \eta_M(\pi^{\delta}) - 2\lambda \delta, \quad \delta \geq \delta_{min}
\end{aligned}$$

Notice this means that

$$\begin{aligned}
\eta_M(\hat{\pi}) &\geq \eta_M(\pi^{\beta}) - 2\lambda \epsilon_u(\pi^{\beta}) \\
\eta_M(\hat{\pi}) &\geq \eta_M(\pi^*) - 2\lambda \epsilon_u(\pi^*)
\end{aligned}$$

where π^* is the optimal policy under M . Since $\epsilon_u(\pi^{\beta})$ has a high chance to be small, $\eta_M(\hat{\pi})$ will likely be better than the behavior policy. Moreover, the lower bound $\eta_M(\hat{\pi})$ gets closer to $\eta_M(\pi^*)$ as the total expected uncertainty penalty gets smaller for π^* . The total expected penalty for π^* will generally be smaller if π^* is similar to π^{β} , which would allow for the trained dynamics model to be accurate over the states visited by π^* .

Note that a good penalty $\lambda u(s, a)$ can be difficult to implement in practice (e.g. we don't know $T(s, a)$ and it is difficult to find a good bound for $\|V_M^{\pi}\|_{\mathcal{H}}$). As a result, we just train an ensemble of dynamics models $\hat{T}^1(s, a), \dots, \hat{T}^N(s, a)$ which each output the mean and log standard deviation of multivariate gaussians with diagonal covariance matrices $\Sigma_{\phi}^i(s, a)$. We then set

$$\lambda u(s, a) = \lambda \max_{i \in \{1, \dots, N\}} \|\Sigma_{\phi}^i(s, a)\|_F$$

6.2.1 COMBO

We can use the idea of conservative Q-learning for policy evaluation on some policy π , where we add a penalty to the Q value evaluation update rule but combine it with model-based RL

$$\hat{Q}^{k+1} \leftarrow \arg \min_Q \beta \left(\mathbb{E}_{s \sim \hat{d}^{\pi}(s), a \sim \pi(a|s)} [Q(s, a)] - \mathbb{E}_{s, a \sim D} [Q(s, a)] \right) + \frac{1}{2} \mathbb{E}_{s \sim d_f^{\pi}(s), a \sim \pi(a|s)} \left[\left(Q(s, a) - \hat{B}^{\pi} \hat{Q}^k(s, a) \right)^2 \right]$$

where $d_f(s)$ represents the state marginal if a state is drawn from the modeled data with probability f and drawn from the offline dataset with probability $(1 - f)$

$$d_f(s) = f d^{\pi^{\beta}}(s) + (1 - f) \hat{d}^{\pi}(s)$$

Just like with Conservative Evaluation from before, we can use the learned critic to $\hat{Q}$ to improve π for example by taking a gradient step.

Using $\mathbb{E}_{s \sim \hat{d}^{\pi}(s), a \sim \pi(a|s)} [Q(s, a)]$ as the penalty pushes Q values down for state action pairs that are likely under the modeled dynamics and allow for the penalty to cancel out with $\mathbb{E}_{s, a \sim D} [Q(s, a)]$, similar to basic "Conservative Evaluation Equation 2". Essentially, basic "Conservative Evaluation" pushes down out of distribution actions for the current policy by setting $\mu = \pi$ but this takes it a step further by also pushing down on Q values for out of distribution states, which can now arise due to generating synthetic data using the learned dynamics model.

6.2.2 Trajectory Transformer

Besides using Dyna style algorithms, where we generate synthetic data and use that data to improve the policy, we can directly just model the dynamics and plan under the predicted dynamics. Essentially, given a dataset of transitions (s, a, s', r) , where states are dimension N , actions are dimension M , and rewards are of dimension 1, we first discretize the states and actions. The paper does this per dimension rather than using something like a VQ VAE. We then train a transformer using trajectories as sequence

$$\tau = (... , s_t^1, \dots, s_t^N, a_t^1, \dots, a_t^M, r_t, \dots)$$

where each token is one dimension of a state or action or a reward. This transformer can then be used for imitation learning using beam search during inference. To modify the use case for goal conditioned RL, we can train the model by prepending the goal state to the beginning of the sequence. For offline RL, for each transition, we can append the discounted reward to go R_t to each transition giving us

$$\tau = (... , s_t^1, \dots, s_t^N, a_t^1, \dots, a_t^M, r_t, R_t, \dots)$$

During inference, we then use beam search with probabilities on intermediate tokens and perform beam search using the cumulative reward plus the predicted reward to go once a full transition has been generated (every $N+M+2$ intermediate tokens). Theoretically the beam search width for intermediate and full transitions can be separate. Modeling the behavior policy implicitly biases the model to generate in distribution actions, so we do not need any special techniques to make sure we are underestimating the estimated rewards.

7 Maximum Margin Optimization

In hard margin SVMs, we want to solve the following optimization problem

$$\begin{aligned} \min_w \quad & \frac{1}{2} \|w\|^2 \\ \text{subject to} \quad & \max_i y_i (w^T x_i) \geq 1 \end{aligned}$$

This is the same problem as finding the largest margin m

$$\begin{aligned} \max_{m, \tilde{w}} \quad & m \\ \text{subject to} \quad & \max_i y_i (\tilde{w}^T x_i) \geq m \\ & \|\tilde{w}\|^2 = 1 \\ & m \geq 0 \end{aligned}$$

To prove this, we know that for any solution, we will prove that solving the first problem gives a solution to the second and vice versa. For any solution w that satisfies the constraints for the first problem

$$\begin{aligned} \tilde{w} &= \frac{w}{\|w\|^2} \\ m &= \frac{1}{\|w\|} \end{aligned}$$

is a feasible point to the second since

$$\begin{aligned} & \max_i y_i (w^T x_i) \geq 1 \\ \implies & y_i \left(\left(\frac{w}{\|w\|} \right)^T x_i \right) \geq \frac{1}{\|w\|} \\ \implies & \max_i y_i (\tilde{w}^T x_i) \geq m \end{aligned}$$

We also know that feasible point $m, \tilde{w}$ for the second problem implies that

$$w = \frac{\tilde{w}}{m}$$

is feasible for the first since

$$\begin{aligned} & \max_i y_i (\tilde{w}^T x_i) \geq m \\ \implies & y_i \left(\left(\frac{\tilde{w}}{m} \right)^T x_i \right) \geq 1 \quad (\text{m is positive so inequality does not flip}) \\ \implies & y_i (w^T x_i) \geq 1 \end{aligned}$$

If w^* is a solution to the first problem, we know that

$$\begin{aligned} \tilde{w}^* &= \frac{w^*}{\|w^*\|} \\ m^* &= \frac{1}{\|w^*\|} \end{aligned}$$

is feasible for the second. Assume for contradiction that there is another solution $m'^*, \tilde{w}'^*$ that satisfies the constraints of problem 2 such that $m'^* \geq m^*$. In this case, we know that

$$w'^* = \frac{\tilde{w}'^*}{m'^*}$$

is feasible for the first. However,

$$\begin{aligned}
\|w'^*\|^2 &= \left\| \frac{\tilde{w}^*}{m'^*} \right\|^2 \\
&= \frac{1}{(m'^*)^2} \\
&\leq \frac{1}{(m^*)^2} \\
&= \left\| \frac{\tilde{w}^*}{m^*} \right\|^2 \\
&= \left\| \frac{\frac{w^*}{\|w^*\|}}{\frac{1}{\|w^*\|}} \right\|^2 \\
&= \|w^*\|^2
\end{aligned}$$

which is a contradiction since w^* was supposed to be the feasible minimizer of problem 1. Similarly. going the other direction, if $\tilde{w}^*, m^*$ is a solution to the second problem

$$w^* = \frac{\tilde{w}^*}{m^*}$$

is a solution to problem 1. Note that this means

$$\begin{aligned}
\|w^*\| &= \left\| \frac{\tilde{w}^*}{m^*} \right\| \\
&= \frac{1}{m^*} \\
\implies m^* &= \frac{1}{\|w^*\|}
\end{aligned}$$

Assume for contradiction that there is another solution w'^* that satisfies the constraints of problem 1 such that $\frac{1}{2}\|w'^*\|^2 \leq \frac{1}{2}\|w^*\|^2$. In this case, we know that

$$\begin{aligned}
\tilde{w}'^* &= \frac{w'^*}{\|w'^*\|} \\
m'^* &= \frac{1}{\|w'^*\|}
\end{aligned}$$

is feasible for the second problem. However,

$$\begin{aligned}
m'^* &= \frac{1}{\|w'^*\|} \\
&\geq \frac{1}{\|w^*\|} \\
&= m^*
\end{aligned}$$

which is a contradiction since m^* was supposed to be the maximal feasible point of problem 2. One older algorithm for inverse RL was perform feature matching on a linear model reward

$$r_\psi(s, a) = \psi^T f(s, a)$$

where $f(s, a)$ is some feature embedding dependent on the state and action. Feature matching tries to make the expected value of $f(s, a)$ of the optimal policy under $r_\psi(s, a)$ similar to the observed policy (e.g. we try to make the policy under the reward visit similar states action pairs as the observed data)

$$\mathbb{E}_{\pi^{r_\psi}}[f(\mathbf{s}, \mathbf{a})] = \mathbb{E}_{\pi^*}[f(\mathbf{s}, \mathbf{a})]$$

where π^* is the observed policy. However, there are many different ψ that could give the same policy, and a heuristic to pick out a specific solution is to pick the policy that makes the expert look better e.g.

$$\begin{aligned} & \max_{\psi, m} \\ \text{such that } & \tilde{\psi}^T \mathbb{E}_{\pi^*}[f(\mathbf{s}, \mathbf{a})] \geq \max_{\pi \in \Pi} \tilde{\psi}^T \mathbb{E}_{\pi}[f(\mathbf{s}, \mathbf{a})] + m \\ & \|\tilde{\psi}\|^2 = 1 \\ & m \geq 0 \end{aligned}$$

where Π is the set of all possible policies.

Just like in the case with the SVM (proof is basically the same), this is equivalent to

$$\begin{aligned} & \min_{\psi} \frac{1}{2} \|\psi\|^2 \\ \text{such that } & \psi^T \mathbb{E}_{\pi^*}[f(\mathbf{s}, \mathbf{a})] \geq \max_{\pi \in \Pi} \psi^T \mathbb{E}_{\pi}[f(\mathbf{s}, \mathbf{a})] + 1 \end{aligned}$$

Moreover, we might also want to give preference to policies close to π^* giving us the following modified problem

$$\begin{aligned} & \min_{\psi} \frac{1}{2} \|\psi\|^2 \\ \text{such that } & \psi^T \mathbb{E}_{\pi^*}[f(\mathbf{s}, \mathbf{a})] \geq \max_{\pi \in \Pi} \psi^T \mathbb{E}_{\pi}[f(\mathbf{s}, \mathbf{a})] + D(\pi^*, \pi) \end{aligned}$$

where D can be something like difference in feature expectations or KL divergence.

7.1 Inference as Control

When performing a task, the expert is typically not deterministic and can make mistakes but those mistakes typically are less likely to impact the outcome of the task. These mistakes can arise from many reasons, such as the expert being lazy. For instance, if making a left turn in a hallway, it might be optimal to make the turn while being closer to the left side wall but people may drift to the right side since the manner the turn is made does not really matter that much in the overarching goal of say reaching another room. To model near-optimal behavior and random behavior, we can use a probabilistic graphical model where the agent will prefer to do tasks that have little impact on the reward at random. For the graphical model, we have the typical state and action variables but also add separate Bernoulli optimality O_t variables for all state actions at each pair time step that represents whether the agent is trying to be optimal at that timestep such that

$$P(O_t|s_t, a_t) = \exp(r(s_t, a_t))$$

where O_t is conditionally independent of other variables given (s_t, a_t) . Note this model assumes that all rewards are negative but additive factors to the reward does not affect optimal behavior (we can subtract the max reward from all rewards to make the reward negative). We can think of this as trying to collect some optimality token at each time step with the probability of collecting a token proportional to the reward at that time step.

One inference problem we can try to solve in this model is

$$\begin{aligned} P(\tau|O_{1:t}) &= \frac{P(\tau, O_{1:t})}{P(O_{1:t})} \\ &= \frac{P(s_0) \prod_t P(a_t|s_t) P(s_{t+1}|s_t, a_t) P(O_t|s_t, a_t)}{P(O_{1:t})} \\ &= \frac{P(s_0) \prod_t P(a_t|s_t) P(s_{t+1}|s_t, a_t) \exp(r(s_t, a_t))}{P(O_{1:t})} \\ &= \frac{P(s_0) \prod_t P(a_t|s_t) P(s_{t+1}|s_t, a_t) \exp(r(s_t, a_t))}{P(O_{1:t})} \\ &= \frac{P(s_0) \prod_t P(a_t|s_t) P(s_{t+1}|s_t, a_t) \prod_t^T \exp(r(s_t, a_t))}{P(O_{1:t})} \\ &= \frac{P(\tau) \exp(\sum_t r(s_t, a_t))}{P(O_{1:t})} \\ &\propto P(\tau) \exp(\sum_t r(s_t, a_t)) \end{aligned}$$

In this model, we typically fix the prior $P(a_t|s_t)$ to be uniform. We also assume an action is conditionally independent of the past given the current state and that s_{t+1} (the dynamics) is conditionally independent of the past given s_t, a_t . Another assumption is that O_t is conditionally independent of all other random variables given s_t, a_t , which implies O_t is conditionally independent of the past given s_t .

$$\begin{aligned} &P(O_t|s_t, a_{t-1}, s_{t-1}, O_{t-1}, \dots, s_0, a_0, O_0) \\ &= \int_a P(O_t, a_t|s_t, a_{t-1}, s_{t-1}, O_{t-1}, \dots, s_0, a_0, O_0) da_t \\ &= \int_a p(a_t|s_t, a_{t-1}, s_{t-1}, O_{t-1}, \dots, s_0, a_0, O_0) P(O_t|a_t, s_t, a_{t-1}, s_{t-1}, O_{t-1}, \dots, s_0, a_0, O_0) da_t \\ &= \int_a p(a_t|s_t) P(O_t|a_t, s_t) da_t \\ &= \int_a P(O_t, a_t|s_t) da_t \\ &= P(O_t|s_t) \end{aligned}$$

7.1.1 Backwards Messages

In this graphical model, we can compute the problem of being optimal in all future timesteps given a state or action, which we call a state action backward message

$$\begin{aligned}
\beta_t(s_t, a_t) &= p(O_{t:T}|s_t, a_t) \\
&= \int_{s_{t+1}} p(O_{t:T}, s_{t+1}|s_t, a_t) ds_{t+1} \\
&= \int_{s_{t+1}} p(O_{t+1:T}|s_{t+1}, s_t, O_t, a_t) p(s_{t+1}|s_t, a_t, O_t) p(O_t|s_t, a_t) ds_{t+1} \\
&= \int_{s_{t+1}} p(O_{t+1:T}|s_{t+1}) p(s_{t+1}|s_t, a_t) p(O_t|s_t, a_t) ds_{t+1} \\
&= \int_{s_{t+1}} \left(\int_{a_{t+1}} p(O_{t+1:T}|s_{t+1}, a_{t+1}) p(a_{t+1}|s_{t+1}) da_{t+1} \right) p(s_{t+1}|s_t, a_t) p(O_t|s_t, a_t) ds_{t+1} \\
&= \int_{s_{t+1}} \left(\int_{a_{t+1}} \beta_{t+1}(s_{t+1}, a_{t+1}) p(a_{t+1}|s_{t+1}) da_{t+1} \right) p(s_{t+1}|s_t, a_t) p(O_t|s_t, a_t) ds_{t+1} \\
&= p(O_t|s_t, a_t) \int_{s_{t+1}} \mathbb{E}_{a_{t+1} \sim p(a_{t+1}|s_{t+1})} [\beta_{t+1}(s_{t+1}, a_{t+1})] p(s_{t+1}|s_t, a_t) ds_{t+1} \\
&= \exp(r(s_t, a_t)) \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\mathbb{E}_{a_{t+1} \sim p(a_{t+1}|s_{t+1})} [\beta_{t+1}(s_{t+1}, a_{t+1})]] \\
&= \exp(r(s_t, a_t)) \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\beta_{t+1}(s_{t+1})]
\end{aligned}$$

where

$$\beta_t(s_t) = \mathbb{E}_{a_t \sim p(a_t|s_t)} [\beta_t(s_t, a_t)]$$

Defining

$$\begin{aligned}
V_t(s_t) &= \log \beta_t(s_t) \\
Q_t(s_t, a_t) &= \log \beta_t(s_t, a_t) \\
&\implies \\
\beta_t(s_t) &= \exp(V_t(s_t)) \\
\beta_t(s_t, a_t) &= \exp(Q_t(s_t, a_t))
\end{aligned}$$

we get that

$$\begin{aligned}
\beta_t(s_t, a_t) &= \exp(r(s_t, a_t)) \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\beta_{t+1}(s_{t+1})] \\
\beta_t(s_t) &= \mathbb{E}_{a_t \sim p(a_t|s_t)} [\beta_t(s_t, a_t)] \\
&\implies \\
\exp(Q_t(s_t, a_t)) &= \exp(r(s_t, a_t)) \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\exp(V_{t+1}(s_{t+1}))] \\
\exp(V_t(s_t)) &= \mathbb{E}_{a_t \sim p(a_t|s_t)} [\beta_t(s_t, a_t)] \\
&\implies \\
Q_t(s_t, a_t) &= r(s_t, a_t) + \log \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\exp(V_{t+1}(s_{t+1}))] \\
V_t(s_t) &= \log \mathbb{E}_{a_t \sim p(a_t|s_t)} [\beta_t(s_t, a_t)] \\
&\implies \\
Q_t(s_t, a_t) &= r(s_t, a_t) + \log \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\exp(V_{t+1}(s_{t+1}))] \\
V_t(s_t) &= \log \int_{a_{t+1}} \exp(Q_t(s_t, a_t) + \log p(a_t|s_t))
\end{aligned}$$

We can then start at the last time step T and recursively compute:

$$\begin{aligned}
1. Q_t(s_t, a_t) &= r(s_t, a_t) + \log \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\exp(V_{t+1}(s_{t+1}))] \\
2. V_t(s_t) &= \log \int_{a_{t+1}} \exp(Q_t(s_t, a_t) + \log p(a_t|s_t))
\end{aligned}$$

where $V_{T+1}(s_{T+1}) = 0 \iff \beta_{T+1}(s_{T+1}) = 1$ Notice how if we set

$$\tilde{r}(s_t, a_t) = r(s_t, a_t) + \log p(a_t|s_t) - \log \tilde{p}(a_t|s_t)$$

we get that

$$\begin{aligned}\tilde{Q}_t(s_t, a_t) &= \tilde{r}(s_t, a_t) + \log \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\exp(V_{t+1}(s_{t+1}))] = Q_t(s_t, a_t) + \log p(a_t|s_t) - \log \tilde{p}(a_t|s_t) \\ &\iff Q_t(s_t, a_t) = \tilde{Q}_t(s_t, a_t) - \log p(a_t|s_t) + \log \tilde{p}(a_t|s_t)\end{aligned}$$

This gives us the following alternative update rule

$$\begin{aligned}1. \tilde{Q}_t(s_t, a_t) &= \tilde{r}(s_t, a_t) + \log \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\exp(V_{t+1}(s_{t+1}))] \\ 2. V_t(s_t) &= \log \int_{a_{t+1}} \exp(\tilde{Q}_t(s_t, a_t) + \log \tilde{p}(a_t|s_t))\end{aligned}$$

meaning we can replace $p(a_t|s_t)$ with any arbitrary function $\tilde{p}(a_t|s_t)$, which does not need to integrate to 1 or be a constant like 0, and still recover the state backward message as usual. We can transform $\tilde{Q}$ to Q to recover the state action backward message. However, in the setting with this graphical model, we typically assume the prior is uniform.

Computing the backward messages requires us to know the transition dynamics and integrate over all possible states/actions.

7.1.2 Forward Messages

In the graphical model, we can also compute the probability of a state given we are optimal for all previous time steps, which we call the forward message

$$\begin{aligned}\alpha_t(s_t) &= p(s_t|\mathcal{O}_{1:t-1}) \\ &= \int_{a_{t-1}} \int_{s_{t-1}} p(s_t, s_{t-1}, a_{t-1}|\mathcal{O}_{1:t-1}) ds_{t-1} da_{t-1} \\ &= \int_{a_{t-1}} \int_{s_{t-1}} p(s_t|s_{t-1}, a_{t-1}, \mathcal{O}_{1:t-1}) p(a_{t-1}|s_{t-1}, \mathcal{O}_{1:t-1}) p(s_{t-1}|\mathcal{O}_{1:t-1}) ds_{t-1} da_{t-1} \\ &= \int_{a_{t-1}} \int_{s_{t-1}} p(s_t|s_{t-1}, a_{t-1}) p(a_{t-1}|s_{t-1}, \mathcal{O}_{1:t-1}) p(s_{t-1}|\mathcal{O}_{1:t-1}) ds_{t-1} da_{t-1} \\ &= \int_{a_{t-1}} \int_{s_{t-1}} p(s_t|s_{t-1}, a_{t-1}) \frac{p(a_{t-1}, s_{t-1}, \mathcal{O}_{1:t-1})}{p(s_{t-1}, \mathcal{O}_{1:t-1})} \frac{p(s_{t-1}, \mathcal{O}_{1:t-1})}{p(\mathcal{O}_{1:t-1})} ds_{t-1} da_{t-1} \\ &= \int_{a_{t-1}} \int_{s_{t-1}} p(s_t|s_{t-1}, a_{t-1}) \frac{p(a_{t-1}, s_{t-1}, \mathcal{O}_{1:t-1})}{p(\mathcal{O}_{1:t-1})} ds_{t-1} da_{t-1} \\ &= \int_{a_{t-1}} \int_{s_{t-1}} p(s_t|s_{t-1}, a_{t-1}) \frac{p(\mathcal{O}_{t-1}|a_{t-1}, s_{t-1}, \mathcal{O}_{1:t-2}) p(a_{t-1}|s_{t-1}, \mathcal{O}_{1:t-2}) p(s_{t-1}|\mathcal{O}_{1:t-2}) p(\mathcal{O}_{1:t-2})}{p(\mathcal{O}_{t-1}|\mathcal{O}_{1:t-2}) p(\mathcal{O}_{1:t-2})} ds_{t-1} da_{t-1} \\ &= \int_{a_{t-1}} \int_{s_{t-1}} p(s_t|s_{t-1}, a_{t-1}) p(\mathcal{O}_{t-1}|a_{t-1}, s_{t-1}, \mathcal{O}_{1:t-2}) p(a_{t-1}|s_{t-1}, \mathcal{O}_{1:t-2}) p(s_{t-1}|\mathcal{O}_{1:t-2}) ds_{t-1} da_{t-1} \\ &= \int_{a_{t-1}} \int_{s_{t-1}} p(s_t|s_{t-1}, a_{t-1}) p(\mathcal{O}_{t-1}|a_{t-1}, s_{t-1}) p(a_{t-1}|s_{t-1}, \mathcal{O}_{1:t-2}) \alpha_{t-1}(s_{t-1}) ds_{t-1} da_{t-1} \\ &= \int_{a_{t-1}} \int_{s_{t-1}} p(s_t|s_{t-1}, a_{t-1}) \exp(r(s_{t-1}, a_{t-1})) p(a_{t-1}|s_{t-1}) \alpha_{t-1}(s_{t-1}) ds_{t-1} da_{t-1}\end{aligned}$$

where we use the assumption given a state, the action is conditionally independent of all previous timesteps just like above when computing the backward messages and in the original MDP. Just like with computing the backward messages, this requires us to know the transition dynamics and integrate over all possible states/actions.

7.1.3 State Marginals Under Optimality

The probability of a state while being optimal can be computed as

$$\begin{aligned}
p(s_t|O_{1:T}) &= \frac{p(s_t, O_{1:T})}{p(O_{1:T})} \\
&= \frac{p(O_{t:T}|s_t, O_{1:t-1})p(s_t, O_{1:t-1})p(O_{1:t-1})}{p(O_{1:T})} \\
&= \frac{p(O_{t:T}|s_t)p(s_t, O_{1:t-1})}{p(O_{t:T}|O_{1:t-1})} \\
&= \frac{\beta_t(s_t)\alpha_t(s_t)}{p(O_{t:T}|O_{1:t-1})} \\
&\propto \beta_t(s_t)\alpha_t(s_t)
\end{aligned}$$

Let's say we have problem where we want to reach a goal state, where the reward is $-\infty$ everywhere except the goal state with reward 0. Start by visualizing a timeline on the x-axis from left to right and the y-axis represents states. $\beta_t(s_t)$ can be thought of as a triangle with a base at the left (time 0) that covers states which can reach the goal state. Note that $\beta_t(s_t)$ is technically decreases at smaller time steps due to have more probability terms being multiplied in, but we can think of this as the backward message being normalized by something accounts for smaller trajectory probabilities at smaller time steps, such as the average probability of a trajectory at the timestep (may not be best, just for intuition. Note that if transitions are deterministic, the probability of a trajectory from a state at a particular time step is just 1 over the number of possible trajectories starting from that state at the time step assuming we have uniform action priors. We can similarly think of $\alpha_t(s_t)$ as a triangle with a base at the last time step (time T) which covers states that can be reached given we act optimally. The intersection of the two triangles can be thought of as states that have a chance of being seen at each timestep, given that we act optimally at all time steps.

7.1.4 Deriving the Policy

Using the backward messages, we can compute the probability of an action given that we are always trying to be optimal, which can be used as a policy. Just like in the original MDP, we assume that given a state, the action is conditionally independent of all previous timesteps.

$$\begin{aligned}
\pi(a_t|s_t) &= p(a_t|s_t, O_{1:T}) \\
&= p(a_t|s_t, O_{t:T}) \\
&= \frac{p(a_t, s_t, O_{t:T})}{p(s_t, O_{t:T})} \\
&= \frac{p(a_t, s_t|O_{t:T})p(O_{t:T})}{p(s_t|O_{t:T})p(O_{t:T})} \\
&= \frac{p(a_t, s_t|O_{t:T})}{p(s_t|O_{t:T})} \\
&= \frac{p(O_{t:T}|a_t, s_t)p(a_t, s_t)/p(O_{t:T})}{p(O_{t:T}|s_t)p(s_t)/p(O_{t:T})} \\
&= \frac{p(O_{t:T}|a_t, s_t)p(a_t, s_t)}{p(O_{t:T}|s_t)p(s_t)} \\
&= \frac{p(O_{t:T}|a_t, s_t)p(a_t|s_t)p(s_t)}{p(O_{t:T}|s_t)p(s_t)} \\
&= \frac{\beta_t(s_t, a_t)}{\beta_t(s_t)}p(a_t|s_t) \\
&= p(a_t|s_t) \exp(Q(s_t, a_t) - V_t(s_t)) \\
&= p(a_t|s_t) \exp(A(s_t, a_t))
\end{aligned}$$

However, this gives us an optimistic policy since the update rule for the backwards message is as follows

$$\begin{aligned} 1. Q_t(s_t, a_t) &= r(s_t, a_t) + \log \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [\exp(V_{t+1}(s_{t+1}))] \\ 2. V_t(s_t) &= \log \int_{a_{t+1}} \exp(Q_t(s_t, a_t) + \log p(a_t|s_t)) \end{aligned}$$

For the Q function update, we take the log expected value for of the exponent rather than just the expected value, which is essentially take a soft max. As a result, the Q value will be higher for actions that will have a chance of reaching a high value state even if that chance is incredibly small (e.g. the next state values are dominated by the luckiest state).

Consider a scenario with 2 time steps, where the action is whether we buy a lottery ticket, where the reward for time step 1 is all the same. Here, we have the same reward whether we choose to buy a ticket or not in time step 1 and will have the reward resulting from our action in time step 1 at time step 2 just for the sake of showing an example of optimistic transitions. Note In time step 1, if we buy a lottery ticket and win, we enter a state where no matter what action we take, we get a large reward. Similarly, we buy a lottery ticket in the first time step and lose, we enter a state where no matter which action we get, we get a small reward. If we do not buy a lottery ticket at the first timestep, we enter a state where no matter which action we get, we get a medium reward. The probability of being optimal is high if the reward is larger. Finding the action with the highest probability given we were optimal in both time steps is like asking what was our action given we earned a lot of money, which would lead us to the incorrect action of buying a lottery ticket. This is because

$$p(s_{t+1}|s_t, a_t, \mathcal{O}_{1:T}) \neq p(s_{t+1}|s_t, a_t)$$

when conditioned on all time steps being optimal, the next state for any action taken uses optimistic transition dynamics.

To mitigate this issue, we want to find a policy $\pi(a|s)$ such that the distribution of trajectories $q(\tau)$ under π has a close to $p(\tau|\mathcal{O}_{1:T})$ (the distribution where the probability of each trajectory is directly proportion to the exponential of the sum of rewards along that trajectory) but uses the original dynamics $p(s_{t+1}|s_t, a_t)$. To make $q(\tau)$ use the original dynamics, we fix the distribution class for q such the probability of a trajectory must depend on only the policy and original dynamics

$$q(s_{1:T}, a_{1:T}) = p(s_1) \prod_t p(s_{t+1}|s_t, a_t) \pi(a_t|s_t)$$

Note that π is dependent on the time step in the finite horizon case as usual and fully characterizes q . We can then try to find the policy π that minimizes the KL divergence between $p(\tau|\mathcal{O}_{1:T})$ and $q(\tau)$. Even though VAEs are typically used to maximize the prior $\log(p(x))$ by maximizing the ELBO with respect to both $p(x|z)$ (parameterized as $p_\theta(x|z)$ and $q(z)$ (parameterized as $q_\phi(z|x)$ in the amortized set up)

$$\begin{aligned} \log(p(x)) &= \mathbb{E}_{z \sim q(z)} [\log(p(x, z))] - \mathbb{E}_{z \sim q(z)} [q(z)] + D_{KL}(q(z)||p(z|x)) \\ &= ELBO + D_{KL}(q(z)||p(z|x)) \end{aligned}$$

we can also minimize $D_{KL}(q(z)||p(z|x))$ by maximizing the ELBO with respect to $q(z)$. In our case, we want to minimize $D_{KL}(q(\tau)||p(\tau|\mathcal{O}_{1:T}))$, which can be done by maximizing

$$ELBO = \mathbb{E}_{\tau \sim q(\tau)} [\log(p(\mathcal{O}_{1:T}, \tau))] - \mathbb{E}_{\tau \sim q(\tau)} [\log(q(\tau))]$$

where we essentially replace z with τ and x with $\mathcal{O}_{1:T}$. This gives us

$$\begin{aligned}
ELBO &= \mathbb{E}_{\tau \sim q(\tau)} [\log(p(\mathcal{O}_{1:T}, \tau))] - \mathbb{E}_{\tau \sim q(\tau)} [q(\tau)] \\
&= \mathbb{E}_{\tau \sim q(\tau)} \left[\log(p(s_1) \prod_{t=1}^T p(a_t|s_t) p(s_{t+1}|s_t, a_t) p(O_t|s_t, a_t)) \right] - \mathbb{E}_{\tau \sim q(\tau)} \left[\log(p(s_1) \prod_{t=1}^T p(s_{t+1}|s_t, a_t) \pi(a_t|s_t)) \right] \\
&= \mathbb{E}_{\tau \sim q(\tau)} \left[\log(p(s_1) \prod_{t=1}^T p(a_t|s_t) p(s_{t+1}|s_t, a_t) p(O_t|s_t, a_t)) \right] - \mathbb{E}_{\tau \sim q(\tau)} \left[\log(p(s_1) \prod_{t=1}^T p(s_{t+1}|s_t, a_t) \pi(a_t|s_t)) \right] \\
&= \mathbb{E}_{\tau \sim q(\tau)} \left[\log p(s_1) + \sum_{t=1}^T \log p(a_t|s_t) + \log p(s_{t+1}|s_t, a_t) + \log p(O_t|s_t, a_t) \right] \\
&\quad - \mathbb{E}_{\tau \sim q(\tau)} \left[\log p(s_1) + \sum_{t=1}^T \log p(s_{t+1}|s_t, a_t) + \log \pi(a_t|s_t) \right] \\
&= \mathbb{E}_{\tau \sim q(\tau)} \left[\sum_{t=1}^T \log p(a_t|s_t) + \log p(O_t|s_t, a_t) - \log \pi(a_t|s_t) \right] \\
&= \mathbb{E}_{\tau \sim q(\tau)} \left[\sum_{t=1}^T \log p(a_t|s_t) + r(s_t, a_t) - \log \pi(a_t|s_t) \right]
\end{aligned}$$

Assuming $p(a_t|s_t)$ is uniform and equal to a constant c , we get that

$$\begin{aligned}
\arg \max_{\pi} ELBO &= \arg \max_{\pi} \mathbb{E}_{\tau \sim q(\tau)} \left[\sum_{t=1}^T \log p(a_t|s_t) + r(s_t, a_t) - \log \pi(a_t|s_t) \right] \\
&= \arg \max_{\pi} \mathbb{E}_{\tau \sim q(\tau)} \left[\sum_{t=1}^T c + r(s_t, a_t) - \log \pi(a_t|s_t) \right] \\
&= \arg \max_{\pi} c + \mathbb{E}_{\tau \sim q(\tau)} \left[\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t|s_t) \right] \\
&= \arg \max_{\pi} \mathbb{E}_{\tau \sim q(\tau)} \left[\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t|s_t) \right] \\
&= \arg \max_{\pi} \mathbb{E}_{\tau \sim q(\tau)} \left[\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t|s_t) \right] \\
&= \arg \max_{\pi} \sum_{t=1}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t) - \log \pi(a_t|s_t)] \\
&= \arg \max_{\pi} \sum_{t=1}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t)] - \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [\log \pi(a_t|s_t)] \\
&= \arg \max_{\pi} \sum_{t=1}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t)] - \mathbb{E}_{s_t \sim q(\tau)} [\mathbb{E}_{a_t \sim \pi(a_t|s_t)} [\log \pi(a_t|s_t)|s_t]] \\
&= \arg \max_{\pi} \sum_{t=1}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t)] + \mathbb{E}_{s_t \sim q(\tau)} [\mathcal{H}(\pi(a|s))] \\
&= \arg \max_{\pi} \sum_{t=1}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t)] + \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [\mathcal{H}(\pi(a_t|s_t))] \\
&= \arg \max_{\pi} \sum_{t=1}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t) + \mathcal{H}(\pi(a_t|s_t))]
\end{aligned}$$

To solve this optimization problem, we can use dynamic programming and start from the last time step

$$\arg \max_{\pi} \mathbb{E}_{(s_T, a_T) \sim q(\tau)} [r(s_T, a_T) + \mathcal{H}(\pi(a_T|s_T))]$$

From the "Explicit Policy Constraints" section with $\lambda = 1$, we get that for any fixed s_T

$$\pi(a_T|s_T) = \frac{1}{Z(s_T)} \frac{1}{|A|} \exp(r(s_T, a_T))$$

where $Z(s_T)$ is a normalization constant

$$\begin{aligned} \pi(a_T|s_T) &= \frac{1}{Z(s_T)} \frac{1}{|A|} \exp(r(s_T, a_T)) \\ \implies \sum_{a_T} \frac{1}{Z(s_T)} \frac{1}{|A|} \exp(r(s_T, a_T)) &= 1 \\ \implies Z(s_T) &= \sum_{a_T} \frac{1}{|A|} \exp(r(s_T, a_T)) \\ \implies \pi(a_T|s_T) &= \frac{\exp(r(s_T, a_T))}{\sum_{a_T} \exp(r(s_T, a_T))} \end{aligned}$$

Plugging in the optimal value of $\pi(a_T|s_T)$ in the maximization problem

$$\begin{aligned} \max_{\pi} \mathbb{E}_{(s_T, a_T) \sim q(\tau)} [r(s_T, a_T) + \mathcal{H}(\pi(a_T|s_T))] &= \mathbb{E}_{(s_T, a_T) \sim q(\tau)} [r(s_T, a_T) - \log \pi(a_T|s_T)] \\ &= \mathbb{E}_{(s_T, a_T) \sim q(\tau)} [r(s_T, a_T) - \log \pi(a_T|s_T)] \\ &= \mathbb{E}_{(s_T, a_T) \sim q(\tau)} \left[r(s_T, a_T) - \log \exp(r(s_T, a_T)) + \log \sum_{a_T} \exp(r(s_T, a_T)) \right] \\ &= \mathbb{E}_{(s_T, a_T) \sim q(\tau)} \left[r(s_T, a_T) - r(s_T, a_T) + \log \sum_{a_T} \exp(r(s_T, a_T)) \right] \\ &= \mathbb{E}_{(s_T, a_T) \sim q(\tau)} \left[\log \sum_{a_T} \exp(r(s_T, a_T)) \right] \end{aligned}$$

Let's define the Q_t as the sum of the reward at a time set plus the expected value of the V_t at the next time step over the transition dynamics (where $V_{T+1} = 0$) and V_t as the log sum exp of Q_t

$$\begin{aligned} Q_T(s_T, a_T) &= r(s_T, a_T) + \mathbb{E}_{s_{T+1} \sim p(s_{T+1}|s_T, a_T)} [V_{T+1}(s_{T+1})|s_T, a_T] = r(s_T, a_T) \\ V_T(s_T) &= \log \sum_{a_T} \exp(Q(s_T, a_T)) \end{aligned}$$

Substituting in $V_T(s_T)$, we get

$$\begin{aligned} \max_{\pi(a_T|s_T)} \mathbb{E}_{(s_T, a_T) \sim q(\tau)} [r(s_T, a_T) + \mathcal{H}(\pi(a_T|s_T))] &= \mathbb{E}_{(s_T, a_T) \sim q(\tau)} \left[\log \sum_{a_T} \exp(r(s_T, a_T)) \right] \\ &= \mathbb{E}_{(s_T) \sim q(\tau)} [V_T(s_T)] \end{aligned}$$

Using induction, we can prove that at any time step k ,

$$\begin{aligned} 1. \pi(a_k|s_k) &= \frac{\exp(Q(s_k, a_k))}{\sum_{a_T} \exp(Q(s_k, a_k))} \\ 2. \max_{\pi} \sum_{t=k}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t) + \mathcal{H}(\pi(a_t|s_t))] &= V_t(s_t) \end{aligned}$$

where

$$\begin{aligned} 1. Q_t(s_t, a_t) &= r(s_t, a_t) + \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [V(s_{t+1})|s_t, a_t] \\ 2. V_t(s_t) &= \log \sum_{a_t} \exp(Q(s_t, a_t)) \end{aligned}$$

The base case holds as shown above when solving for the last time step above.
In the inductive step, we have that

$$\begin{aligned}
\max_{\pi} \sum_{t=k-1}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t) + \mathcal{H}(\pi(a_t|s_t))] &= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} [r(s_{k-1}, a_{k-1}) + \mathcal{H}(\pi(a_{k-1}|s_{k-1}))] \\
&\quad + \max_{\pi} \sum_{t=k}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t) + \mathcal{H}(\pi(a_t|s_t))] \\
&= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} [r(s_{k-1}, a_{k-1}) + \mathcal{H}(\pi(a_{k-1}|s_{k-1}))] \\
&\quad + \mathbb{E}_{s_k \sim q(\tau)} [V_k(s_k)] \quad (\text{inductive hypothesis}) \\
&= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}, s_k) \sim q(\tau)} [r(s_{k-1}, a_{k-1}) + V_k(s_k) + \mathcal{H}(\pi(a_{k-1}|s_{k-1}))] \\
&= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} [\mathbb{E}_{s_k \sim p(s_k|s_{k-1}, a_{k-1})} [r(s_{k-1}, a_{k-1}) \\
&\quad + V_k(s_k) + \mathcal{H}(\pi(a_{k-1}|s_{k-1})) | s_{k-1}, a_{k-1}]] \\
&= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} [r(s_{k-1}, a_{k-1}) + \mathbb{E}_{s_k \sim p(s_k|s_{k-1}, a_{k-1})} [\\
&\quad + V_k(s_k) | s_{k-1}, a_{k-1}] + \mathcal{H}(\pi(a_{k-1}|s_{k-1}))] \\
&= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} [Q_{k-1}(s_{k-1}, a_{k-1}) + \mathcal{H}(\pi(a_{k-1}|s_{k-1}))]
\end{aligned}$$

Just like above, we get that for any fixed s_{k-1} , the maximizing policy at time step $k-1$ satisfies

$$\pi(a_{k-1}|s_{k-1}) = \frac{\exp(Q(s_{k-1}, a_{k-1}))}{\sum_{a_{k-1}} \exp(Q(s_{k-1}, a_{k-1}))}$$

Plugging this in, we get

$$\begin{aligned}
\max_{\pi} \sum_{t=k-1}^T \mathbb{E}_{(s_t, a_t) \sim q(\tau)} [r(s_t, a_t) + \mathcal{H}(\pi(a_t|s_t))] &= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} [Q_{k-1}(s_{k-1}, a_{k-1}) + \mathcal{H}(\pi(a_{k-1}|s_{k-1}))] \\
&= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} [Q_{k-1}(s_{k-1}, a_{k-1}) - \log \pi(a_{k-1}|s_{k-1})] \\
&= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} \left[Q_{k-1}(s_{k-1}, a_{k-1}) - \log \frac{\exp(Q(s_{k-1}, a_{k-1}))}{\sum_{a_T} \exp(Q(s_{k-1}, a_{k-1}))} \right] \\
&= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} [Q_{k-1}(s_{k-1}, a_{k-1}) - Q(s_{k-1}, a_{k-1}) \\
&\quad + \log \sum_{a_{k-1}} \exp(Q(s_{k-1}, a_{k-1}))] \\
&= \max_{\pi} \mathbb{E}_{(s_{k-1}, a_{k-1}) \sim q(\tau)} \left[\log \sum_{a_{k-1}} \exp(Q(s_{k-1}, a_{k-1})) \right] \\
&= \max_{\pi} \mathbb{E}_{s_{k-1} \sim q(\tau)} [V_{k-1}(s_{k-1})]
\end{aligned}$$

which completes the inductive step. This means we can solve for the optimal π by recursive repeating the following backwards through time

$$\begin{aligned}
1. Q_t(s_t, a_t) &= r(s_t, a_t) + \mathbb{E}_{s_{t+1} \sim p(s_{t+1}|s_t, a_t)} [V_{t+1}(s_{t+1})] \\
2. V_t(s_t) &= \log \sum_{a_t} Q_t(s_t, a_t)
\end{aligned}$$

This optimal π being proportional to the exponential Q makes the probability of each trajectory proportional to the exp of the sum of rewards times the probability of the transition dynamics of the trajectory.

We can turn this into the following Q value iteration algorithm

$$\begin{aligned}
1. Q(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim p(s'|s, a)} [V(s')] \\
2. V(s) &= \text{soft} \max_a Q(s, a)
\end{aligned}$$

This is essentially the original Q value iteration algorithm, but we replace the max with a soft max. We can do the same with fitted Q -iteration with a replay buffer and replace the max with a soft max.

We can also derive a policy gradient as follows

$$\begin{aligned}
\nabla_{\theta} J(\theta) &= \nabla_{\theta} \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\sum_{t=1}^T r(s_t, a_t) + \mathcal{H}(\pi_{\theta}(a_t | s_t)) \right] \\
&= \nabla_{\theta} \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\sum_{t=1}^T r(s_t, a_t) - \log \pi_{\theta}(a_t | s_t) \right] \\
&= \int_{\tau} \nabla_{\theta} \left(p_{\theta}(\tau) \left(\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) \right) d\tau \\
&= \int_{\tau} (\nabla_{\theta} (p_{\theta}(\tau))) \left(\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) + (p_{\theta}(\tau)) \nabla_{\theta} \left(\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) d\tau \quad (\text{chain rule}) \\
&= \int_{\tau} p_{\theta}(\tau) \nabla_{\theta} \log p_{\theta}(\tau) \left(\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) - p_{\theta}(\tau) \left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) d\tau \\
&= \int_{\tau} p_{\theta}(\tau) \left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) \left(\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) - p_{\theta}(\tau) \left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) d\tau \\
&= \int_{\tau} p_{\theta}(\tau) \left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) \left(\left(\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) - 1 \right) d\tau \\
&= \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) \left(\left(\sum_{t=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) - 1 \right) \right] \\
&= \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(\left(\sum_{t'=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) - 1 \right) \right] \\
&= \sum_{t=1}^T \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(\left(\sum_{t'=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) - 1 \right) \right] \\
&= \sum_{t=1}^T \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(\sum_{t'=1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) \right] \quad (\text{constant baseline does not affect gradient}) \\
&= \sum_{t=1}^T \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(\sum_{t'=t}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) \right] \quad (\text{apply reward to to go}) \\
&= \sum_{t=1}^T \mathbb{E}_{\tau_{1:t+1} \sim p_{\theta}(\tau)} \left[\mathbb{E}_{\tau_{t+2:T} \sim p_{\theta}(\tau)} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(\sum_{t'=t}^T r(s_t, a_t) - \log \pi(a_t | s_t) \right) \middle| \tau_{1:t+1} \right] \right] \\
&= \sum_{t=1}^T \mathbb{E}_{\tau_{1:t+1} \sim p_{\theta}(\tau)} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \mathbb{E}_{\tau_{t+2:T} \sim p_{\theta}(\tau)} \left[\sum_{t'=t}^T r(s_t, a_t) - \log \pi(a_t | s_t) \middle| \tau_{1:t+1} \right] \right] \\
&= \sum_{t=1}^T \mathbb{E}_{\tau_{1:t+1} \sim p_{\theta}(\tau)} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(r(s_t, a_t) - \log \pi(a_t | s_t) + \mathbb{E}_{\tau_{t+2:T} \sim p_{\theta}(\tau)} \left[\sum_{t'=t+1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \middle| \tau_{1:t+1} \right] \right) \right] \\
&= \sum_{t=1}^T \mathbb{E}_{\tau_{1:t+1} \sim p_{\theta}(\tau)} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left(r(s_t, a_t) - \log \pi(a_t | s_t) + \mathbb{E}_{\tau_{t+2:T} \sim p_{\theta}(\tau)} \left[\sum_{t'=t+1}^T r(s_t, a_t) - \log \pi(a_t | s_t) \middle| s_{t+1}, a_{t+1} \right] \right) \right] \\
&= \sum_{t=1}^T \mathbb{E}_{\tau_{1:t+1} \sim p_{\theta}(\tau)} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) (r(s_t, a_t) - \log \pi(a_t | s_t) + Q_{\theta}(s_{t+1}, a_{t+1}))]
\end{aligned}$$

Using this in a policy gradient algorithm, we can where we fit a Q value network and set the policy as

$$\pi(a|s) = \exp(Q_{\theta}(s, a) - V_{\theta}(s))$$

where $V(s)$ can be a separate network or estimated by averaging samples from $Q_\theta(s, a)$ under the current policy. Substituting this in, we get

$$\begin{aligned}
\nabla_\theta J(\theta) &= \sum_{t=1}^T \mathbb{E}_{\tau_{1:t+1} \sim p_\theta(\tau)} [\nabla_\theta \log \pi_\theta(a_t | s_t) (r(s_t, a_t) - \log \pi(a_t | s_t) + Q_\theta(s_{t+1}, a_{t+1}))] \\
&= \sum_{t=1}^T \mathbb{E}_{\tau_{1:t+1} \sim p_\theta(\tau)} [(\nabla_\theta Q_\theta(s_t, a_t) - \nabla_\theta V_\theta(s)) (r(s_t, a_t) - Q_\theta(s_t, a_t) + V_\theta(s) + Q_\theta(s_{t+1}, a_{t+1}))] \\
&= \sum_{t=1}^T \mathbb{E}_{\tau_{1:t+1} \sim p_\theta(\tau)} [(\nabla_\theta Q_\theta(s_t, a_t) - \nabla_\theta V_\theta(s)) (r(s_t, a_t) + Q_\theta(s_{t+1}, a_{t+1}) - Q_\theta(s_t, a_t))] \quad (\text{state dependent baseline})
\end{aligned}$$

Compare this to soft max Q learning algorithm from above

$$\begin{aligned}
1. Q(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim p(s' | s, a)} [V(s')] \\
2. V(s) &= \text{soft} \max_a Q(s, a) \\
&\iff \\
1. Q(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim p(s' | s, a)} [\text{soft} \max_a Q(s', a)]
\end{aligned}$$

which can be optimized with a Q_θ network with the gradient of the squared error between $Q_\theta(s_t, a_t)$ and $(r(s_t, a_t) + \text{soft} \max_a Q(s_{t+1}, a))$

$$\sum_{t=1}^T \mathbb{E}_{\tau_{1:t+1} \sim p_\theta(\tau)} \left[\nabla_\theta Q_\theta(s_t, a_t) \left(r(s_t, a_t) + \text{soft} \max_a Q(s_{t+1}, a) - Q_\theta(s_t, a_t) \right) \right]$$

Notice that we can get rid of the softmax if we are doing on-policy Q-learning (e.g. the sample trajectories are from a soft optimal policy using the existing Q_θ where the Q_θ includes the entropy penalty at all time steps including the first).

7.2 Soft Actor Critic

Instead of having an implicit policy given by the Q value network as in policy gradient with soft optimality (it was technically actor critic but the actor is determined by the critic)

$$\pi(a|s) = \exp(Q_\theta(s, a) - V_\theta(s))$$

we can have a separate network for the policy π_θ and repeat the following

$$\begin{aligned} 1. Q^\pi(s, a) &= r(s, a) + \mathbb{E}_{s' \sim p(s'|s, a), a' \sim \pi(a'|s')} [Q^\pi(s', a') - \log \pi(a'|s')] \quad (\text{policy evaluation}) \\ 2. \pi &= \arg \min_{\pi'} D_{KL} \left(\pi'(\cdot|s) \parallel \frac{1}{Z(s)} \exp(Q_\pi(s, \cdot)) \right) \end{aligned}$$

We can think of step 1 as performing variational inference in the graphical model to learn the Q value associated with the ELBO. The Q value learned has the entropy bonus added at every time step except the first unlike the policy gradient with soft optimality which has the Q value also include the entropy bonus at the first time step. We can rewrite the above steps as the following loss function updates with the a Q_ϕ and π_θ network as follows

$$\begin{aligned} 1a. \phi &\leftarrow \phi - \nabla_\phi \left(\frac{1}{2} \mathbb{E}_{(s, a, s') \in \mathcal{D}} Q_\phi(s, a) (Q_\phi(s, a) - y(s, a, s')) \right) \\ &\approx \phi - \nabla_\phi Q_\phi(s, a) (Q_\phi(s, a) - y(s, a, s')) \quad (\text{single sample estimator}) \\ 2. \theta &\leftarrow \theta - \nabla_\theta \mathbb{E}_{(s, a, s') \in \mathcal{D}} \left[\mathbb{E}_{\tilde{a} \sim \pi_\theta(\tilde{a}|s)} \left[\left(\log \pi_\theta(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_\phi(s, \tilde{a})) \right) \right) \right] \right] \\ &\approx \theta - \nabla_\theta \mathbb{E}_{(s, a, s') \in \mathcal{D}} \left[\mathbb{E}_{\tilde{a} \sim \pi_\theta(\tilde{a}|s)} \left[\left(\log \pi_\theta(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_\phi(s, \tilde{a})) \right) \right) \right] \right] \\ &\approx \theta - \nabla_\theta \mathbb{E}_{\tilde{a} \sim \pi_\theta(\tilde{a}|s)} \left[\left(\log \pi_\theta(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_\phi(s, \tilde{a})) \right) \right) \right] \quad (\text{single sample estimator}) \end{aligned}$$

where

$$y(s, a, s') = \text{StopGradient} (r(s, a) + (Q^\phi(s', a') - \log \pi_\theta(a'|s')))$$

If we do not use the reparameterization trick, we can simplify 2 as

$$\begin{aligned} 2. \theta &\leftarrow \theta - \nabla_\theta \mathbb{E}_{\tilde{a} \sim \pi_\theta(\tilde{a}|s)} \left[\left(\log \pi_\theta(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_\phi(s, \tilde{a})) \right) \right) \right] \\ &= \theta - \int_{\tilde{a}} \nabla_\theta \left(\pi_\theta(\tilde{a}|s) \left(\log \pi_\theta(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_\phi(s, \tilde{a})) \right) \right) \right) d\tilde{a} \\ &= \theta - \int_{\tilde{a}} \pi_\theta(\tilde{a}|s) \nabla_\theta \log \pi_\theta(\tilde{a}|s) \left(\log \pi_\theta(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_\phi(s, \tilde{a})) \right) \right) + \pi_\theta(\tilde{a}|s) (\nabla_\theta \log \pi_\theta(\tilde{a}|s)) d\tilde{a} \quad (\text{chain rule}) \\ &= \theta - \int_{\tilde{a}} \pi_\theta(\tilde{a}|s) \nabla_\theta \log \pi_\theta(\tilde{a}|s) \left(\log \pi_\theta(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_\phi(s, \tilde{a})) \right) + 1 \right) d\tilde{a} \\ &= \theta - \mathbb{E}_{\tilde{a} \sim \pi_\theta(\tilde{a}|s)} \left[\nabla_\theta \log \pi_\theta(\tilde{a}|s) \left(\log \pi_\theta(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_\phi(s, \tilde{a})) \right) + 1 \right) \right] \\ &= \theta - \mathbb{E}_{\tilde{a} \sim \pi_\theta(\tilde{a}|s)} \left[\nabla_\theta \log \pi_\theta(\tilde{a}|s) \left(\log \pi_\theta(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_\phi(s, \tilde{a})) \right) \right) \right] \quad (\text{constant baseline}) \\ &= \theta - \mathbb{E}_{\tilde{a} \sim \pi_\theta(\tilde{a}|s)} [\nabla_\theta \log \pi_\theta(\tilde{a}|s) (\log \pi_\theta(\tilde{a}|s) - Q_\phi(s, \tilde{a}) + \log Z(s))] \\ &= \theta - \mathbb{E}_{\tilde{a} \sim \pi_\theta(\tilde{a}|s)} [\nabla_\theta \log \pi_\theta(\tilde{a}|s) (\log \pi_\theta(\tilde{a}|s) - Q_\phi(s, \tilde{a}))] \quad (\text{state dependent baseline}) \\ &\approx \theta - \nabla_\theta \log \pi_\theta(\tilde{a}|s) (\log \pi_\theta(\tilde{a}|s) - Q_\phi(s, \tilde{a})) \quad (\text{single sample estimator}) \end{aligned}$$

If we do use the parameterization trick with a Gaussian by setting $\pi_\theta(\tilde{a}|s) = \mathcal{N}(\mu_\theta(s), \sigma(s))$, we can instead sample $\epsilon \sim \mathcal{N}(0, I)$ to get a sample for $\tilde{a}$ using

$$\tilde{a} = \mu_\theta(s) + \sigma_\theta(s) \odot \epsilon = f(\epsilon, \theta)$$

This let's us simplify 2 as

$$\begin{aligned}
2.\theta &\leftarrow \theta - \nabla_{\theta} \mathbb{E}_{\tilde{a} \sim \pi_{\theta}(\tilde{a}|s)} \left[\left(\log \pi_{\theta}(\tilde{a}|s) - \log \left(\frac{1}{Z(s)} \exp(Q_{\phi}(s, \tilde{a})) \right) \right) \right] \\
&= \theta - \nabla_{\theta} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} \left[\left(\log \pi_{\theta}(f(\epsilon, \theta)|s) - \log \left(\frac{1}{Z(s)} \exp(Q_{\phi}(s, f(\epsilon, \theta))) \right) \right) \right] \\
&= \theta - \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} \left[\nabla_{\theta} \left(\log \pi_{\theta}(f(\epsilon, \theta)|s) - \log \left(\frac{1}{Z(s)} \exp(Q_{\phi}(s, f(\epsilon, \theta))) \right) \right) \right] \\
&= \theta - \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} [\nabla_{\theta} (\log \pi_{\theta}(f(\epsilon, \theta)|s) - Q_{\phi}(s, f(\epsilon, \theta)) + Z(s))] \\
&= \theta - \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} [\nabla_{\theta} (\log \pi_{\theta}(f(\epsilon, \theta)|s) - Q_{\phi}(s, f(\epsilon, \theta)))] \quad (Z(s) \text{ is not dependent on } \theta) \\
&\approx \theta - \nabla_{\theta} (\log \pi_{\theta}(f(\epsilon, \theta)|s) - Q_{\phi}(s, f(\epsilon, \theta))) \quad (\text{single sample estimator})
\end{aligned}$$

Plugging in these derivations for 2, adding a discount factor, and adding a temperature α , we get that the soft actor critic essentially repeats

$$\begin{aligned}
1a.\phi &\leftarrow \phi - \nabla_{\phi} Q_{\phi}(s, a) (Q_{\phi}(s, a) - y(s, a, s')) \\
2.\theta &\leftarrow \theta - \nabla_{\theta} \log \pi_{\theta}(\tilde{a}|s) (\alpha \log \pi_{\theta}(\tilde{a}|s) - Q_{\phi}(s, \tilde{a}))
\end{aligned}$$

if we do not use the reparameterization trick or

$$\begin{aligned}
1a.\phi &\leftarrow \phi - \nabla_{\phi} Q_{\phi}(s, a) (Q_{\phi}(s, a) - y(s, a, s')) \\
2.\theta &\leftarrow \theta - \nabla_{\theta} (\alpha \log \pi_{\theta}(f(\epsilon, \theta)|s) - Q_{\phi}(s, f(\epsilon, \theta)))
\end{aligned}$$

if we do use the reparameterization trick where

$$y(s, a, s') = \text{StopGradient} (r(s, a) + \gamma (Q^{\phi}(s', a') - \alpha \log \pi_{\theta}(a'|s')))$$

Note that we typically use a different $f(\epsilon, \theta)$ than $\mu_{\theta}(s) + \sigma_{\theta}(s) \odot \epsilon$ and can average over a batch of data points per iteration that are used for both updates in 1 and 2 instead of using a single sample estimator. The original SAC paper also learns a separate value function network V_{ψ} and uses the following update rule before step 1 in each iteration

$$V^{\pi}(s) = \mathbb{E}_{\tilde{a} \sim \pi(\tilde{a}|s)} [Q^{\pi}(s, \tilde{a}) - \log \pi(\tilde{a}|s)]$$

This V_{π} includes the entropy bonus in the first time step. The reason we do not need the entropy bonus in the first timestep for Q is because are minimizing $D_{KL} \left(\pi'(\cdot|s) \parallel \frac{1}{Z(s)} \exp(Q_{\pi}(s, \cdot)) \right)$ where

$$\frac{1}{Z(s)} \exp(Q_{\pi}(s, \cdot))$$

is the optimal policy given we add the entropy bonus to the first time step (e.g. the optimal policy we are using as the target assumes we added an entropy bonus to the first timestep on top of whatever Q function is learned).

7.3 Inverse RL

7.4 Max Ent IRL

Given the graphical model that incorporates whether the agent is trying to be optimal at each time step, we know that

$$P(\tau|O_{1:t}) \propto P(\tau) \exp\left(\sum_t r(s_t, a_t)\right)$$

In the inverse RL setting, we can try to explain agent behavior using this graphical model by finding a reward function that maximizes the probability of the observed trajectories. This essentially minimizes $D_{KL}(p_{\pi^*}(\tau) || P(\tau|O_{1:t}, \psi))$ where π^* is the expert policy that we have samples for

$$\begin{aligned} \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\log P(\tau|O_{1:t}, \psi)] &= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{P(\tau) \exp(\sum_t r_{\psi}(s_t, a_t))}{P(O_{1:t})} \right] \\ &= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{P(\tau) \exp(\sum_t r_{\psi}(s_t, a_t))}{\int_{\tau'} P(\tau') P(O_{1:t}|\tau') d\tau'} \right] \\ &= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{P(\tau) \exp(\sum_t r_{\psi}(s_t, a_t))}{\int_{\tau'} P(\tau') \exp(\sum_t r_{\psi}(s_t, a_t)) d\tau'} \right] \quad (P(\tau) \text{ does not depend on } \psi) \\ &= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{\exp(\sum_t r_{\psi}(s_t, a_t))}{\int_{\tau'} P(\tau') \exp(\sum_t r_{\psi}(s_t, a_t)) d\tau'} \right] \\ &= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \exp\left(\sum_t r_{\psi}(s_t, a_t)\right) \right] - \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \int_{\tau'} P(\tau') \exp\left(\sum_t r_{\psi}(s_t, a_t)\right) d\tau' \right] \\ &= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t r_{\psi}(s_t, a_t) \right] - \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \int_{\tau'} P(\tau') \exp\left(\sum_t r_{\psi}(s_t, a_t)\right) d\tau' \right] \\ &= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t r_{\psi}(s_t, a_t) \right] - \log \int_{\tau'} P(\tau') \exp\left(\sum_t r_{\psi}(s_t, a_t)\right) d\tau' \end{aligned}$$

Taking the gradient, we have that

$$\begin{aligned} &\nabla_{\psi} \left(\mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t r_{\psi}(s_t, a_t) \right] - \log \int_{\tau'} P(\tau') \exp\left(\sum_t r_{\psi}(s_t, a_t)\right) d\tau' \right) \\ &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right] - \frac{1}{\int_{\tau'} P(\tau') \exp(\sum_t r_{\psi}(s_t, a_t)) d\tau'} \int_{\tau'} P(\tau') \nabla_{\psi} \exp\left(\sum_t r_{\psi}(s_t, a_t)\right) d\tau' \\ &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right] - \frac{1}{P(O_{1:t}|\psi)} \int_{\tau'} P(\tau') \nabla_{\psi} \exp\left(\sum_t r_{\psi}(s_t, a_t)\right) d\tau' \\ &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right] - \frac{1}{P(O_{1:t}|\psi)} \int_{\tau'} P(\tau') \exp\left(\sum_t r_{\psi}(s_t, a_t)\right) \nabla_{\psi} \left(\sum_t r_{\psi}(s_t, a_t) \right) d\tau' \\ &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right] - \frac{1}{P(O_{1:t}|\psi)} \int_{\tau'} P(\tau, O_{1:t}|\psi) \nabla_{\psi} \left(\sum_t r_{\psi}(s_t, a_t) \right) d\tau' \\ &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right] - \int_{\tau'} \frac{P(\tau, O_{1:t}|\psi)}{P(O_{1:t}|\psi)} \left(\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right) d\tau' \\ &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right] - \int_{\tau'} P(\tau|O_{1:t}, \psi) \left(\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right) d\tau' \\ &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right] - \mathbb{E}_{\tau \sim P(\tau|O_{1:t}, \psi)} \left[\sum_t \nabla_{\psi} r_{\psi}(s_t, a_t) \right] \\ &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\nabla_{\psi} r(\tau)] - \mathbb{E}_{\tau \sim P(\tau|O_{1:t}, \psi)} [\nabla_{\psi} r(\tau)] \end{aligned}$$

To approximate this gradient, we can do the following

1. Run a max entropy RL algorithm to convergence to learn $p(a_t|s_t\mathcal{O}_{1:T}, \psi)$
2. Run $p(a_t|s_t\mathcal{O}_{1:T}, \psi)$ to get samples
3. $\psi = \psi - \sum_{i=1}^N \nabla_{\psi} r_{\psi}(\tau_i) - \sum_{j=1}^M \sum_t \nabla_{\psi} \nabla_{\psi} r_{\psi}(\tau_j)$

where τ_i are samples from the expert policy and τ_j are samples from 2. However, this requires us to run max entropy RL algorithm like soft Q learning, entropy regularized policy gradient, soft actor critic, etc., to convergence for every gradient step of the reward. Note that these three algorithms do not actually learn $p(a_t|s_t\mathcal{O}_{1:T}, \psi)$ (which assumes optimistic transitions) but learns the policy that tries to minimize the KL divergence between the distribution of trajectories under that policy and $P(\tau|O_{1:t}, \psi)$, the distribution of trajectories that is proportional to the exponential of the sum of rewards along the trajectory and the probability of the trajectory according to the dynamics. Learning $p(a_t|s_t\mathcal{O}_{1:T}, \psi)$ exactly requires a finite state space and knowledge of transition dynamics to compute the backwards messages.

7.4.1 Guided Cost Learning

Just like above

$$\begin{aligned}
\arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\log P(\tau|O_{1:t}, \psi)] &= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{P(\tau) \exp(r_{\psi}(\tau))}{P(O_{1:t})} \right] \\
&= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{P(\tau) \exp(r_{\psi}(\tau))}{\int_{\tau} P(\tau, O_{1:t}) d\tau} \right] \\
&= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{P(\tau) \exp(r_{\psi}(\tau))}{\int_{\tau} P(\tau) \exp(\sum_t r_{\psi}(s_t, a_t)) d\tau} \right] \\
&= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{P(\tau) \exp(r_{\psi}(\tau))}{Z_{\psi}} \right] \\
&= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\log P(\tau) + r_{\psi}(\tau) - \log Z] \\
&= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{P(\tau) \exp(r_{\psi}(\tau))}{Z_{\psi}} \right] \\
&= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\log P(\tau)] - \log Z_{\psi} \\
&= \arg \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] - \log Z_{\psi} \quad (\text{term does not depend on } \psi)
\end{aligned}$$

Instead of taking the gradient first, we can use importance sampling for Z_ψ before taking the gradient

$$\begin{aligned}
Z_\psi &= \int_\tau P(\tau) \exp\left(\sum_t r_\psi(s_t, a_t)\right) \\
&= \int_\tau q(\tau) \frac{P(\tau) \exp\left(\sum_t r_\psi(s_t, a_t)\right)}{q(\tau)} \\
&= \int_\tau q(\tau) \frac{p_0(s_0) \prod_{t=0}^T P(s_{t+1}|s_t, a_t) p(a_t|s_t) \exp\left(\sum_t r_\psi(s_t, a_t)\right)}{p_0(s_0) \prod_{t=0}^T P(s_{t+1}|s_t, a_t) \pi(a_t|s_t)} \\
&= \int_\tau q(\tau) \frac{\prod_{t=0}^T p(a_t|s_t) \exp\left(\sum_t r_\psi(s_t, a_t)\right)}{\prod_{t=0}^T \pi(a_t|s_t)} \\
&= \int_\tau q(\tau) \frac{\prod_{t=0}^T c \exp\left(\sum_t r_\psi(s_t, a_t)\right)}{\prod_{t=0}^T \pi(a_t|s_t)} \quad (\text{action prior is uniform in model}) \\
&= c^T \int_\tau q(\tau) \frac{\exp(r_\psi(\tau))}{\prod_{t=0}^T \pi(a_t|s_t)} \\
&= c^T \mathbb{E}_{\tau \sim q(\tau)} \left[\frac{\exp(r_\psi(\tau))}{\prod_{t=0}^T \pi(a_t|s_t)} \right] \\
&= \mathbb{E}_{\tau \sim q(\tau)} \left[\frac{\exp(r_\psi(\tau) + T \log c)}{\prod_{t=0}^T \pi(a_t|s_t)} \right] \\
&= \mathbb{E}_{\tau \sim q(\tau)} \left[\frac{\exp(r_\psi(\tau))}{\prod_{t=0}^T \pi(a_t|s_t)} \right] \quad (\text{if } c = 1)
\end{aligned}$$

The paper assumes an action prior $c = 1$, which is always possible if we just add $\log c$ to all rewards (which does not change the optimal policy). Moreover, the paper assumes deterministic starting state and transition dynamics, which means that $\prod_{t=0}^T \pi(a_t|s_t) = P(\tau)$. We can then estimate Z using sample trajectories from some distribution $q(\tau)$, which we denote as $\mathcal{D}_q$

$$Z \approx \frac{1}{M} \sum_{\tau_j \in \mathcal{D}_q} \frac{\exp(r_\psi(\tau))}{\prod_{t=0}^T \pi(a_t|s_t)}$$

Taking the gradient of the objective

$$\begin{aligned}
\nabla_\psi (\mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \log Z_\psi) &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\nabla_\psi r_\psi(\tau)] - \frac{1}{Z} \mathbb{E}_{\tau \sim q(\tau)} \left[\nabla_\psi \frac{\exp(r_\psi(\tau))}{\prod_{t=0}^T \pi(a_t|s_t)} \right] \\
&= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\nabla_\psi r_\psi(\tau)] - \frac{1}{Z} \mathbb{E}_{\tau \sim q(\tau)} \left[\frac{\exp(r_\psi(\tau))}{\prod_{t=0}^T \pi(a_t|s_t)} \nabla_\psi r_\psi(\tau) \right] \\
&= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\nabla_\psi r_\psi(\tau)] - \frac{1}{Z} \mathbb{E}_{\tau \sim q(\tau)} [w_\psi(\tau) \nabla_\psi r_\psi(\tau)] \\
&= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\nabla_\psi r_\psi(\tau)] - \frac{1}{Z} \mathbb{E}_{\tau \sim q(\tau)} [w_\psi(\tau) \nabla_\psi r_\psi(\tau)]
\end{aligned}$$

where

$$w_\psi(\tau) = \frac{\exp(r_\psi(\tau))}{\prod_{t=0}^T \pi(a_t|s_t)}$$

This is the same gradient we derived previously.

$$\begin{aligned}
\nabla_\psi (\mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \log Z_\psi) &= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\nabla_\psi r_\psi(\tau)] - \frac{1}{Z} \mathbb{E}_{\tau \sim q(\tau)} [w_\psi(\tau) \nabla_\psi r_\psi(\tau)] \\
&= \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\nabla_\psi r_\psi(\tau)] - \mathbb{E}_{\tau \sim P(\tau|O_{1:t}, \psi)} [\nabla_\psi r_\psi(\tau)]
\end{aligned}$$

We can perform gradient ascent to maximize the objective using by repeating

$$\begin{aligned}
\psi &\leftarrow \psi - \frac{1}{N} \sum_{\tau_i \in \mathcal{D}_{\pi^*}} \nabla_{\psi} r_{\psi}(\tau_i) - \frac{1}{\frac{1}{M} \sum_{\tau_j \in \mathcal{D}_q} \frac{\exp(r_{\psi}(\tau_j))}{\prod_{t=0}^T \pi(a_t|s_t)}} \frac{1}{M} \sum_{\tau_j} w_{\psi}(\tau_j) \nabla_{\psi} r_{\psi}(\tau_j) \\
&= \psi - \frac{1}{N} \sum_{\tau_i \in \mathcal{D}_{\pi^*}} \nabla_{\psi} r_{\psi}(\tau_i) - \frac{1}{\frac{1}{M} \sum_{\tau_j \in \mathcal{D}_q} \frac{\exp(r_{\psi}(\tau_j))}{\prod_{t=0}^T \pi(a_t|s_t)}} \frac{1}{M} \sum_{\tau_j} w_{\psi}(\tau_j) \nabla_{\psi} r_{\psi}(\tau_j) \\
&= \psi - \frac{1}{N} \sum_{\tau_i \in \mathcal{D}_{\pi^*}} \nabla_{\psi} r_{\psi}(\tau_i) - \frac{1}{\sum_{\tau_j \in \mathcal{D}_q} w_{\psi}(\tau_j)} \sum_{\tau_j} w_{\psi}(\tau_j) \nabla_{\psi} r_{\psi}(\tau_j)
\end{aligned}$$

When estimating

$$\mathbb{E}_{x \sim p(x)} [f(x)]$$

with importance sampling using

$$\mathbb{E}_{x \sim q(x)} \left[\frac{p(x)f(x)}{q(x)} \right]$$

the variance is minimized when $q(x) \propto |f(x)p(x)|$ (proof at <https://artowen.su.domains/mc/Ch-var-is.pdf>). Thus, the variance of the importance sampled estimate of Z_{ψ} is minimized when $q(\tau) \propto P(\tau) \exp(r_{\psi}(\tau))$, which is distribution of the soft optimal policy.

Guided cost learning repeats the following

$$1. \psi \leftarrow \psi - \frac{1}{N} \sum_{\tau_i \in \mathcal{D}_{\pi^*}} \nabla_{\psi} r_{\psi}(\tau_i) - \frac{1}{\sum_{\tau_j \in \mathcal{D}_q} w_{\psi}(\tau_j)} \sum_{\tau_j} w_{\psi}(\tau_j) \nabla_{\psi} r_{\psi}(\tau_j) \quad 2. \text{Update the policy using max entropy RL and } r_{\psi} \text{ (not neces}$$

For 2, we can do something like soft actor critic or policy gradient with soft optimality (paper uses guided policy search).

Note that importance sampling distribution was set to a mixture policy of the demonstration distribution p^* and the samples from the learned policy q rather than just q in the paper, which means

$$Z_{\psi} = \mathbb{E}_{\tau \sim q(\tau)} \left[\frac{\exp(r_{\psi}(\tau))}{\frac{1}{2} \prod_{t=0}^T \pi(a_t|s_t) + \frac{1}{2} \prod_{t=0}^T \pi^*(a_t|s_t)} \right]$$

Note that whether we use the mixture policy or the learned policy, as the learned policy gets closer to the soft optimal policy, the variance of the importance sampling estimator decreases. Note that p is trained using behavioral cloning. The paper used LQR (assumed linear dynamics with Gaussian noise) to learn an approximate $\pi^*(a_t|s_t)$.

7.4.2 GANs and IRL

With GANs, we have a generator that generates inputs $x = G(z)$ where z is some prior distribution we sample from and there is a discriminator that tries to output the probability any given sample is real. We typically feed equal number of generated and fake samples during training. The objective that the discriminator maximizes is given by

$$L_D = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))]$$

and the loss function of the generator given by

$$L_G = \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))]$$

which is just the original discriminator objective without the first term, which is not impacted by the generator. The optimal discriminator gives the probability the input x is real

$$\begin{aligned}
L_D &= \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))] \\
&= \int_x p_{\text{data}}(x) \log D(x) dx + \int_z p(z) \log(1 - D(G(z))) dz \\
&= \int_x p_{\text{data}}(x) \log D(x) dx + \int_x p_{\text{generator}}(x) \log(1 - D(x)) dx \\
&= \int_x p_{\text{data}}(x) \log D(x) p_{\text{generator}}(x) \log(1 - D(x)) dx
\end{aligned}$$

In the case where $D(x)$ can be any function, we can find the optimal value of $D(x)$ for each fixed input x by taking the derivative of the integrand with respect to $D(x)$ and setting it to 0

$$\begin{aligned}
0 &= \frac{\partial}{\partial D^*(x)} (p_{\text{data}}(x) \log D(x) + p_{\text{generator}}(x) \log(1 - D^*(x))) \\
&= \frac{p_{\text{data}}(x)}{D(x)} - \frac{p_{\text{generator}}(x)}{1 - D^*(x)} \\
\implies p_{\text{data}}(x)(1 - D^*(x)) &= p_{\text{generator}} D^*(x) \\
\implies p_{\text{data}}(x) &= (p_{\text{generator}}(x) + p_{\text{data}}(x)) D^*(x) \\
\implies D^*(x) &= \frac{p_{\text{data}}(x)}{p_{\text{generator}}(x) + p_{\text{data}}(x)}
\end{aligned}$$

This is precisely the probability that an input is real

$$\begin{aligned}
p(r|x) &= \frac{p(x|r)p(r)}{p(x|r)p(r) + p(x|\bar{r})p(\bar{r})} \\
&= \frac{p(x|r)}{p(x|r) + p(x|\bar{r})} \\
&= \frac{p_{\text{data}}(x)}{p_{\text{generator}}(x) + p_{\text{data}}(x)} \\
&= D^*(x)
\end{aligned}$$

where we enforce $p(r) = p(\bar{r})$ by using an equal number of real and fake training examples.

Guided cost learning can be thought of as a game where one step maximizes the reward of the expert distributions and minimizes the reward of the current soft optimal policy to make them easier to distinguish while another step increases the reward of the current soft optimal policy to make harder to distinguish from the expert distribution. We can extend this to GANs by parameterizing the discriminator as

$$\begin{aligned}
\frac{p^*(\tau)}{p^*(\tau) + q(\tau)} &= \frac{\frac{1}{Z} p_0(s_0) \prod_{t=0}^T P(s_{t+1}|s_t, a_t) p(a_t|s_t) \exp(r_\psi(\tau))}{\frac{1}{Z} p_0(s_0) \prod_{t=0}^T P(s_{t+1}|s_t, a_t) p(a_t|s_t) \exp(r_\psi(\tau)) + p_0(s_0) \prod_{t=0}^T P(s_{t+1}|s_t, a_t) \pi_\theta(a_t|s_t)} \\
&= \frac{\frac{1}{Z} \prod_{t=0}^T p(a_t|s_t) \exp(r_\psi(\tau))}{\frac{1}{Z} \prod_{t=0}^T p(a_t|s_t) \exp(r_\psi(\tau)) + \prod_{t=0}^T \pi_\theta(a_t|s_t)} \\
&= \frac{\frac{1}{Z} \exp(r_\psi(\tau))}{\frac{1}{Z} \exp(r_\psi(\tau)) + \prod_{t=0}^T \pi_\theta(a_t|s_t)} \quad (\text{assuming } p(a_t|s_t) = 1)
\end{aligned}$$

for a particular ψ that defines a reward. Essentially, we can think of ψ as the parameters the discriminator is trying to optimize. The discriminator takes an input τ and uses ψ to output $\exp(r_\psi(\tau))$. Note that this parameterization is possible and can be applied whenever the generator distribution $q(\tau)$ can be evaluated and we can just let the discriminator learn $p^*(\tau)$ for real and generate samples τ . Note that in this case, maximizing the discriminator objective is essentially just directly maximizing $p^*(\tau)$, which means we do not technically need a GAN when $q(\tau)$ can be evaluated.

The Discriminator takes in a sample τ and returns

$$\begin{aligned}
\sigma(r_\psi(\tau) - \log(Z) - \log q(\tau)) &= \frac{1}{1 + \exp(-(r_\psi(\tau) + \log(Z) - \log q(\tau)))} \\
&= \frac{\exp(r_\psi(\tau) - \log Z)}{\exp(r_\psi(\tau) - \log Z) + \exp(\log q(\tau))} \\
&= \frac{\frac{1}{Z} \exp(r_\psi(\tau))}{\frac{1}{Z} \exp(r_\psi(\tau)) + q(\tau)}
\end{aligned}$$

where ψ and Z are learnable parameters. Note that even though the partition function Z is dependent on ψ , the Z here can be thought of as an independent parameter from ψ .

During training, the discriminator and generator objectives are as follows

$$\begin{aligned}
1. \arg \max_{\psi, Z} L_D &= \arg \max_{\psi, Z} \mathbb{E}_{\tau \sim p^*(\tau)} [-\log D(\tau)] + \mathbb{E}_{\tau \sim q_\theta(\tau)} [-\log(1 - D(\tau))] \\
&= \arg \max_{\psi, Z} \mathbb{E}_{\tau \sim p^*(\tau)} \left[-\log \frac{\frac{1}{Z} \exp(r_\psi(\tau))}{\frac{1}{Z} \exp(r_\psi(\tau)) + q_\theta(\tau)} \right] + \mathbb{E}_{\tau \sim q_\theta(\tau)} \left[-\log \frac{q_\theta(\tau)}{\frac{1}{Z} \exp(r_\psi(\tau)) + q_\theta(\tau)} \right] \\
&= \arg \max_{\psi, Z} \frac{1}{2} \mathbb{E}_{\tau \sim p^*(\tau)} \left[-\log \frac{\frac{1}{Z} \exp(r_\psi(\tau))}{\frac{1}{Z} \exp(r_\psi(\tau)) + q_\theta(\tau)} \right] + \frac{1}{2} \mathbb{E}_{\tau \sim q_\theta(\tau)} \left[-\log \frac{q_\theta(\tau)}{\frac{1}{Z} \exp(r_\psi(\tau)) + q_\theta(\tau)} \right] \\
&= \arg \max_{\psi, Z} \mathbb{E}_{\tau \sim p^*(\tau)} \left[-\log \frac{\frac{1}{Z} \exp(r_\psi(\tau))}{\frac{1}{2Z} \exp(r_\psi(\tau)) + \frac{1}{2} q_\theta(\tau)} \right] + \mathbb{E}_{\tau \sim q_\theta(\tau)} \left[-\log \frac{q_\theta(\tau)}{\frac{1}{2Z} \exp(r_\psi(\tau)) + \frac{1}{2} q_\theta(\tau)} \right] \\
2. \arg \min_{\theta} L_G &= \arg \min_{\theta} \mathbb{E}_{\tau \sim q_\theta(\tau)} [-\log D(\tau)] + \mathbb{E}_{\tau \sim q_\theta(\tau)} [\log(1 - D(\tau))]
\end{aligned}$$

We multiplied the objective for 1. by 2 since that is what the paper does (<https://arxiv.org/pdf/1611.03852>) in order to have a cleaner proof to show that maximizing L_D with respect to Z gives us a Z that turns out to be equal to the partition function Z_ψ at convergence and maximizing L_D with respect to ψ is the same as maximizing max ent RL objective for q_θ (this is the opposite as in guided cost learning where ψ updates maximized the max ent IRL objective instead). Similarly, it can be show that minimizing L_G with respect to θ is the same as maximizing the max ent IRL objective (this is opposite to guided cost learning where θ updates maximized the max ent RL objective for q_θ instead. Another thing to notice is that the paper adds an additional

$$\mathbb{E}_{\tau \sim q_\theta(\tau)} [-\log D(\tau)]$$

to the standard generator loss to give more training signal when the discriminator is very accurate (typically at the start of training), which causes the gradient of the vanilla generator objective to be small.

Note that if replace the discriminator with a normal discriminator that is a neural network without the special parameterization and interleave optimizing a policy to maximize the Discriminator prediction on generated samples (basically just policy gradient but with $D(\tau)$ as the reward), we can recover a policy that is hard to distinguish from the expert demonstrations, essentially performing behavioral cloning.

7.4.3 Relationship Between Max Ent IRL and Feature Matching

In the case where the reward is linear in ψ , the problem

$$\max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\sum_t r_\psi(s_t, a_t) \right] - \log \int_{\tau'} P(\tau') \exp(\sum_t r_\psi(s_t, a_t)) d\tau'$$

is equivalent to

$$\begin{aligned}
&\min_{P(\tau)} -H(P(\tau)) \\
&\text{subject to } \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [f(\tau)] = \mathbb{E}_{\tau \sim P(\tau)} [f(\tau)] \\
&\sum_{\tau} P(\tau) = 1 \\
&P(\tau) \geq 0 \quad \forall \tau
\end{aligned}$$

where $f(\tau) = \sum_{(s_t, a_t) \in \tau} f(s, a)$. The variables being optimized are $P(\tau)$ for every possible trajectory τ , which we group in $\vec{P}(\tau)$. Since $-H(P(\tau))$ is convex and the constraints are affine in τ (convex constraints), we just need any feasible point for strong duality to hold by Slater's condition. One such solution is to allocate all probability to one trajectory.

The dual of the maximum entropy (min negative entropy) problem can be written as

$$\begin{aligned}
&\max_{\psi, \nu, \lambda \geq 0} \min_{P(\tau)} \sum_{\tau} P(\tau) \log P(\tau) + \psi^T \left(\mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [f(\tau)] - \sum_{\tau} P(\tau) f(\tau) \right) + \nu \left(\sum_{\tau} P(\tau) - 1 \right) - \lambda^T P(\vec{\tau}) \\
&= \max_{\psi, \nu, \lambda \geq 0} \min_{P(\tau)} \sum_{\tau} P(\tau) \log P(\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\psi^T f(\tau)] - \sum_{\tau} P(\tau) \psi^T f(\tau) + \nu \left(\sum_{\tau} P(\tau) - 1 \right) - \lambda^T P(\vec{\tau}) \\
&= \max_{\psi, \nu, \lambda \geq 0} \min_{P(\tau)} \sum_{\tau} P(\tau) \log P(\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \sum_{\tau} P(\tau) r_\psi(\tau) + \nu \left(\sum_{\tau} P(\tau) - 1 \right) - \lambda^T P(\vec{\tau})
\end{aligned}$$

where $r_\psi(\tau) = \sum_{(s_t, a_t) \in \tau} r_\psi(s_t, a_t)$. The Lagrangian is convex in $P(\tau)$, so taking the gradient and setting it equal to 0, we get that

$$\begin{aligned} & \nabla_{P(\vec{\tau})} \left(\sum_{\tau} P(\tau) \log P(\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \sum_{\tau} P(\tau) r_\psi(\tau) + \nu (\sum_{\tau} P(\tau) - 1) - \lambda^T P(\vec{\tau}) \right) \\ &= \vec{g} \end{aligned}$$

where $\vec{g}$ has length equal to the number of trajectories $|\tau|$ and the index of $\vec{g}$ corresponding to a particular trajectory τ is equal to

$$(1 + \log P(\tau)) - r_\psi(\tau) + \nu - \lambda_\tau$$

Setting $\vec{g}$ equal to $\vec{0}$, we get the following element-wise equation

$$\begin{aligned} (1 + \log P(\tau)) - r_\psi(\tau) + \nu - \lambda_\tau &= 0 \\ \implies \log P(\tau) &= r_\psi(\tau) - \nu - 1 + \lambda_\tau \\ \implies P(\tau) &= \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) \\ \implies P(\vec{\tau}) &= \exp(r_\psi(\vec{\tau}) - \nu \vec{1} - \vec{1} + \lambda) \end{aligned}$$

Plugging this back into the dual, we get that

$$\begin{aligned} & \max_{\psi, \nu, \lambda \geq \vec{0}} \min_{P(\tau)} \sum_{\tau} P(\tau) \log P(\tau) + \psi^T \left(\mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [f(\tau)] - \sum_{\tau} P(\tau) f(\tau) \right) + \nu (\sum_{\tau} P(\tau) - 1) - \lambda^T P(\vec{\tau}) \\ &= \max_{\psi, \nu, \lambda \geq \vec{0}} \min_{P(\tau)} \sum_{\tau} P(\tau) \log P(\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \sum_{\tau} P(\tau) r_\psi(\tau) + \nu (\sum_{\tau} P(\tau) - 1) - \lambda^T P(\vec{\tau}) \\ &= \max_{\psi, \nu, \lambda \geq \vec{0}} \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) \log \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] \\ & \quad - \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) r_\psi(\tau) + \nu (\sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) - 1) - \lambda^T \exp(r_\psi(\vec{\tau}) - \nu \vec{1} - \vec{1} + \lambda) \\ &= \max_{\psi, \nu, \lambda \geq \vec{0}} \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) (r_\psi(\tau) - \nu - 1 + \lambda_\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] \\ & \quad - \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) r_\psi(\tau) + \nu (\sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) - 1) - \lambda^T \exp(r_\psi(\vec{\tau}) - \nu \vec{1} - \vec{1} + \lambda) \\ &= \max_{\psi, \nu, \lambda \geq \vec{0}} \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) (0 - 0 - 1 + \lambda_\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] \\ & \quad - 0 + \nu(0 - 1) - \lambda^T \exp(r_\psi(\vec{\tau}) - \nu \vec{1} - \vec{1} + \lambda) \\ &= \max_{\psi, \nu, \lambda \geq \vec{0}} \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) (-1 + \lambda_\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] \\ & \quad - \nu - \lambda^T \exp(r_\psi(\vec{\tau}) - \nu \vec{1} - \vec{1} + \lambda) \\ &= \max_{\psi, \nu, \lambda \geq \vec{0}} \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) (-1 + \lambda_\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \nu - \sum_{\tau} \lambda_\tau \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) \\ &= \max_{\psi, \nu, \lambda \geq \vec{0}} \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) (-1 + 0) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \nu - 0 \\ &= \max_{\psi, \nu, \lambda \geq \vec{0}} - \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \nu \end{aligned}$$

Here, we see that the objective increases as lambda increases, so the maximizing $\lambda^* = \vec{0}$, giving us

$$\begin{aligned} & \max_{\psi, \nu, \lambda \geq \vec{0}} \min_{P(\tau)} \sum_{\tau} P(\tau) \log P(\tau) + \psi^T \left(\mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [f(\tau)] - \sum_{\tau} P(\tau) f(\tau) \right) + \nu (\sum_{\tau} P(\tau) - 1) - \lambda^T P(\vec{\tau}) \\ &= \max_{\psi, \nu, \lambda \geq \vec{0}} - \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1 + \lambda_\tau) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \nu \\ &= \max_{\psi, \nu} - \sum_{\tau} \exp(r_\psi(\tau) - \nu - 1) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_\psi(\tau)] - \nu \end{aligned}$$

Since the objective is convex in ν , we can also set the derivative with respect ν to 0 and solve for the optimal ν^*

$$\begin{aligned}
& \sum_{\tau} \exp(r_{\psi}(\tau) - \nu^* - 1) - 1 = 0 \\
\implies & \sum_{\tau} \exp(r_{\psi}(\tau) - \nu^* - 1) = 1 \\
\implies & \exp(-\nu^*) = \frac{1}{\sum_{\tau} \exp(r_{\psi}(\tau) - 1)} \\
\implies & \nu^* = -\log \frac{1}{\sum_{\tau} \exp(r_{\psi}(\tau) - 1)} \\
& = \log \sum_{\tau} \exp(r_{\psi}(\tau) - 1)
\end{aligned}$$

Plugging in the optimal ν^* , the problem becomes

$$\begin{aligned}
& \max_{\psi, \nu, \lambda \geq 0} \min_{P(\tau)} \sum_{\tau} P(\tau) \log P(\tau) + \psi^T \left(\mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [f(\tau)] - \sum_{\tau} P(\tau) f(\tau) \right) + \nu (\sum_{\tau} P(\tau) - 1) - \lambda^T P(\vec{\tau}) \\
& = \max_{\psi, \nu} - \sum_{\tau} \exp(r_{\psi}(\tau) - \nu - 1) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] - \nu \\
& = \max_{\psi} - \sum_{\tau} \exp \left(r_{\psi}(\tau) - \log \left(\sum_{\tau'} \exp(r_{\psi}(\tau') - 1) \right) - 1 \right) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] - \log \sum_{\tau} \exp(r_{\psi}(\tau) - 1) \\
& = \max_{\psi} - \sum_{\tau} \exp \left(r_{\psi}(\tau) - \log \left(\sum_{\tau'} \exp(r_{\psi}(\tau') - 1) \right) - 1 \right) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] - \log \sum_{\tau} \exp(r_{\psi}(\tau) - 1) \\
& = \max_{\psi} - \exp \left(-\log \sum_{\tau'} \exp(r_{\psi}(\tau') - 1) \right) \sum_{\tau} \exp(r_{\psi}(\tau) - 1) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] - \log \sum_{\tau} \exp(r_{\psi}(\tau) - 1) \\
& = \max_{\psi} - \frac{1}{\sum_{\tau'} \exp(r_{\psi}(\tau') - 1)} \sum_{\tau} \exp(r_{\psi}(\tau) - 1) + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] - \log \sum_{\tau} \exp(r_{\psi}(\tau) - 1) \\
& = \max_{\psi} -1 + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] - \log \left(\exp(-1) \sum_{\tau} \exp(r_{\psi}(\tau)) \right) \\
& = \max_{\psi} -1 + \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] + 1 - \log \sum_{\tau} \exp(r_{\psi}(\tau)) \\
& = \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [r_{\psi}(\tau)] - \log \sum_{\tau} \exp(r_{\psi}(\tau)) \\
& = \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\log \exp(r_{\psi}(\tau))] - \log \sum_{\tau} \exp(r_{\psi}(\tau)) \\
& = \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \exp(r_{\psi}(\tau)) - \log \sum_{\tau'} \exp(r_{\psi}(\tau')) \right] \\
& = \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{\exp(r_{\psi}(\tau))}{\sum_{\tau'} \exp(r_{\psi}(\tau'))} \right]
\end{aligned}$$

Assuming deterministic dynamics and a uniform prior over actions, we get that

$$\begin{aligned}
& \max_{\psi, \nu, \lambda \geq 0} \min_{P(\tau)} \sum_{\tau} P(\tau) \log P(\tau) + \psi^T \left(\mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [f(\tau)] - \sum_{\tau} P(\tau) f(\tau) \right) + \nu (\sum_{\tau} P(\tau) - 1) - \lambda^T P(\vec{\tau}) \\
& = \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} \left[\log \frac{\exp(r_{\psi}(\tau))}{\sum_{\tau'} \exp(r_{\psi}(\tau'))} \right] \\
& = \max_{\psi} \mathbb{E}_{\tau \sim p_{\pi^*}(\tau)} [\log P(\tau | O_{1:t}, \psi)]
\end{aligned}$$

which was our original objective. So basically, maximizing the probability of the trajectories in the graphical model (e.g. making the soft optimal trajectories look as close as the expert trajectories) is the same as feature matching

while making the policy as random as possible when disambiguating between multiple solutions when the reward is linear in the features and the dynamics are deterministic.